

**Test Report on the Supplementary Test of
the OPSIS SM 200 Emission Measuring
Instrument with PM10 Pre-separator
marketed by OPSIS AB for the PM10
Suspended Particles Fraction**

**TÜV Report No.: 936/21201592/A, TÜV Immission Control and
Power Systems GmbH's Immission Control Unit, Cologne,
Germany, 5 July 2005**

(English translation)

Any publication of this report in excerpts requires our written approval.
TÜV Immission Control and Power Systems GmbH
TÜV Rheinland Group
51105 Köln, Am Grauen Stein, Phone 0221/806-2756, Fax 0221/806-1349

**TÜV IMMISSION CONTROL
AND POWER SYSTEMS GMBH**

Report on the Supplementary Test of the OPSIS
SM 200 Emission Measuring Instrument with
PM10 Pre-separator marketed by OPSIS AB
for the PM10 Suspended Particles Fraction

TÜV Report: 936/21201592/A
Cologne, 5 July 2005

TÜV Immission Control and Power Systems GmbH's Immission Control Unit
is DIN EN ISO/IEC 17025 and DIN EN ISO 9002 accredited for the following working areas:

- Determination of air pollutant and odorant emissions/immissions
- Inspection for proper installation, operation and calibration of continuous emission measuring devices including data analyzing and remote emission measurement equipment
- Suitability testing of measuring instruments for continuous emission / immission monitoring and of electronic data analyzing and remote emission measurement systems.

The aforementioned accreditation is valid until 4 December 2005.
German Accreditation Council (DAR) registration No.: DAP-PL-3856.99

TÜV Immission Control and Power Systems GmbH
Air Quality Management

Report on the Supplementary Tests of the OPSIS SM 200 Emission
Measuring Instrument with PM10 Pre-separator marketed by OPSIS AB
for the PM10 Suspended Particulates Fraction, Report No.:
936/21201592/A

936/21201592/A





Report on the Supplementary Test of the OPSIS SM 200 Emission
Measuring Instrument with PM10 Pre-separator marketed by OPSIS AB
for the PM10 Suspended Particulates Fraction

Measuring instrument tested:	OPSIS SM 200 Emission Measuring Instrument with PM10 Pre-separator
Equipment manufacturer:	OPSIS AB Box 244 SE-244 02 Furulund Sweden
Testing period:	from: January 2004 until: April 2005
Report date:	July 5, 2005
Report number:	936/21201592/A
Scope of report:	Total: 306 pages Appendices starting on page 138 Manual (98 pages) starting on page 208.

TÜV Immission Control and Power Systems GmbH
Air Quality Management

Report on the Supplementary Tests of the OPSIS SM 200 Emission
Measuring Instrument with PM10 Pre-separator marketed by OPSIS AB
for the PM10 Suspended Particulates Fraction, Report No.:
936/21201592/A

936/21201592/A



TABLE OF CONTENTS

1	SUMMARY AND ANNOUNCEMENT PROPOSAL	4
1.1	Summary	4
1.2	Announcement proposal	7
1.3	Summary presentation of test results	9
2	OBJECTIVE	15
2.1	Test type	15
2.2	Purpose	15
3	DESCRIPTION OF THE TESTED INSTRUMENT	18
3.1	Measuring principle	18
3.2	Operation of the instrument	18
3.3	Scope and structure of the instrument	20
3.4	Comparison between OPSIS SM 200 (New) and OPSIS SM 200 (Old)	26
4	TEST PROGRAM	30
4.1	General	30
4.2	Laboratory test	31
4.3	Field test	32
5	REFERENCE MEASURING METHOD	43
6	TEST RESULTS	45
6.1	4.1.1 Display of measured values	45
6.1	4.1.2 Maintenance friendliness	46
6.1	4.1.3 Functional test	47
6.1	4.1.4 Setup and run-in times	49
6.1	4.1.5 Design	51
6.1	4.1.6 Unauthorized change of settings	52

6.1	4.1.7	Measuring signal output.....	53
6.1	4.2	Requirements on measuring systems for mobile use	55
6.1	5.1	General	56
6.1	5.2.1	Measuring range	57
6.1	5.2.2	Negative measuring signals.....	58
6.1	5.2.3	Analytical function.....	59
6.1	5.2.4	Linearity	63
6.1	5.2.5	Limit of detection.....	64
6.1	5.2.6	Response time	66
6.1	5.2.7	Zero point dependence on ambient temperature.....	67
6.1	5.2.8	Dependence of the measured value on ambient temperature.....	69
6.1	5.2.9	Zero point drift.....	71
6.1	5.2.10	Measured value drift	76
6.1	5.2.11	Cross-sensitivity	81
6.1	5.2.12	Repeatability	82
6.1	5.2.13	Hourly values	84
6.1	5.2.14	Mains voltage and mains frequency	85
6.1	5.2.15	Power failure.....	87
6.1	5.2.16	Equipment functions	88
6.1	5.2.17	Mode switching	89
6.1	5.2.18	Availability	90
6.1	5.2.19	Converter efficiency	93
6.1	5.2.20	Maintenance interval.....	94
6.1	5.2.21	Overall uncertainty.....	95
6.1	5.3.1	Equivalence of the sampling system	104
6.1	5.3.2	Sampling system comparability	111
6.1	5.3.3	Calibration.....	117

6.1	5.3.4	Cross-sensitivity.....	118
6.1	5.3.5	Daily average.....	121
6.1	5.3.6	Constancy of the sampling flow rate.....	122
6.1	5.3.7	Tightness of the sampling system	125
6.1	5.4	Requirements on multi-constituent measuring systems	127
7		RECOMMENDATIONS ON PRACTICAL USE	136
7.1		Work in the maintenance interval	136
7.2		Functional test and calibration.....	136
8		LITERATURE	137
9		APPENDICES.....	138



TÜV Rheinland Group

Page 8 of 306

TÜV Immission Control and Power Systems GmbH
Air Quality Management

Report on the Supplementary Tests of the OPSIS SM 200 Emission Measuring
Instrument with PM10 Pre-separator marketed by OPSIS AB for the PM10
Suspended Particulates Fraction, Report No.: 936/21201592/A

1 Summary and Announcement Proposal

1.1 Summary

Pursuant to the first 1st Daughter Directive 1999/30/EC of 22 April 1999 "relating to limit values for sulphur dioxide, nitrogen dioxide and oxides of nitrogen, particulate matter and lead in ambient air" to the Air Quality Directive 96/62/EC of 27 September 1996 "on ambient air quality assessment and management," the reference methods to be employed for PM10 concentration measurements are those described in EN 12341 "Determination of the PM10 fraction of suspended particulate matter - Reference method and field test procedure to demonstrate reference equivalence of measurement methods". However, a member state is free to use any other method which can be demonstrated to "display a consistent relationship to the reference method. In that event the results achieved by that method must be corrected by a relevant factor to produce results equivalent to those that would have been achieved by using the reference method" (1999/30/EC, Annex IX, Art. IV, 2nd paragraph).

The guideline VDI 4202, Sheet 1, of June 2002 describes the "Minimum requirements on automatic air quality measuring instruments in the suitability test". The general boundary conditions for the associated tests are detailed in the guideline VDI 4203, Sheet 1, "Test procedures for point-related ambient air quality measuring systems of gaseous and particulate pollutants" of October. A more detailed presentation of these boundary conditions is provided in VDI 4203, Sheet 3, Testing of automatic measuring systems – Test procedures for point-related ambient air measuring systems of gaseous and particulate pollutants", (August 2004).

TÜV Immission Control and Power Systems GmbH had been commissioned by OPSIS AB to conduct suitability tests of the OPSIS SM 200 measuring instrument for the PM10 Suspended Particulates fraction. The Suitability Announcement for this measuring instrument was published in the official part of the Federal Gazette [*Bundesanzeiger*] of May 15, 2003.

The measuring instrument was then modified by OPSIS AB subsequent to this announcement. This made it necessary to determine and, if applicable, to confirm the conformity of the modified OPSIS SM 200 measuring instrument with PM10 pre-separator (referred to as SM200 (New) for clarity's sake hereinbelow) with the original measuring instrument (referred to as SM200 (Old) for clarity's sake hereinbelow) by means of a supplementary test.

The report on the suitability tests of the original measuring instrument (Report No. 936/801013/A of January 29, 2003) has been used as the basis for the present Supplementary Test Report.

The supplementary test was carried out in compliance with the following regulations and requirements:

- VDI Guideline 4202, Sheet 1, „Minimum requirements for suitability tests of automated ambient air quality measuring systems - Point-related measurement methods of gaseous and particulate pollutants“, June 2002
- VDI Guideline 4203, Sheet 3, „Testing of automated measuring systems - Test procedures for point-related ambient air quality measuring systems of gaseous and particulate pollutants“, August 2004
- European Standard EN 12341, „Air quality - Determination of the PM10 fraction of suspended particulate matter - Reference method and field test procedure to demonstrate reference equivalence of measurement methods“, German version EN 12341: 1998

The OPSIS SM 200 measuring instrument draws ambient air onto a filter via a PM10 sampling head by means of a pump. The dust mass deposited on the filter is then determined, upon completion of the sampling operation, by the radiometric beta absorption measuring principle. In addition, the filter can be subjected to gravimetric measurements and remains available for additional analytical procedures, e.g., an analysis of heavy metals.

The investigations were carried out in the laboratory and in the course of a field test of several months' duration.

Field testing took place over several months at the sites summarized in Table 1 and Table 2.

Table1: Description of the measuring sites

	Parking lot area, Cologne	Wesseling	Mechernich	Brühl
Time period	12/2000 – 02/2001	04/2002 – 07/2002	07/2002 – 11/2002	11/2002 – 01/2003
Number of measurements	39	62	114	45
Description	urban background	industrial area	rural structure	gravel pit
Level of air pollution	average to high	high	low	average

As part of the supplementary test scheme, a field test lasting several months was also carried out with the modified measuring instruments at a location in Sweden:

Table 2: Description of the site used for the supplementary tests

	Furulund
Time period	12/2004 – 04/2005
Number of measurements	104
Description	rural structure
Level of air pollution	low to average

The conditions laid down in the Minimum Requirements were met both in the suitability tests and during the supplementary testing campaign.

TÜV Immission Control and Power Systems GmbH therefore proposes an announcement of this device as a suitability-tested measuring instrument for the continuous measurement of PM10 suspended particulate immissions.

1.2 Announcement proposal

In view of the favourable results achieved, the following recommendation for announcement as a suitability-tested measuring instrument is herewith issued:

- | | | | |
|---------------|--|---|---|
| 1.2.1 | Measuring task | : | Continuous recording of immissions of PM10 suspended particulates |
| 1.2.2 | Device name | : | OPSIS SM 200 with PM10 Pre-separator |
| 1.2.3 | Measured fractions | : | PM10 suspended particulates |
| 1.2.4 | Manufacturer | : | OPSIS AB
Box 244, SE-244 02 Furulund, Sweden |
| 1.2.5 | Suitability | : | for continuous immissions measurement of PM10 fraction
suspended particulates in stationary use |
| 1.2.6 | Measuring ranges covered in the suitability tests | : | 0 to 200 µg/m ³ |
| 1.2.7 | Software version | : | Version 1.03 (OPSIS SM 200 (New)) |
| 1.2.8 | Restrictions | : | none |
| 1.2.9 | Notes | : | <ol style="list-style-type: none">1. Supplementary test to the suitability announcement made in the Federal Gazette 2003, No. 90, p. 107422. The two versions of the measuring instruments can be reliably distinguished by their serial number:
SN < 1000 = OPSIS SM 200, old version
TÜV Report No.: 936/801013A
SN > 1000 = OPSIS SM 200, new version
TÜV Report No.: 936/21201592/A3. The measuring instrument is also marketed under the name Aeris AB, Box 244, 244 02 Furulund, Sweden.4. The measuring instrument must be operated in a lockable measuring container.5. Linearity testing of the radiometric measuring function requires various reference films made available by the equipment manufacturer.6. The inlet tube must be flushed with ambient air up to the analyzer port (Option C). |
| 1.2.10 | Testing institute | : | TÜV Immission Control & Power Systems GmbH, Cologne
TÜV Rheinland Group
Responsible Examiner: Karsten Pletscher, Dipl.-Ing. |
| 1.2.11 | Test report | : | 936/21201592/A dated July 5, 2005 |

1.3 Summary presentation of test results

Minimum requirement	Requirement	Test result	Fulfilled	Page
4. Design requirements				
4.1 General requirements				
4.1.1 Measured value display	Must be provided.	The instrument possesses a measured value display.	yes	49
4.1.2 Maintenance-friendliness	Maintenance work should be possible from outside the instrument without excessive effort.	Maintenance work can be performed with common tools and reasonable effort from outside the instrument.	yes	50
4.1.3 Functional check	Any special devices intended for this purpose shall be deemed part of the instrument and shall be used and assessed in the respective sub-tests. Test gas generating systems must report their readiness for operation to the measuring instrument via status signals, and must be controllable either directly or telemetrically. The uncertainty of this test gas generating system shall not exceed 1% of B2 in three months.	All equipment functions described in the operating manual are provided, available and operable.	yes	51
4.1.4 Setup and run-in times	Information on this subject must be provided in the Operating Instructions.	Setup and run-in times were determined.	yes	54
4.1.5 Design	Information on this subject must be provided in the Operating Instructions	The design information given in the Operating Manual is complete and correct.	yes	55
4.1.6 Unauthorized manipulation	A safeguard against tampering must be provided	The instrument must be installed in a locked measuring container.	yes	56
4.1.7 Measurement signal output	Must be supplied in digital and/ or analog form	Measurement signals are made available in analog (0-10V) and digital (via RS232) format.	yes	57
4.2 Requirements for mobile use	Continuous readiness for operation must be ensured; requirements for stationary use shall be similarly met in mobile service.	The instrument was operated at various sites as part of the field testing program. However, it cannot be run out of a moving vehicle.	no	59

Minimum requirement	Requirement	Test result	Fulfilled	Page
5. Performance requirements				
5.1 General	Manufacturer's information given in the Operating Instructions shall not be inconsistent with results of the suitability test.	No differences between equipment features and the Manuals were identified.	yes	60
5.2 General requirements				
5.2.1 Measuring range	Measuring range must exceed B ₂ .	The instrument is set to a 0-200 µg/m ³ or 0-1000 µg/m ² measuring range.	yes	61
5.2.2 Negative measurement signals	Must not be suppressed (live zero)	Negative measurement signals are not directly indicated by the instrument. However, a negative measurement value can be calculated on the basis of stored data in the individual case where necessary for quality assurance purposes.	yes	62
5.2.3 Analytical function	Relationship between output signal and measured variable must be expressible by an analytical function and determinable by regression analysis.	Formation of hourly values of the PM10 fine dust fraction is not necessary for monitoring the relevant threshold values.	yes	88
5.2.4 Linearity	Group averages of the readings shall not deviate from the calibration function by more than 5% of B ₁ in the 0 ... B ₁ range, nor by more than 1% of B ₂ in the 0 ... B ₂ range.	For dust measuring instruments this test is to be performed in accordance with minimum requirement 5.3.1, "Sampling System Equivalence".	yes	67
5.2.5 Detection limit	Must not exceed B ₀ .	The tests yielded a detection limit of 0.41 µg/m ³ for instrument 1 (SN 285) and of 1.08 mg/m ³ for instrument 2 (SN 288) (based on instrument output values). Calculations based on cpm gave a detection limit of 0.96 µg/m ³ for instrument 1 (SN 285) and of 1.29 µg/m ³ for instrument 2 (SN 288).	yes	67
5.2.6 Response time	Must not exceed 5% of averaging period (= 180 seconds)	Not applicable	-	70
5.2.7 Zero point dependence on ambient temperature	Zero point reading shall not exceed B ₀ in response to a temperature change ΔT _u of 15 K between +5°C und +20 °C, or of 20 K between +20°C und +40°C	No ambient temperature influence on the zero point was observed on the basis of instrument output. An analysis based on values calculated from cpm showed maximum deviations of 0.42 µg/m ³ for instrument 1 (SN 285) and of 0.48% µg/m ³ for instrument 2 (SN 288).	yes	71

Minimum requirement	Requirement	Test result	Fulfilled	Page
5.2.8 Measured value dependence on ambient temperature	Measured value in the range of B ₁ shall not exceed $\pm 5^{\circ}\text{C}$ in the case of a temperature change (ΔT_u) by 15 K between $+5^{\circ}\text{C}$ und $+20^{\circ}\text{C}$, or by 20 K between $+20^{\circ}\text{C}$ and $+40^{\circ}\text{C}$	Deviations from the baseline value at 20°C were found not to exceed $> -0.58\%$ (instrument 1 (SN 285)) and $> -1.2\%$ (instrument 2 (SN 288)).	yes	73
5.2.9 Zero point drift	Not to exceed B ₀ over 24 hrs. and during the maintenance interval	All measurements taken lie within permitted limits.	yes	75
5.2.10 Measured value drift	Not to exceed 5% of B ₁ over 24 hrs. and during the maintenance interval	Measured value drift amounted to max. 3.14% (SN 285) and 2.4% (SN 288), respectively, during the maintenance interval. In the supplementary test, the maximum measured value drift was found to be 0.34% (SN 1110) and -0.80% (SN 1112), respectively.	yes	80
5.2.11 Cross-sensitivity	Must exceed neither B ₀ in the zero range, nor 3% of B ₂ in the range of B ₂ .	Not applicable	-	85
5.2.12 Repeatability	$R_D \geq 10$ related to B ₁	Repeatability in the field test was at least 10-16. In the supplementary test, a repeatability of at least 12 was determined.	yes	86
5.2.13 Hourly values	Formation must be possible.	Formation of hourly values of the PM10 fine dust fraction is not necessary for monitoring the relevant threshold values.	not applicable	88
5.2.14 Power supply voltage and frequency	Measured value at B ₁ shall not change by more than B ₀ in the 230_{-20}^{+15}V voltage range. In mobile use, the measured value shall not change by more than B ₀ in the $(50 \pm 2)\text{Hz}$ frequency range.	Power supply fluctuations did not cause deviations from the previous value by $> 0.56\%$ in instrument 1 (SN 285) or $> 0.77\%$ in instrument 2 (SN 288). The impact of power supply frequency changes on the measurement signal was found not to exceed -0.45% for instrument 1 (SN 285) and -0.57% for instrument 2 (SN 288) of the previous value.	yes	89
5.2.15 Power failure	Uncontrolled release of operating and calibrating gas must be prevented; instrument parameters must be buffer-protected against loss; instrument must be ready to resume measurement upon power recovery, any ongoing measurement must be continued.	All instrument parameters are protected against loss by buffering. Instrument is ready to operate and in error-free state upon power recovery.	yes	91

Minimum requirement	Requirement	Test result	Fulfilled	Page
5.2.16 Instrument functions	Monitoring by telemetrically transmittable status signals must be possible.	The instruments can be modem-controlled from a remote computer in the same manner as if operated locally.	yes	92
5.2.17 Mode switching	Measuring, functional control and/or calibration modes must be triggerable telemetrically and manually.	As a rule, all functional checking and calibration steps can be performed either directly on the instrument or by telemetric remote control.	yes	93
5.2.18 Availability	At least 90 %	The availability of both instruments was determined at 97.8% (without test-related downtime) and 90.6% (including test-related downtime), respectively. In the supplementary test, the availability of both instruments was determined at 100% (without test-related downtime) and 89.7% (including test-related downtime), respectively.	yes	94
5.2.19 Converter effectiveness	At least 95%	Not applicable	-	97
5.2.20 Maintenance interval	If possible, 28 days. Not less than 14 days.	The maintenance interval is determined by the necessary maintenance steps and has a duration of 1 month.	yes	98
5.2.21 Overall uncertainty	Compliance with data quality requirements [G10 through G12]	The overall uncertainty adds up to 15.4% and 15.1% for U(c), and to 10.6% for both systems for U(c). In the supplementary test, the overall uncertainty was found to be 8.6% or 9.1%, respectively, for U(c), and 9.0% for both systems for U(c).	yes	99
5.3	Requirements on measuring instruments for particulate air pollutants			
5.3.1 Sampling system equivalence	Equivalence to the reference method according to DIN EN 12 341 [Part 2] shall be demonstrated.	The reference equivalence functions lie within the limits of the relevant acceptance range. The variation coefficient R^2 of the calculated reference equivalence functions is $\geq 0,95$ in the relevant concentration range. The above holds true for both the OPSIS 200 (old) and the OPSIS SM 200 (new) systems.	yes	109

Minimum requirement	Requirement	Test result	Fulfilled	Page
5.3.2 Sampling system comparability	To be demonstrated in the field test per DIN EN 12341 [Part 2] using two sampling systems of identical design.	The two-sided confidence interval Cl_{95} was $\leq 3.96 \mu\text{g}/\text{m}^3$, i.e., below the specified value of $5 \mu\text{g}/\text{m}^3$. In the supplementary test the two-sided confidence intervals Cl_{95} varied between a minimum of $2.28 \mu\text{g}/\text{m}^3$ and a maximum of $3.29 \mu\text{g}/\text{m}^3$, thus remaining below the specified limit of $5 \mu\text{g}/\text{m}^3$ for all possible instrument combinations.	yes	119
5.3.3 Calibration	Via comparative measurements in the field test against the DIN EN 12 341 [Part 2] reference method; the correlation between the measurement signal and the gravimetric reference concentration is to be determined as a continuous function.	Refer to module 5.2.3	-	125
5.3.4 Cross-sensitivity	Not to exceed 10% of B_1	Impact of air humidity in the sample flow on the zero signal was not found to exceed $0,2 \mu\text{g}/\text{m}^3$ deviation from the target value (determined on the basis of instrument output). The impact of air humidity in the sample flow on instrument sensitivity could not be tested. However, changing relative air humidities during the field tests had no observable effect on instrument readings.	yes	126
5.3.5 One-day averages	Instrument must be capable of forming 24-hr. averages; filter change time not to exceed 1% of averaging period.	The formation of one-day averages is possible. The maximum filter change time in 24-hour sampling mode is approx. 0,28 % of the averaging period.	yes	129
5.3.6 Sampling volume constancy	$\pm 3 \%$ of target value over the sampling time, transients not to exceed $\pm 5 \%$ of target value over the sampling time.	No daily average deviates from the target value by $\pm 3\%$ or more. Transient values were always within 5% of the target.	yes	130
5.3.7 Tightness of the sampling system	Leakage shall not exceed 1 % of the sampled volume.	The maximum determined leak flow was 0.79% for instrument 1 (SN 285) and 0.82% for instrument 2. In the supplementary test, the maximum leakage rate on the OPSIS SM 200 (new) was found to be 0.98%.	yes	134
5.4 Requirements on multi-constituent measuring instruments	Must be met for each individual constituent when all measuring channels are operated simultaneously; in sequential mode the formation of hourly averages must be possible	Not applicable	-	135

2 Objective

2.1 Test type

TÜV Immission Control and Power Systems GmbH had been commissioned by OPSIS AB to perform a suitability test of the OPSIS SM 200 measuring instrument with PM10 pre-separator. The test was carried out in the form of a supplementary test based on TÜV Report No. 936/801013/A dated January 29, 2003.

2.2 Purpose

Following a modification of the SM 200 measuring instrument with PM10 pre-separator, a new suitability test of this instrument was carried out on behalf of OPSIS AB of Sweden. The test was performed as a supplementary test, i.e., it supplements the suitability examination of the previous SM 200 measuring instrument with PM10 pre-separator.

The modification of the measuring instrument was effected in the context of OPSIS AB's changeover from a licensed manufacture of this measuring instrument to its fully independent production.

Suitability testing of the modified measuring system was executed in the form of a supplement to the previous suitability tests, given that the structure and concept of the existing measuring instrument were not changed in any way whatsoever. All relevant parts of the modified instrument have been taken from its predecessor unit and are used in the same manner. The modifications are limited mainly to exterior design changes, the addition of quality assurance sensors, and a re-design of the instrument's electronic circuitry (man/machine interface, new menu structure, additional functions). All functionalities available in the prior version are also available on the current one. The differences between the new version, hereinbelow referred to as SM 200 (New), and its predecessor, hereinbelow referred to as SM 200 (Old), are described in detail in Chapter 3 of this report.

The present report is based partially on results of the previous suitability test, as set forth in the suitability test report for the SM 200 measuring instrument with PM10 pre-separator (TÜV Report No. 936/801013/A of January 29, 2003 [10]. These prior results were extracted where applicable and adapted as appropriate for the purposes of the present report. Where entire sections were taken from the previous document, these have been highlighted in colour. All relevant test criteria calling for renewed testing of the modified instrument, or for additions to the tests described in [10], are covered by the present report.

Scope of the supplementary test

The supplementary test of the modified OPSIS SM 200 (New) measuring instrument were carried out in Sweden using two measuring instruments of identical design, viz.

Instrument 1: No. SN 1110 (SM 200 (New))

Instrument 2: No. SN 1112 (SM 200 (New))

Moreover, one OPSIS SM 200 unit of the original (old) type was run in parallel with the two modified units during the tests carried out in Sweden.

Instrument 3: No. SN 276 (Predecessor model SM 200 (Old))

The field tests in Sweden comprised testing for the following criteria:

- Comparability of the tested instruments (new versus new, new versus old)
- Comparability of the tested instrument with the reference method
- Calibration capability, creation of the analytical function
- Repeatability (new versus new, new versus old)
- Zero point drift and sensitivity variation over time
- Tightness of the sampling system
- Duration of the maintenance interval
- Availability rate
- Overall uncertainty of the tested instrument

The option to conduct extensive re-testing of the measuring instrument in the laboratory was waived since no changes in instrument performance are to be expected, given that the concept and principal structure of the device were taken in full from the previous model and that all components relevant to its performance (mechanical, pneumatic and radiometric systems) are of identical type and execution. As part of the test program, checks were conducted to establish that none of the original functions have deteriorated or been eliminated. Any improvements and/or additions to the instrument's functionalities are listed under the respective criteria.

Where entire test items or portions thereof have been copied in unchanged form from the report on the predecessor model, these passages are highlighted in colour herein.

The measuring instrument is intended to determine the PM10 fine dust content of ambient air in the 0 - 200 (1000) $\mu\text{g}/\text{m}^3$ concentration range.

The supplementary test had to be carried out in line with current suitability testing guidelines / directives and taking into account the most recent developments.

The test was carried out in compliance with the following:

- VDI Guideline 4202, Sheet 1, „Minimum requirements for suitability tests of automated ambient air quality measuring systems - Point-related measurement methods of gaseous and particulate pollutants“, June 2002 [1]
- VDI Guideline 4203, Sheet 3, „Testing of automated measuring systems - Test procedures for point-related ambient air quality measuring systems of gaseous and particulate pollutants“, August 2004 [2]
- European Standard EN 12341, „Air quality - Determination of the PM10 fraction of suspended particulate matter - Reference method and field test procedure to demonstrate reference equivalence of measurement methods“, German version EN 12341: 1998 [3]

3 Description of the tested instrument

3.1 Measuring principle

The mass determination principle is based on the physical law whereby beta rays are attenuated as they pass through a thin layer of material.

If one defines x_f as the value of the filter's mass density and x_p as the value of the mass density of the collected particles, the following equation applies:

$$F_{\text{collect}} = F_{\text{blank}} * e^{-\frac{x_p}{K(x_f)}} \quad \text{or} \quad x_p = K(x_f) \ln \frac{F_{\text{blank}}}{F_{\text{collect}}}$$

where $K(x_f)$ = mass absorption coefficient and F_{blank} and F_{collect} represent the beta ray flow before and after the sampling cycle.

Using the value of the sampling surface area S ($=11.95 \text{ cm}^2$, $\varnothing = 3.9 \text{ cm}$), the mass of the suspended particles m_p deposited on the filter can be computed as follows:

$$m_p = S * x_p$$

The function $K(x_f)$ was determined by the instrument manufacturer and is programmed into the system. After each power-up of the instrument, the stability of the initial calibration is verified by means of two reference orifices placed in the passage of the beta rays between the source and the Geiger counter. This function can also be triggered manually and is available as a regular Auto Test feature. The result of the last "beta test" can be viewed anytime.

3.2 Operation of the instrument

The following is a description of the internal processes taking place in the OPSIS SM200 instrument in its standard dust monitor version. For more detailed information on this subject the reader is referred to the Operating Manual enclosed herewith. Fig. 1 is a schematic diagram of the filter change sequence and associated operations. The content of each step is explained in the sections following the illustration.

For monitoring and ensuring the instrument's proper operation, the latter features various internal quality control tests:

- Check for calibration of the beta absorption mass determination process
- Check for calibration of the flow rate measurement
- Check of the pneumatic circuit for air-tightness
- Check for flow rate constancy
- Check of the Geiger counter and pressure transducers
- Use of optical sensors for monitoring correct mechanical operation
- Alarm messages in the case of problems/non-compliances during operation of the measuring system

A more detailed description of the various test procedures, with notes on the interpretation of test results, are given in Chapter 3 of the instrument Operating Manual.

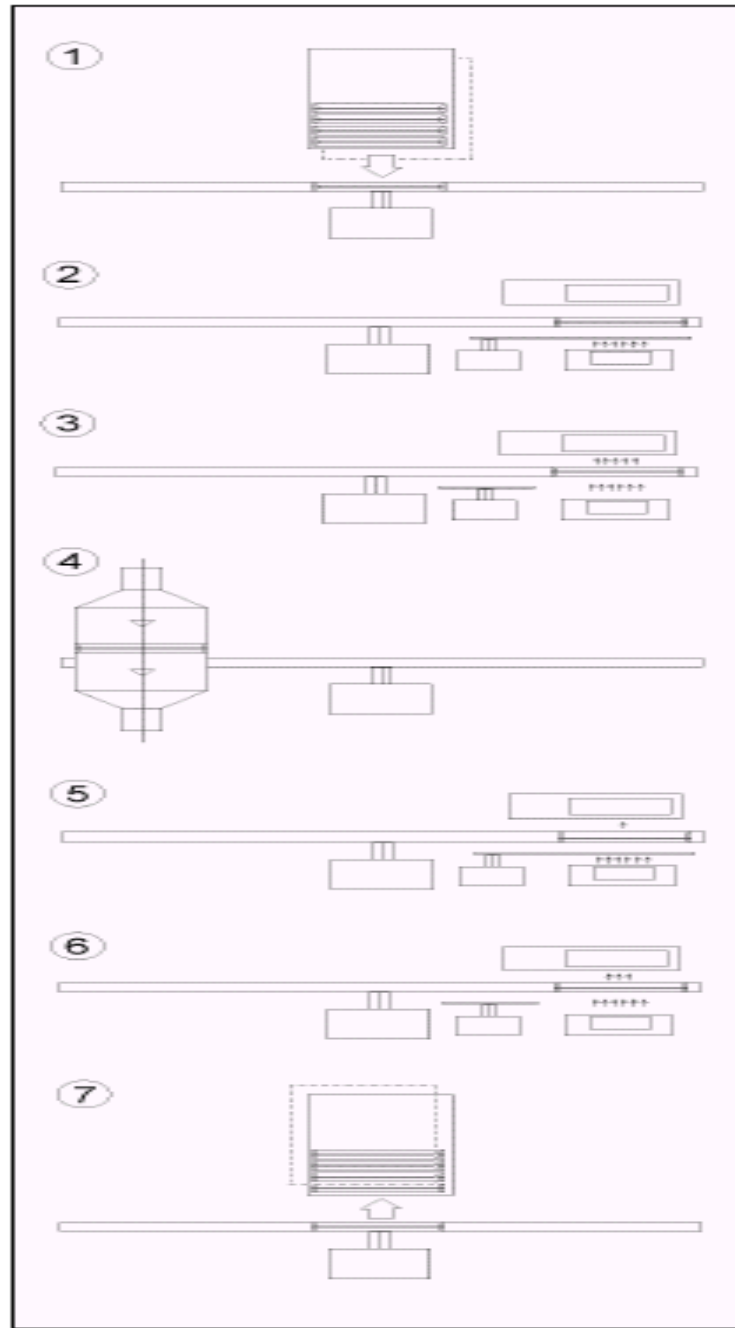


Fig. 1: Instrument operating description (passage of one filter through the instrument)

- A clean filter is taken from the clean filter container and placed on the carousel.
- The carousel (a rotating disc) moves the clean filter from Container 1 to the sampling position, then on to the measuring position, and finally to its point of unloading into Container 2 (sampled filter container). While the instrument is in operation there are always two filters on the carousel at any one time – one in the sampling position and one in the measuring position. When a new filter is placed on the carousel, the old filter is discharged at the same time.
- The background radiation of the incoming clean filter is measured by means of a Geiger counter. For this operation, the beta radiation source is covered with a screen (dark measurement).
- The screen is then moved sideways so that beta radiation reaches the Geiger counter. The new filter is measured prior to the sampling cycle (blank measurement).
- The carousel moves the filter into the sampling position. The filter is enclosed by the sampling chamber; it remains in this position throughout the sampling cycle.
- At the end of the sampling cycle the dust-laden filter is returned to the Geiger counter. The beta source is screened off and the natural radiation of the collected particles is measured (natural measurement).
- The screen is moved away so that the beta rays can pass through the filter. The Geiger counter detects the beta rays which are not absorbed by the filter and its dust load ("collect measurement").
- Upon completion of the Geiger counter measurement the sampled filter is moved to the unloading position and placed in the filter container for sampled filters.
- From the four Geiger counter measurements (items 2,3,5,6) the mass of the dust layer on the filter can be computed. Taking into account the volume flow drawn through the system, one obtains the dust concentration.

3.3 Scope and structure of the instrument

The OPSIS SM200 (new) dust monitoring unit permits the collection of suspended particulates on membrane filters and provides the option to subject the collected sample to subsequent further qualitative and quantitative analyses. In addition, the particle mass deposited on the membrane filter during sampling is determined by the beta absorption method inside the system, and the concentration of suspended dust is calculated in $\mu\text{g}/\text{m}^3$ on the basis of the volumetric flow rate.

The measuring system consists of the sampling head with inlet tube, the pump module, the sampling and measuring module, and the filter containers for the storage of clean and sampled filters. These filter containers can hold up to 40 filters.

The sampling head is a PM10 sample intake unit functioning as a pre-separator for the suspended particles drawn in from the ambient air. The system operates with a constant controlled volumetric flow rate of $16.67 \text{ l/min} = 1.0 \text{ m}^3/\text{h}$. Alternatively, the instrument can be run with TSP, PM2.5 and PM1 sample intake units.

The inlet tube links the sampling head to the sampling and measuring module. To prevent condensation inside the tube at its point of passage through the measuring container roof and to avoid losses of volatile dust constituents due to temperature variations in the inlet line to the sampling and measuring module, the inlet tube was installed with a surrounding tubular lead-through device passing through the roof. The interspace between this lead-

through pipe and the inlet tube of the measuring module is flushed with ambient air. This ensures that the inlet air will maintain its original temperature inside the inlet tube all the way to the filter. To prevent a transmission of mechanical forces (e.g., shock) to the sampling and measuring module via the inlet tube, a stabilizer firmly enclosing the tube was fitted at the inlet to the sampling and measuring module.

The pump module (refer to Fig. 2) is connected to the sampling and measuring module (Fig. 3) via two flexible hoses (inlet / outlet). The sampling and measuring module controls the pump; it also contains the internal filter handling mechanism, large parts of the pneumatic system, the actual measuring device, and all electronic circuitry and microprocessors necessary for controlling and supervising the measuring system.

The measuring system is operated from a membrane-type keyboard in the front panel (Fig. 4). All requisite parameters (e.g., sampling time, volumetric throughput, etc.) are set from here. A number of quality assurance functions can also be activated.



Fig. 2.: Pump module



Fig. 3: Sampling and measuring module



Fig. 4: Front view of the OPSIS SM 200 (New)

Current data from an ongoing sampling cycle, data stored from past samples and measurements, and numerous additional quality assurance data can be invoked on the display at any time without interfering with the sampling process. The internal memory content can be transferred to an external computer via the instrument's RS323 interface, or polled remotely by modem. The instrument can also be remote-controlled via this route.

For control of the volumetric air flow rate, pressure and temperature sensors are fitted inside and outside the instrument. The flow rate is determined from the pressure loss across a built-in orifice, the absolute pressure, and the temperature of the measuring chamber. The current throughput is continuously compared with the preset target value, with compensation – e.g., for ambient air temperature and pressure variations or filter loading – being provided by a regulator valve. During the test the output value of the air intake volume over the sampling cycle was related to normal conditions (273 K, 101.3 kPa). Other settings can be selected.

A detailed description of the instrument's internal processes is given in chapter 3.1

Filters are changed fully automatically. For storage of clean and sampled filters the sampling and measuring module is equipped with two filter containers. These filter containers can accommodate up to 40 filters, each filter being mounted in a special filter holder. At the usual 24-hour sampling period this gives the instrument an operating autonomy of 40 days without requiring a manual filter change. The filters are of the Teflon type and measure 47 mm in diameter (RZPJ047 or R2P J047, respectively; Pall Gelman Laboratories).

Fig. 5 gives an overall view of the structure of the measuring system. Table 3 lists some key specifications of the OPSIS SM200 Dust Monitor (New and Old).

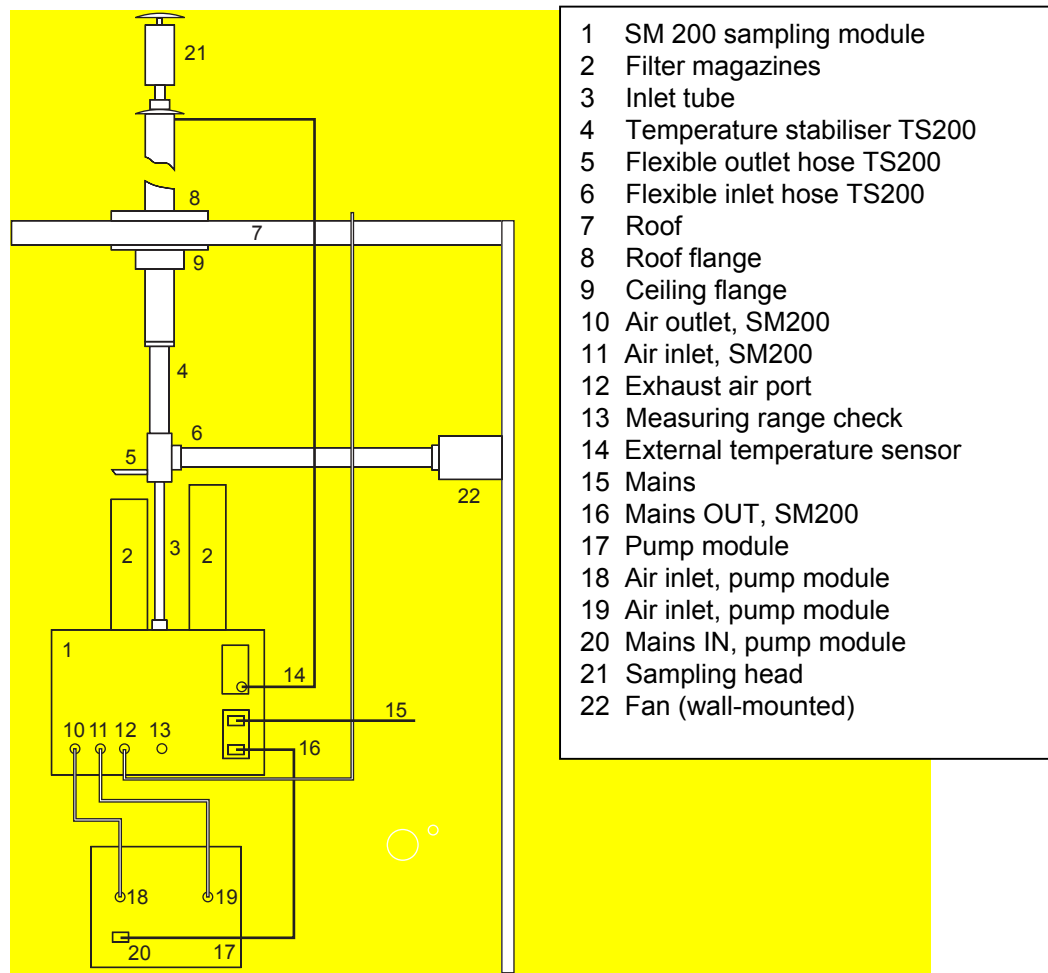


Fig. 5: Structure of the OPSIS SM 200 (New) instrument

Table 3 Specifications of the OPSIS SM 200 system (manufacturer's data) – old vs. new

Dimensions / weight	OPSIS SM 200 (old)	OPSIS SM 200 (new)
Sampling & measuring module	430 x 600 x 260 mm / 45 kg	440 x 630 x 300 mm / 42 kg
Pump module	320 x 220 x 300 mm / 20 kg	310 x 280 x 250 mm / 20 kg
Inlet tube	1.5 m	1.5 m
Sampling head	depending on manufacturer	depending on manufacturer
Operating voltage	230 V ($\pm 10\%$), 50/60 Hz	230 V (+6%, -10%), 50/60 Hz
Power consumption	800 W	800 W
Ambient conditions		
Temperature	+5 ... +40°C	+5 ... +40°C
Moisture	max. 80% relative humidity	max. 80% relative humidity
Sample flow rate	8 to 25 l/min, nominal: 16.67 l/min = 1 m ³ /h	8 to 25 l/min, nominal: 16.67 l/min = 1 m ³ /h
Beta source		
Type	¹⁴ C polymethyl methacrylate	¹⁴ C polymethyl methacrylate
Decay type	Beta radiation	Beta radiation
Activity	13.4 MBq, 361 μ Ci	9.9 MBq, 267 μ Ci
Specific activity	55.5 MBq/g	55.5 MBq/g
Mass determination		
Measuring range(according to Operating Manual)	0 ... 60 mg	0 - 60 mg
Sampling time (cycle time)	8 h ... 100 days	6 h ... 100 days
Time required per beta measurement	120 min	120 min
Data storage capacity	400 data records	> 100 data records
Data storage per measurement	24 different parameters (listed in the Operating Manual, page A1)	45 different parameters (listed in the Operating Manual, page A1)
Analog output	0 ... 10 V (0 ... 1000 μ g/m ³)	0 ... 10 V or 0 ... 20 mA (freely presettable concentration range)
Digital output	RS 232 interface for data transfer and remote control	3 x RS232 interfaces for data transfer and remote control
Status signals	26 status signals (listed in the Operating Manual, page A2 et seq.)	Information available in measuring mode (see Operating Manual, page A1) or in the Service Menu / Analog Sensors

3.4 Comparison between OPSIS SM 200 (New) and OPSIS SM 200 (Old)

As part of the supplementary test, the two equipment versions were thoroughly examined for potential differences and for any modifications causing the OPSIS SM 200 (New) model to differ from its predecessor.

It was found that the modifications have not altered the fundamental structure of the measuring instrument, nor the relevant system components, in any way whatsoever. The main differences lie in the altered exterior design, the restructured electronic module, the use of more dependable or more accurate sub-components (e.g., analog sensors), and the fact that new information and features have been made available to the user. No relevant functionalities of the predecessor model have been abolished.

The following table 4 shows a comparison between the OPSIS SM 200 (New) and OPSIS SM 200 (Old) instruments.

Table 4: OPSIS SM 200 (New) vs. OPSIS SM 200 (Old)

	Evaluation criterion	OPSIS SM 200 (New) vs. OPSIS SM 200 (Alt)
Construction / Design	Sampling and measuring module	New exterior design
	Pump module	Identical
	Filter container	Identical, additional perspex enclosure
	Filter holder	Identical
	Inlet tube	Identical
	Sampling head	Identical
	Prevention of moisture condensation in inlet tube	TS200 temperature stabilizer, identical, Option C

Report on the Supplementary Tests of the OPSIS SM 200 Emission Measuring Instrument with PM10 Pre-separator marketed by OPSIS AB for the PM10 Suspended Particulates Fraction, Report No.: 936/21201592/A

Pneumatic system	Operating mode	Identical
	General design	Identical
	Inlet area of the sampling and measuring module	Identical
	Filter chamber	Identical
	Valves / nozzles / orifices in the pneumatic system	Identical
	Flow control (needle valve)	Identical
	Measuring chamber throughput	Identical
	Pump module	Identical, additional installation of a separate ON/OFF switch
	Tightness / measuring range test	Identical
	Pressure sensors	Identical
	Ambient pressure / temperature sensors	Identical
	QA/QC procedures	Identical

Mechanical system	Operating mode	Identical
	General design	Identical
	Loading mechanism	Identical + 1 additional sensor
	Carousel charging system	Identical
	Carousel	Identical + 4 position monitoring sensors
	Carousel locking / release system	Identical
	Filter wheel	Identical
	Measuring chamber locking mechanism	New stepper motor
	Unloading mechanism	New stepper motor

Radiometrical system	Operating mode	Identical
	General design	Identical
	Geiger counter	Identical
	Electronic circuitry	Identical
	Beta radiation source	Same type / same specific activity
	Shield	Identical
	Temperature in measuring chamber = 40 °C	Identical
	Beta test	Identical, can now also be regularly carried out as an Auto Test
	QA/QC procedures	Identical

Electronic circuitry	Operating mode	Identical
	Fan	Identical
	Main cards	New design, identical functions, additional 220 / 110 V preselection feature plus more accurate monitoring of interior instrument temperature.
	High-voltage part	More accurate stabilization
	Analog output	0-10 V or 0-20 mA - $\mu\text{g}/\text{m}^3$ freely selectable
	Mainboard	Identical functions
	Key pad / display	Fully redesigned
	RS 232	3 instead of 1
	Menu structure	Fully redesigned
	Measured values stored in memory	Identical + additional information
	Status messages	Identical
	Additional features (excerpt):	• Direct calibration of the pneumatic / beta system now implemented as an option in the hidden service menu; no extra software (Long) required. • Beta system auto test can be performed on regular basis. • Correction of measured values (normal or operation) • Direct manual control / check of various instrument components can be performed from instrument
Maintenance	As described in the Manual	Identical

4 Test program

4.1 General

The original suitability test as described in [10] was performed on two identical instruments bearing the serial numbers SN 285 and SN 288, respectively. Both instruments had the MN009n software version installed throughout the entire test.

In compliance with BMI Guideline, No. 1.1.2, the manufacturer was advised at the start of the test that he must notify the Test Institute of any modifications made to the tested model of the instrument.

The laboratory test designed to determine the process parameters is followed by a field test of several months' duration, carried out at various field testing sites.

All concentrations determined were stated in $\mu\text{g}/\text{m}^3$ (273 K, 101.3 kPa).

The design of the tested instruments was not changed during the test period.

In the following report on the supplementary test, the minimum requirement defined in the guidelines considered [1,2,3] is stated in the header for each test criterion, along with the number and verbatim text passage.

For the supplementary test, additional field testing was carried out at the "Furulund" site.

The option to conduct extensive re-testing of the measuring instrument in the laboratory was waived since no changes in instrument performance are to be expected, given that the concept and principal structure of the device were taken in full from the previous model, and that all components relevant to its performance (mechanical, pneumatic and radiometric systems) are of identical type and execution. It was checked as part of the test programme that none of the original functions have deteriorated or been eliminated. Any improvements and/or additions to the instrument's functionalities are listed under the respective criteria.

Where entire test items or portions thereof have been copied in unchanged form from the report on the predecessor model, these passages are highlighted in colour in this report.

4.2 Laboratory test

The laboratory tests were conducted with two identical OPSIS SM200 (Old) systems bearing the serial numbers SN 285 and SN 288. In accordance with the guidelines [1,2,3], the following laboratory test program was established:

- Description of instrument functions
- Determination of the detection limit
- Determination of ambient temperature impact on the instrument's zero point / sensitivity
- Determination of mains voltage impact on instrument sensitivity
- Determination of sampling air humidity impact on the instrument's zero point

In the laboratory test aimed at determining the process parameters, the following equipment was used:

- Climatic chamber (-20 °C ... +50°C temperature range, accuracy better than 1°C)
- Apparatus simulating dust-free air, consisting of a sampling head enclosure with dust-free compressed air supply
- Hovacal steam generator
- Isolating variable ratio transformer
- Generator permitting variation of the mains frequency
- Filter – refer to Chapter 4.3
- Weighscale - refer to Chapter 4.3
- Reference films & Eprom (Long software, version 1_8) (OPSIS)

The results of the laboratory tests are summarized in Chapter 6.

4.3 Field test

The field test was conducted with two measuring instruments of identical design, identified as follows:

Instrument 1: No. 1 SN 285 - SM 200 (Old)
Instrument 2: No. 2 SN 288 - SM 200 (Old)

For the field test the following test program had been drawn up:

- Evaluation of comparability between the instruments under test
- Evaluation of comparability of the tested instrument against the reference method
- Evaluation of the constancy of the sampling flow rate
- Determination of the system's calibration capability and analysis function
- Linearity check
- Determination of the dead time & response time
- Determination of reproducibility
- Determination of zero point drift and sensitivity drift over time
- Evaluation of the tightness of the sampling system
- Determination of the maintenance interval
- Determination of the system's availability
- Determination of the overall uncertainty of the tested instruments

The supplementary tests with the modified OPSIS SM 200 (New) measuring instruments were carried out in Sweden, again with two systems of identical design, viz.

Instrument 1: No. SN 1110 (SM 200 (New), software version 1.03)
Instrument 2: No. SN 1112 (SM 200 (New), software version 1.03)

In addition, an OPSIS SM 200 (Old) measuring system of the original (previous) design was run in parallel with the two upgraded units during the tests in Sweden.

Instrument 3: No. SN 276 (predecessor model SM 200 (Old)),
software version MN009n

As part of the supplementary test, the instruments were re-examined for the following criteria during the field tests in Sweden:

- Evaluation of comparability between the instruments under test (new vs. old, old vs. new)
- Evaluation of comparability of the tested instrument against the reference method
- Determination of the system's calibration capability and analysis function
- Determination of zero point drift and sensitivity drift over time
- Evaluation of the tightness of the sampling system
- Determination of the maintenance interval
- Determination of the system's availability
- Determination of the overall uncertainty of the tested instruments

For the original field test, the following equipment had been used:

- Measuring container (property of TÜV Immission Control and Power Systems GmbH, air conditioned to approx. 20°C)
- Meteorological station (CONRAD ELECTRONIC, PC-based version) for detecting meteorological parameters such as air temperature, air pressure, air humidity, wind speed, wind direction and rainfall
- 2 reference measuring instruments as described in Chapter 5
- 2 classifier instruments as described in Chapter 5
- All-metal small flow meter DK 37 E (manufacturer: Krohne Co.)
- Power consumption measuring instrument
- Reference film + EPROM (Long software, version 1_8) (OPSIS)
- Lead-through device for passing the inlet tube through the container roof, with ambient air flushing, to prevent sample temperature variations on the way to the filter (device consists of a tube structure with fan, manufacturer: OPSIS)

For the field test carried out as part of the supplementary test, the following was employed:

- Measuring shed, air conditioned to approx. 20-25 °C
- Meteorological station (CONRAD ELECTRONIC, PC-based version) for detecting meteorological parameters such as air temperature, air pressure, air humidity, wind speed, wind direction and rainfall
- 1 reference measuring unit as described in Chapter 5
- Reference film (software for testing reference film is implemented in the instrument)
- Lead-through device for passing the inlet tube through the container roof, with ambient air flushing, to prevent sample temperature variations on the way to the filter (device consists of a tube structure with fan, Temperature stabilizer TS200, manufacturer: OPSIS)

In the original field tests, two OPSIS SM200 systems as well as one reference and one classifier unit had been operated simultaneously in 24-hour cycles. The reference and classifier units were run discontinuously, i.e., the filter had to be changed manually after each sampling process. For this reason, two reference instruments and two classifiers were employed to permit a precisely timed alternating change-over after every 24 hours without requiring operator attendance at exactly this point in time.

For the supplementary tests, three OPSIS SM 200 systems (two new models, one old model) and one reference instrument were run simultaneously for 24 hours at a time. It was decided to carry out the supplementary test without a classifier unit.

The instrument parameter settings were as follows:

Table 5: Sampling parameter settings for the instruments used

Instrument	Parameter	Preset value
OP SIS SM 200	Sampling time	24 h
	Flow rate	16.7 l/min = 1 m³/h
	Minimum acceptable pressure loss across filter ^{*)}	0 kPa
	Minimum acceptable flow rate	14.7 l/min (2 l/min < target value)
	Pneumatic system test (QA) ^{**)}	every 2 days
Low Volume Sampler LVS3	Sampling time	24 h
	Flow rate	2.3 m³/h
GS 050	Sampling time	24 h

^{*)} An alarm is given if the pressure loss across the filter falls below this threshold value. A perceptible pressure drop at the filter is normal since the filter presents a certain amount of resistance to the flow of the medium. When this pressure loss disappears, a malfunction (e.g., filter fracture) is likely. In practice this alarm threshold should be set to about 5 kPa below the pressure loss that is normal for the filter material used. During the field tests this alarm function was deactivated (set to 0 kPa) since adequate monitoring was ensured by frequent visits to the field testing sites.

^{**)} The pneumatic system test is part of the instrument's built-in quality assurance function. The flow rate is monitored via the pressure loss across an orifice and controlled by means of a needle valve. The pressure loss is related to the flow rate by a specific factor which has been factory-programmed into the instrument. Comparing the current factor with the stored (factory-programmed) one yields a percent deviation value which should ideally be zero. In addition, the system is internally checked for air-tightness of its circuit.

The impaction plates of the PM10 sampling heads were cleaned every 2 – 4 weeks and lubricated with silicone grease to ensure a reliable separation of particles.

In the case of the reference instruments and classifier units, flow rate control measurements were made before and after the field test and before and after each site change using a bellows-type gas flow meter that was connected to the instrument's air inlet orifice via a length of flexible tube.

Measuring locations and instrument operating sites

In the field tests [10] the measuring instruments were installed such that only their sampling heads projected over the roof of the measuring container. The main modules of the two instruments under test and those of the reference and classifier units were mounted inside the air-conditioned measuring container. The main modules were connected to the sampling heads via the intake pipe in the case of the OPSIS SM 200 systems, and via flexible hose connections in the case of the reference and classifier units.

In the supplementary test the OPSIS instruments were installed in a building in Furulund in a manner closely resembling the original field test [10] configuration. Only the reference unit was set up entirely outdoors on a roof.

Since there exists an inherent risk of condensation forming inside the SM200 inlet pipe, especially at high air humidity levels and at ambient temperatures exceeding the temperature at the instrument installation site, various approaches have been devised to avoid condensation and minimize the associated volatile particle loss. These options are summarized in Fig. 6 below, in accordance with section 2.3 of the Operating Manual. During the test the inlet pipe was additionally placed in an ambient air flow (option C). The relevant device was made available by OPSIS.

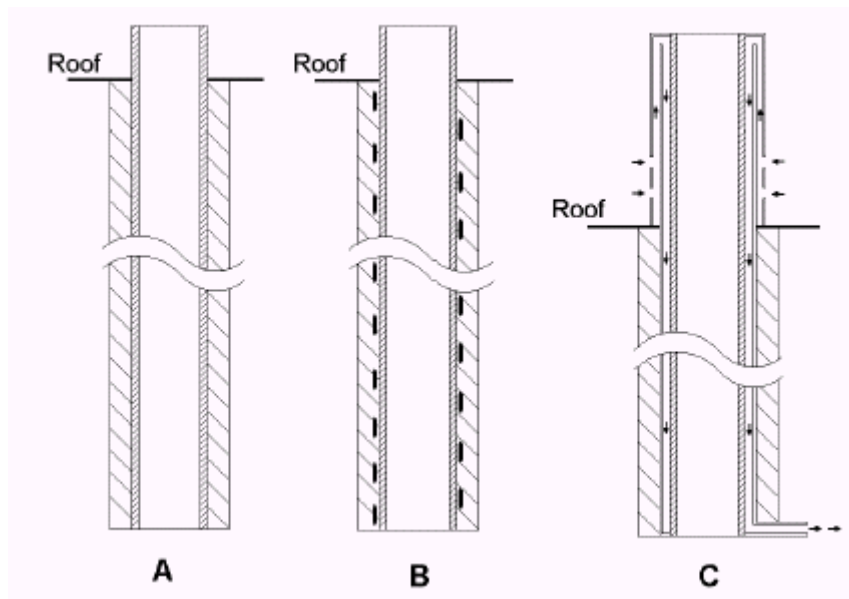


Fig. 6: Various methods of avoiding condensation in the inlet pipe.

For the field test the following measuring sites were selected:

Table 6: Field test sites

No.	Test site	Time period	Description
1	Parking lot area, Cologne	12/2000 – 02/2001	Urban background
2	Wesseling	04/2002 – 07/2002	Industrial area
3	Mechernich / Eifel	07/2002 – 11/2002	Rural setting
4	Brühl	11/2002 – 01/2003	Gravel pit
5	Furulund	12/2004 – 04/2005	Rural setting

Figs. 7 through 10 document the PM concentration distribution at the field test sites as recorded with the reference measuring systems.

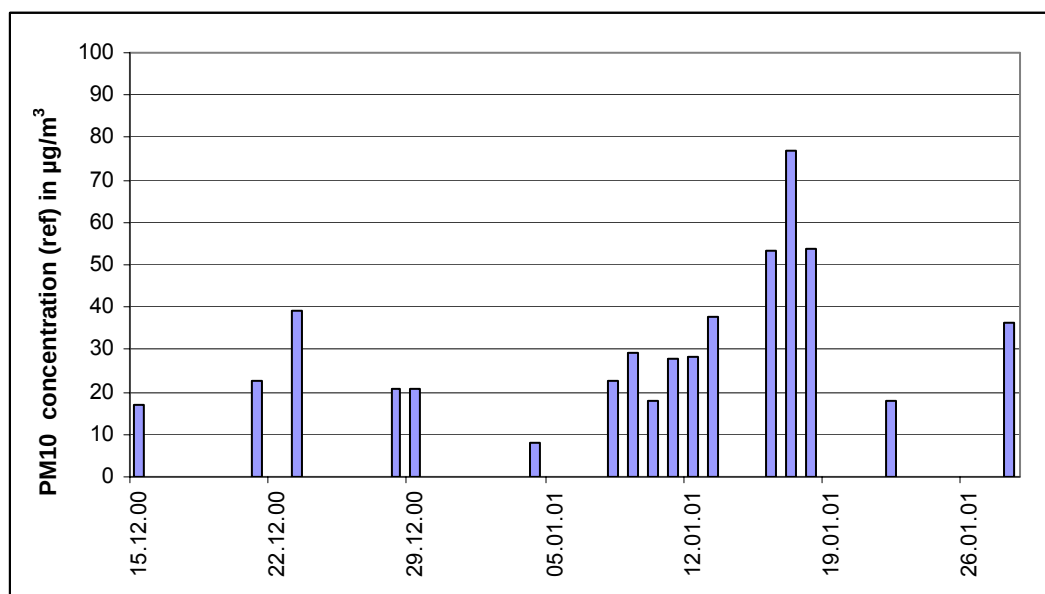


Fig. 7: PM10 concentrations determined with a reference instrument at the "Cologne parking lot" site

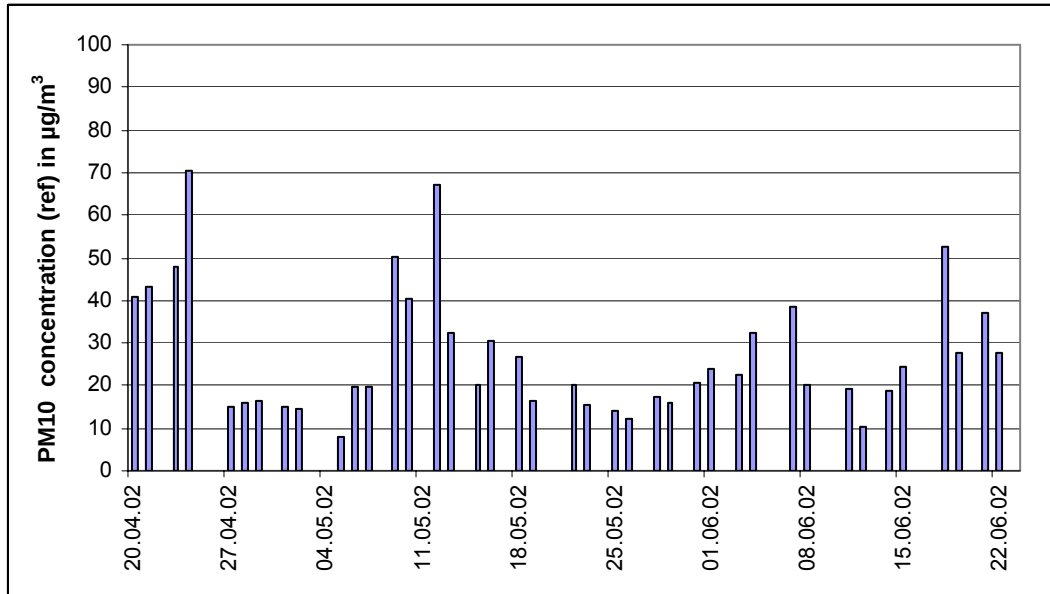


Fig. 8: PM10 concentrations determined with a reference instrument at the "Wesseling" site

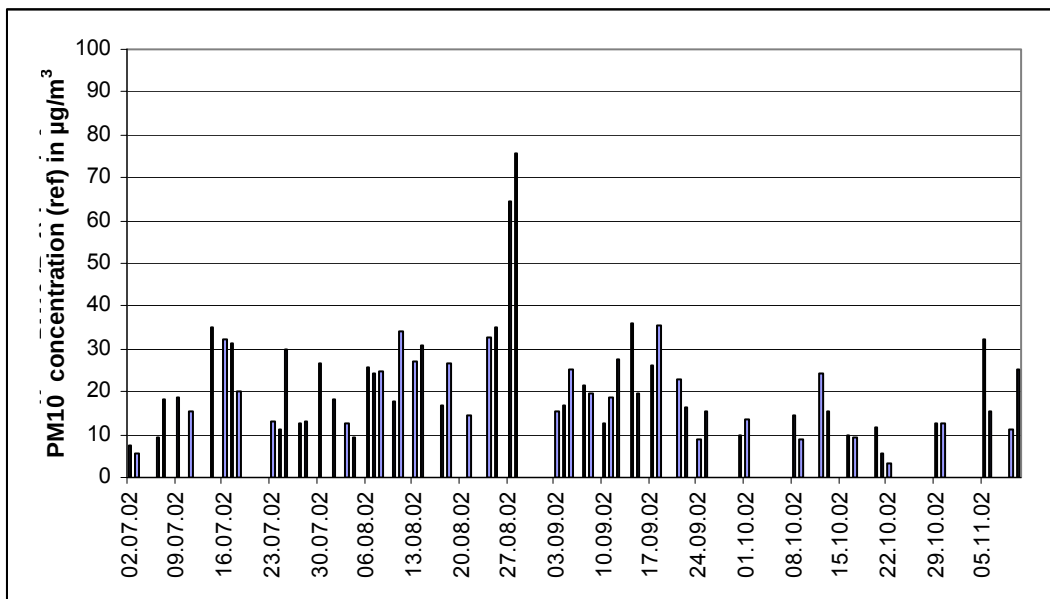


Fig. 9: PM10 concentrations determined with a reference instrument at the "Mechernich" site

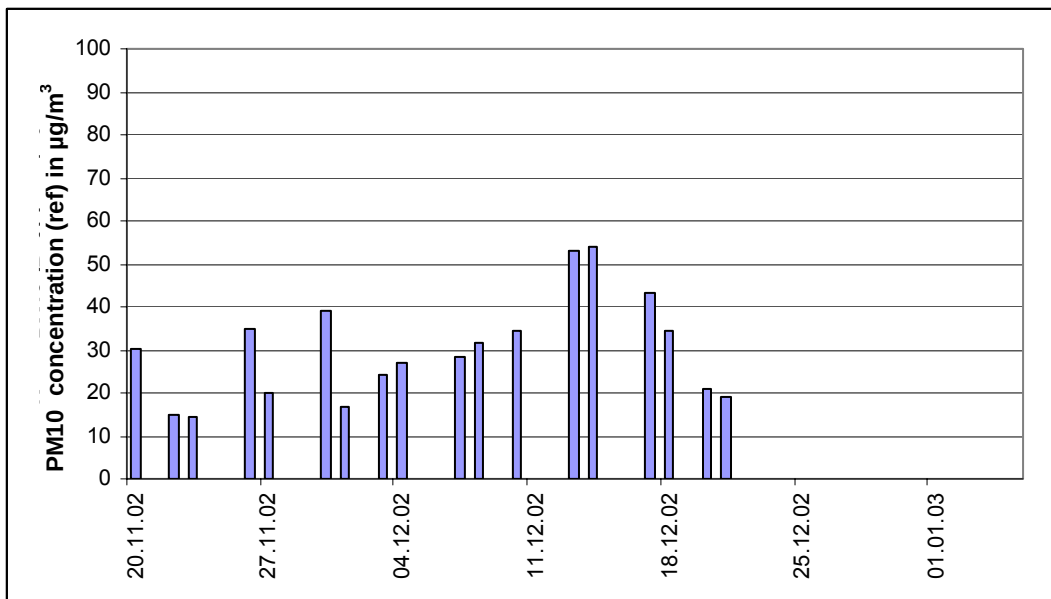


Fig. 10: PM10 concentrations determined with a reference instrument at the "Brühl" site

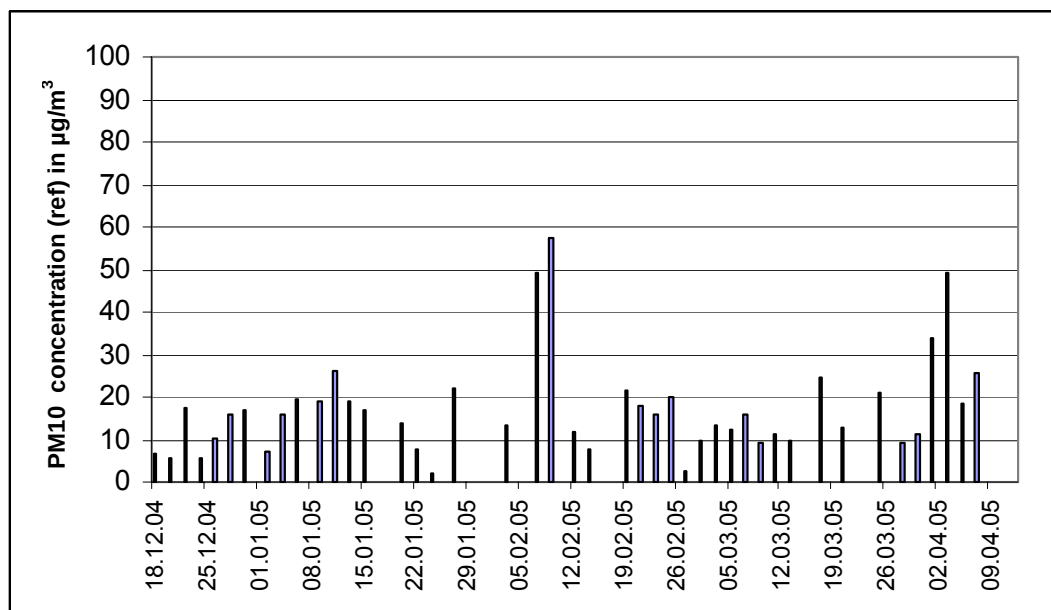


Fig. 11: PM10 concentrations determined with a reference instrument at the "Furulund" site.

The following illustrations show the measuring container at the Wesseling, Mechernich and Brühl field test sites, as well as the setup used at the Furulund site.

Report on the Supplementary Tests of the OPSIS SM 200 Emission Measuring Instrument with PM10 Pre-separator marketed by OPSIS AB for the PM10 Suspended Particulates Fraction, Report No.: 936/21201592/A



Fig. 12: Wesseling field test site



Fig. 13: Mechernich field test site



Fig. 14: Brühl field test site



Fig. 15: Furulund field test site

In addition to the dust monitors under test, the container was equipped with a meteorological sensing station. This station supplied a continuous stream of air temperature, air pressure, air humidity, wind speed, wind direction and rainfall data. These parameters were stored in the form of 10-minute averages.

The container configuration and arrangement of the sampling probes was characterized by the following dimensional data:

- Container roof height: 2.7 m
- Height of the test/reference sampling point 1.0 / 0.8 / 0.7 m above container roof
- Classifier location: 3.7 / 3.5 / 3.4 m above ground
- Air vane mounting height 4.2 m above ground

The height of the sampling intake for the two instruments under test was physically limited by the length of their inlet pipe. The installation heights of the reference and classifier units were adapted accordingly.

- Distance between tested instruments: 0.7 m
- Distance from test instrument 1 to reference unit 1 1.0 m
- Distance from test instrument 1 to reference unit 2 1.6 m
- Distance from test instrument 2 to reference unit 1 1.6 m
- Distance from test instrument 2 to reference unit 2 1.0 m
- Distance from test instrument 1 to classifier 1 1.7 m
- Distance from test instrument 1 to classifier 2 2.2 m
- Distance from test instrument 2 to classifier 1 2.2 m
- Distance from test instrument 2 to classifier 2 1.7 m

In the supplementary test, the sampling probe arrangement was characterized by the following dimensional data:

- Roof height approx. 2.5 m
- Height of test/reference sampling point approx. 1.5 / 1.5 m above roof
approx. 4.0 / 4.0 m above ground
- Air vane mounting height: approx. 4.0 m above ground

The height of the sampling intake for the two instruments under test was physically limited by the length of the inlet pipe. The installation height of the reference unit was adapted accordingly.

- Distance from test instrument 1110 to 1112: approx. 2.0 m
- Distance from test instrument 1110 to 276: approx. 4.0 m
- Distance from test instrument 1112 to 276: approx. 2.0 m
- Distance from test instrument 1110 to reference unit approx. 1.5 m
- Distance from test instrument 1112 to reference unit approx. 1.5 m
- Distance from test instrument 276 to reference unit approx. 3.2 m

The following Table 7 shows the key meteorological values determined during the measurements at the four field test sites. All individual readings are given in Appendices 17, 8 and 13.

Table 7: Ambient conditions at the field test sites, expressed as daily averages

	Parking Lot / Cologne	Wesseling	Mechernich	Brühl	Furulund
Number of value pairs	39	62	114	45	104
PM 10 concentration [%]					
Range	45.7 – 90.8	38.3 – 94.9	44.1 – 98.1	47.9 - 99.2	not determined
Average	75.5	70.3	79.3	84.3	
Air temperature [°C]					
Range	-2.5 – 9.3	9.3 – 29.8	4,7 – 25.7	-3.5 - 13.1	-7.8 - 11.4
Average	3.2	17.9	14.4	6.2	2.1
Air pressure [hPa]					
Range	982 – 1026	992 – 1028	954 – 977	980 - 1023	970 - 1031
Average	1003	1008	965	1004	1009
Relative air humidity [%]					
Range	39.7 – 98.6	33.9 – 92.5	42 - 98.7	28.1 - 98.9	44.8 - 91.6
Average	74.8	62.7	78	77.1	70.5
Wind speed [m/s]					
Range	0.2 – 5.1	0.9 – 5.8	0.2 – 6.5	0.3 - 6.6	0.0 - 10.6
Average	2.2	3.2	1.5	3.1	2.7
Rainfall [mm]					
Range	0 – 13	0 – 39	0 – 55	0 - 18	0 - 22
Average	2.2	2.1	3.4	2.9	1.5

Sampling duration

DIN EN 12341 specifies a 24-hour sampling time. However, the sampling time is allowed to be longer for higher concentrations or shorter for lower concentrations, respectively.

While the field test sampling times were always set to 24 hours for all instruments, the laboratory sampling time was reduced to 8 hours in the case of some tests in order to obtain a larger number of readings.

Filter handling - mass determination

The following filters were used in the suitability tests:

Table 8: Filter materials used

Instrument	Filter material / type	Manufacturer
OPSIS SM 200	Teflon, Ø 47mm	Pall Gelman
Reference unit LVS3	Quartz fiber, Ø 50mm	Munktell
Classifier GS 050	Quartz fiber, Ø 50mm	Munktell

Clean filters for the reference and classifier instruments were stored in a perspex chamber in the weighing room to protect them against dust. In this chamber the filters were conditioned to a temperature of $20 \pm 1^\circ\text{C}$ and a constant relative air humidity of about 60% for at least 48 hours before use. Weighing was carried out using an MC 210P type scale (manufacturer: Sartorius Co.) having an absolute resolution of 10 µg. The filters for the reference unit were moved to the measuring location and back in closed glass dishes. The classifier filters were placed directly in the TSP sampling heads in the weighing room, i.e., the entire sampling head was changed on site. Sampled filters were brought back to the laboratory inside the sampling head to be removed only in the weighing room.

Sampled filters were handled in the weighing room in an equivalent manner.

Filter handling procedures were thus in accordance with DIN EN 12341, Appendix C.

During the supplementary test, clean and sampled filters were shipped back and forth by courier between the field test site in Sweden and the laboratory in Germany.

5 Reference measuring method

The following instruments were used during the field tests in accordance with DIN EN 12341:

1. Reference units: Low Volume Sampler LVS3 (small filter instrument)
 Manufacturer: Ingenieurbüro Sven Leckel, Leberstraße 63, Berlin,
 Germany
 Manufacturing date: 2000
 PM10 sampling head
2. Classifier units: GS 050 small filter instrument
 Manufacturer: Derenda Co., Xantener Str. 22, Berlin, Germany
 Manufacturing date: 1992
 TSP sampling head

Two reference instruments were operated alternately or concurrently at a controlled flow rate of 2.3 m³/h throughout the tests. The flow rate control accuracy under real-life operating conditions is < 1 % of the nominal flow rate.

Sample air is drawn through the sampling head by a sliding-vane type rotary vacuum pump. In the process, the sample air flow rate is determined via an orifice between the filter and the pump. From the pump the inlet air passes to a separator designed to remove sliding vane abrasion particles before reaching the air outlet.

Upon completion of the sampling cycle the electronic measuring system displays the sample air intake volume in m³ (normalized or related to operating conditions).

The PM10 concentration was determined by dividing the suspended particle mass per filter, determined gravimetrically in the laboratory, by the associated volumetric air flow rate in m³ (normalized),.

The sample intake and filter holder are connected in a gas-tight manner by a screw coupling. Usually the sampling head is linked to the sampler by a straight pipe. For the field test the sampler was fitted inside the container while the sampling head was mounted on the container roof. The connection was made using an adapter and a length of flexible tubing.

The classifier instruments detected suspended particulates in air in accordance with VDI Guideline 2463, Sheet 7. They cover the entire grain size range (TSP = **T**otal **S**uspended **P**articulate **M**atter).

The operating principle of the classifier instrument is basically that of the reference unit in uncontrolled mode. Its flow rate determination is based on a vane-type anemometer attached to an electro-mechanical counter taking measurements in m³ with a read accuracy of 0.01 m³. The nominal flow rate is 2.7 to 2.8 m³/h. The flow rate should not drop below an hourly 2.6 m³/h at any point of the sampling period. The sampling air volume is obtained from the difference between the meter readings between the start and end of the cycle.

A conversion of the sample air volume to normalized conditions (273 K, 101.3 kPa) was achieved using the air temperature and air pressure values continuously determined in the meteorological station installed at the container, as well as the negative pressure present at the gas meter.

The TSP concentration was again obtained by dividing the suspended particle load on each filter, determined gravimetrically in the laboratory, by the associated sample air volume in m³ (normalized). The PM10 fraction was calculated by dividing the PM10 concentration measured by the reference instrument by the respective TSP concentration.

The sampling time was set on an electronic timer clock.

For the field test the sampling instrument was again arranged in the container, with its sampling head mounted on the container roof. The connection was made via an adapter and a length of flexible tubing

In the supplementary test, only one reference unit was used. No classifier instrument was run during the supplementary test.

For the supplementary test the OPSIS instruments were installed in a building in Furulund in a manner closely resembling the original field test [10] configuration. Only the reference unit was set up entirely outdoors on a roof.

6 Test results

6.1 4.1.1 Display of measured values

The measuring instrument must have an indicating device displaying the measured values.

6.2 Equipment

No additional equipment is needed.

6.3 Implementation of the test

The presence of a measured value display on the instrument was verified.

6.4 Evaluation

The instrument possesses a measured value display. The measured value can be read from memory and displayed at any time in the form of an average over the sampling time.

6.5 Findings

The instrument possesses a display on which measurements are indicated.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Fig. 16 shows a front view of the instrument's operating unit. Measured values retrieved from memory can be shown on the display.



Fig. 16: Front view of the measuring instrument's operating unit.

6.1 4.1.2 Maintenance friendliness

Necessary instrument maintenance shall not require an excessive input of time or effort. Where possible, it should be achievable by working outside the instrument.

6.2 Equipment

No additional equipment is needed.

6.3 Implementation of the test

The requisite periodic maintenance steps were carried out in accordance with the Operating Instructions.

6.4 Evaluation

The following maintenance work must be carried out by the user:

1. Equipment status verification
The instrument status can be monitored and verified by checking the instrument itself, or via an online connection
2. Monthly cleaning of the instrument. In any case the measuring instrument must be cleaned after each measuring job.
3. The sampling head must be cleaned and its impaction plate re-greased at intervals not exceeding four weeks
4. The pneumatic, electrical and mechanical components can be checked via built-in test procedures. This must be performed monthly, but in any case after each measuring job.
5. All connecting lines, O-rings and fuses must be checked every three months.
6. The filter in the water trap of the pump module must be changed every three months.
7. The pump membrane must be replaced every 6 months
8. Cleaning of the inlet pipe is a requirement to be observed on the occasion of the annual basic maintenance procedure.

All work can be carried out using standard tools..

6.5 Findings

Maintenance can be performed using standard tools from outside the instrument; the amount of time and effort involved is reasonable. The steps listed at items 4 through 7 must be carried out with the instrument shut down. These steps become necessary at monthly, quarterly, semi-annual or annual intervals. For the remaining period, maintenance may be limited to checking for dirt, error messages, or any non-compliances identified in the quality assurance tests.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Throughout the test the instruments were serviced following the job descriptions and workflows stated in the Manual. No problems were encountered when the instructions in the Manual were observed. To date, it has been possible to perform all maintenance work easily using common tools.

6.1 4.1.3 Functional test

Where special equipment is necessary to operate the measuring instrument or to check its functions, such equipment shall be deemed to belong to the instrument and shall be utilized in the relevant sub-tests and included in the evaluation.

Any test gas generating systems associated with the measuring instrument must report their readiness for operation to the measuring instrument via a status signal, and must be controllable directly via the measuring instrument as well as telemetrically.

The uncertainty of the test gas generating system associated with the measuring instrument shall not exceed 1% of the B_2 reference value during a three-month period.

6.2 Equipment

Operating Manual, reference films.

6.3 Implementation of the test

The presence, availability and operability of all equipment functions listed in the Operating Manual was verified. To check the stability of the beta measurement, the instrument carries out the following internal test after every re-start and, optionally, as an auto test at regular intervals (SM 200 (New) only): Two reference screens of different thickness are inserted briefly between the beta radiation source and the Geiger counter. The manufacturer has recorded the Geiger counter's response to the interposition of the two reference screens. This response record provides reference values with which the results of the internal "beta span test" are compared. The result is a percentage difference.

6.4 Evaluation

All equipment functions listed in the Operating Manual are present or available.

Table 9 lists the results of the internal "beta span test". The beta tests are carried out after each re-start of the instrument; they took place over a period of approx. 1.5 years. The Geiger counter signals were found to be highly constant over time in these tests. The evaluated parameter is the change (%) in the ratio of the two measured values (CountA and CountB from the built-in reference diaphragms) against the respective predecessor value. CountA and CountB are expressed as Geiger counter signals in "counts per minute", or "cpm". According to the Operating Instructions (section 5.7.3) the ratio of the two values must be highly constant.

6.5 Findings

All equipment functions listed in the Operating Manual are present, available and functional.

The measuring system performs a functional check after every new start (and, optionally, as an auto test at regular intervals). Table 9 shows the results of the beta span test over the observation period of about 1.5 years. The results are highly constant; no deviations by more than 0.45% from the factory-logged value were identified. Deviations exceeding 3% will trigger a status signal and maintenance request.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 9 below shows the results of the beta test for the two specimen instruments:

Table 9: Results of the beta span test

SN 285				
Re-start date	A [cpm]	B [cpm]	A/B	Deviation [%] from previous value
05-07-2001	95904	73647	1.3022	-
08-08-2001	96297	74015	1.3010	-0.09
30-08-2001	96523	73952	1.3052	0.32
20-09-2001	98041	75111	1.3053	0.01
18-10-2001	96557	73755	1.3092	0.30
05-12-2001	95522	73249	1.3041	-0.39
19-04-2002	95846	73482	1.3043	0.02
10-06-2002	96210	73862	1.3026	-0.13
29-09-2002	97428	74597	1.3061	0.27
19-11-2002	96322	73748	1.3061	0.00

SN 288				
Re-start date	A [cpm]	B [cpm]	A/B	Deviation [%] from previous value
05-07-2001	97609	74837	1.3043	-
08-08-2001	97994	75079	1.3052	0.07
30-08-2001	97946	74997	1.3060	0.06
20-09-2001	96475	73981	1.3041	-0.15
18-10-2001	97762	74932	1.3047	0.05
05-12-2001	97498	74554	1.3078	0.24
19-04-2002	97640	74738	1.3064	-0.10
10-06-2002	97925	75020	1.3053	-0.09
29-09-2002	99246	75690	1.3112	0.45
19-11-2002	97864	74947	1.3058	-0.41

6.1 4.1.4 Setup and run-in times

Instrument setup and run-in times must be stated in the Operating Instructions

6.2 Equipment

An additional clock was used for testing this minimum requirement.

6.3 Implementation of the test

The instruments were started up in accordance with the manufacturer's description. The requisite setup and run-in times were determined separately

Requisite civil engineering measures preceding the installation (e.g., creation of a break-through in the container roof) were not evaluated here.

6.4 Evaluation

The setup time covers the period required to set up the instrument, up to the point where it is put into operation.

The OPSIS system must be installed at a weather-independent location, e.g., inside an air-conditioned measuring container. In addition, the inlet pipe must be passed through the roof and provided with ventilation. Both steps involve a major construction effort at the measuring site. A mobile use of the instrument is therefore assumed to take place always in conjunction with the associated peripherals.

Setting up the measuring instrument involves the following basic steps:

- Connection to the power supply
- Connection of the inlet pipe, stabilizer and PM10 sampling head
- Connection of the pump
- Installation of external temperature sensors (near the sampling head)
- Loading filters into the instrument
- Checking for leaks

Performing these steps (= setup time) requires a period of 1 hour.

The run-in time is the period required from the time the instrument is taken into operation until it is ready to commence measurements.

Upon power-up, the system initially starts a warm-up cycle in which the measuring chamber is heated to operating temperature. The duration of this warm-up phase depends on the instrument and ambient temperatures; it may last up to 1 hour. The warm-up phase can be skipped by pressing "ESC", but this should be done only if no directly subsequent measurements are intended. Heating to operating temperature will continue in the background even after the "ESC" key is pressed.

After the warm-up cycle the system displays its main menu. Here the operator can set and store the sampling parameters (flow rate, sampling time, frequency of the internal pneumatic and/or radiometric system check, etc.). An operator familiar with the instrument can do this within a few minutes' time.

Given the approx. 2-hour measuring time of the beta absorption process, the desired start time for a normal start-up of the instrument must be set about 2.5 - 3 hours before the current time (depending on the time of the last beta check) to allow for completion of the test procedures. Sampling can only be commenced after this interval. If the selected time is too close to the current time, an error message will be given and the start-up procedure must be repeated. The start time is normally set so that the instrument will start measurements at 00:00 o'clock (midnight).

6.5 Findings

The setup and run-in times were determined..

The measuring system can be operated in different measuring locations with a reasonable expenditure of time and effort. The setup time is 1 hour. The run-in time is approx. 2.5 hours minimum, and approx. 3.5 hours for a normal start.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not required in this case.

6.1 4.1.5 Design

The Operating Instructions must contain manufacturer information concerning the design of the instrument. This must essentially include indications regarding the following:

Instrument type/design (e.g., benchtop, panel mounting or independent instrument)

Installation position/orientation (e.g., horizontal, vertical installation)

Safety requirements

Dimensions

Weight

Power consumption

6.2 Equipment

An additional power consumption measuring device as well as a weighscale were used for this test.

6.3 Implementation of the test

The structure of the instruments submitted was compared to the description in the Manual. The stated power consumption is determined over 24 hours in normal operation on three days during the field tests.

6.4 Evaluation

The measuring instrument must be installed in a horizontal position at a weather-independent location. It should be placed on a horizontal surface (e.g., a table). It can also be mounted favourably in a 19" module rack.

The dimensions and weights of the instrument conform to the indications Operating Instructions.

The manufacturer states a maximum of 800 W as the measuring system's power requirement. This statement was verified in three tests lasting 24 hours each. The stated value was not exceeded in any of these tests. The average power consumption over 24 hours was approx. 400 W.

The power consumption measurements were not repeated in the course of the supplementary tests because the design of the measuring system has not changed substantially and the power requirement of the predecessor model was far lower than stated in the Manual.

6.5 Findings

The instrument design information provided in the operating instructions is complete and correct.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not necessary with this minimum requirement.

6.1 4.1.6 Unauthorized change of settings

A feature must be provided which allows the settings of the measuring system to be blocked against inadvertent or unauthorized manipulation.

6.2 Equipment

This minimum requirement can be verified without additional equipment.

6.3 Implementation of the test

The instrument is operated from keypad in the front panel, or via the RS232 interface plus modem from a remote computer. Changing the instrument's sampling parameters always involves pressing multiple key sequences..

Since the instrument cannot be installed outdoors, additional protection is provided by its installation at a location secured against unauthorized access (e.g., in a locked measuring container).

6.4 Evaluation

An unauthorized or inadvertent change of instrument parameters is prevented by requiring a sequence of several keys to be pressed for its operation. The installation in a locked measuring container acts as an additional safeguard against unauthorized manipulation.

6.5 Findings

The measuring instrument is to be installed in a locked measuring container.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not necessary with this minimum requirement.

6.1 4.1.7 Measuring signal output

The measuring signals must be made available in digital (e.g., RS232) and/or analog (e.g., 4 to 20 mA) form.

6.2 Equipment

Digital datalogger, Mobidas datalogger (for analog signals).

6.3 Implementation of the test

The test was carried out using a Mobidas type electronic data acquisition system (analog output) in addition to a digital datalogger.

The dataloggers were connected to the analog and digital outputs ports. The test was conducted by comparing the measurement readings from the instrument's memory (as shown in the display) with the analog and digital outputs during the field tests.

6.4 Evaluation

The measurement signals are made available at the back of the instrument as follows:

Analog: 0 ...10 V or 0-20 mA, respectively

Digital: via an RS 232 interface. Using a direct or modem link to a computer, the instrument can be operated in the same way as locally, e.g., its memory content (with all data on previous measurements) can be read out.

The measurement readings were delivered in analog and digital format. Both were consistent with the displayed value read from the instrument's memory.

6.5 Findings

Measuring signals are made available in both an analog (0-10 V or 0-20 mA, respectively) and a digital format (via RS 232).

Additional measuring units and peripheral equipment can be connected to the instrument via appropriate ports.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Fig. 17 shows a view of the instrument's rear panel with the various measuring signal output ports.



Fig. 17 Rear panel of the OPSIS SM 200 (New) instrument

6.1 4.2 Requirements on measuring systems for mobile use

Measuring instruments for mobile use must meet the requirements for stationary-type measuring instruments, even when in mobile use. It must be ensured during mobile use (e.g., measurements on flowing traffic, temporary measurements at differing sites, aircraft measurements) that the instrument is continuously ready for operation.

6.2 Equipment

Not needed with this minimum requirement.

6.3 Implementation of the test

The measuring system was tested at various field test sites in the course of the field testing program.

6.4 Evaluation

The instruments are designed for stationary installation in a measuring station / measuring container. For mobile service they can only be deployed in conjunction with a measuring container.

Their continuous readiness for operation in temporary measurements at diverse sites is ensured provided the installation conditions (measuring site selection, infrastructure) are met.

In addition to these installation conditions, the system's setup and run-in times must be observed in mobile use.

6.5 Findings

In the course of the field testing program the measuring system was tested at various test sites. However, it cannot be operated out of a moving vehicle.

Minimum requirement fulfilled? no

6.6 Comprehensive presentation of the test result

Not necessary with this minimum requirement.

6.1 5.1 General

Manufacturer's statements in the Operating Manual must not be inconsistent with the results of the suitability test.

6.2 Equipment

Not necessary with this minimum requirement.

6.3 Implementation of the test

The test results are compared with the information given in the Manual.

6.4 Evaluation

Deviations found to exist between the first draft of the Manual and the actual instrument design were duly remedied.

6.5 Findings

No inconsistencies between instrument equipment and the manuals were observed.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Refer to item 6.4 of this module.

6.1 5.2.1 Measuring range

The measuring system's end-of-scale value must be larger than or at least equal to the reference value B_2 .

6.2 Equipment

No further equipment is necessary for verifying compliance with this requirement.

6.3 Implementation of the test

It was verified whether the measuring system's end-of-scale value is larger than or equal to the B_2 reference value.

6.4 Evaluation

The end-of-scale value of the mass quantitation by beta ray absorption is theoretically 60 mg (factory calibration of the beta measurement system). Assuming a 24-hour sampling period this would correspond to a dust concentration of approx. 2500 $\mu\text{g}/\text{m}^3$. Since the ambient air PM10 levels remained below 1000 $\mu\text{g}/\text{m}^3$ throughout the test period, the analog output signal of the instruments under test was set to a 0 ... 1000 $\mu\text{g}/\text{m}^3$ range. Other settings are selectable.

Measuring range: 0 – 20 $\mu\text{g}/\text{m}^3$ and 0 – 1,000 $\mu\text{g}/\text{m}^3$

Reference value: VDI: $B_2 = 200 \mu\text{g}/\text{m}^3$

6.5 Findings

The system is set to a 0-200 $\mu\text{g}/\text{m}^3$ or 0-1000 $\mu\text{g}/\text{m}^3$ measuring range, respectively.

The end-of-scale value of the measuring system is larger than the B_2 reference value

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not necessary with this minimum requirement.

6.1 5.2.2 Negative measuring signals

Negative measuring signals or readings, respectively, must not be suppressed (live zero).

6.2 Equipment

No further equipment is necessary to verify compliance with this requirement.

6.3 Implementation of the test

The instruments were operated without the pump for an 8-hour period. The "deposited mass" on the filter, which should ideally be 0 mg in this case, was determined as during normal sampling.

According to the equations in Fig. 3.1, the mass deposited on the filter can be calculated from the beta radiation intensity before and after the sampling cycle (F_{blank} , F_{collect} , stored in the instrument's memory as "blank" [cpm] and "collect" [cpm]) for a known impingement surface area and known $K(x)$. (Both of these values were supplied by the manufacturer).

6.4 Evaluation

If the F_{collect} reading exceeds F_{blank} the mass will assume a negative mass value; the measuring instrument then returns a concentration of 0 $\mu\text{g}/\text{m}^3$.

6.5 Findings

Negative measuring signals not directly displayed by the measuring system. However, a negative measurement can be calculated from stored data on a case-by-case basis where this is necessary for quality assurance purposes.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 10 illustrates the example of a "zero level" test which, according to the equations in section 3.1, returns a negative mass value although the instrument issues a "0 mg" reading.

Table 10 Negative measurements

F_{blank}	F_{collect}	Calculated mass [mg]	Indicated mass [mg]
195702	195779	-0.02	0

6.1 5.2.3 Analytical function

The analytical function must permit a determination by regression computing of the correlation between the measuring signal and the value of the air quality parameter.

6.2 Equipment

The equipment listed in section 5 of the present report was additionally used for the calibration measurements.

6.3 Implementation of the test

Comparative measurements against the reference instruments described in section 5 were carried out at the "Cologne parking lot" and "Furulund" sites. The measured data were processed by regression analysis, in a manner similar to the comparability comparison between the specimen and reference instruments (module 5.3.1).

6.4 Evaluation

A total of 17 individual measurements were conducted between December 15, 2000 and January 28, 2001 to calibrate the two instruments under test using the reference method. The measurements were taken in a parking lot in Cologne.

As part of the supplementary tests, 15 individual measurements were taken between December 18, 2004, and January 20, 2005, to calibrate the two OPSIS SM 200 (New) instruments by the reference method. These measurements were carried out at the "Furulund" site.

The parameters of the calibration function

$$y = m * x + b$$

were determined by linear regression. The analytical function is the inversion of the calibration function. It reads thus:

$$x = 1/m * y - b/m$$

The gradient m of the regression line characterizes the sensitivity of the measuring instrument; the ordinate intercept b indicates the zero point.

The PM10 concentrations x (in $\mu\text{g}/\text{m}^3$) determined by the reference method were correlated with the corresponding measurement signals y (in $\mu\text{g}/\text{m}^3$) supplied by the instrument. This process yielded the characteristic values listed in Tables 11 and 12.

Table 4: Calibration and analytical function parameters determined at Cologne parking lot site

Instrument No.	Calibration function		Analytical function	
	$Y = m \cdot x + b$		$x = 1/m \cdot y - b/m$	
	m	b	1/m	b/m
	$\mu\text{g}/\text{m}^3 / \mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3 / \mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
No. 1 (SN 285)	1.0374	-1.8928	0.9639	-1.8246
No. 2 (SN 288)	1.0829	-2.697	0.9234	-2.4905

Table 5: Calibration and analytical function parameters determined at Furulund

Instrument No.	Calibration function		Analytical function	
	$Y = m \cdot x + b$		$x = 1/m \cdot y - b/m$	
	m	b	1/m	b/m
	$\mu\text{g}/\text{m}^3 / \mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3 / \mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
No. 1 (SN 1110)	0.9911	0.6756	1.0090	-0.6817
No. 2 (SN 1112)	0.9545	1.7957	1.0477	1.8813

The calibration curves obtained for the two instruments under test are shown in Fig. 18 through Fig. 21.

6.5 Findings

A statistically supported correlation between the reference method and the test instrument readings was demonstrated.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

The individual readings obtained in the comparative measurements at the "Cologne parking lot" and "Furulund" field test sites are listed in Appendix 2 and Appendix 11, respectively.

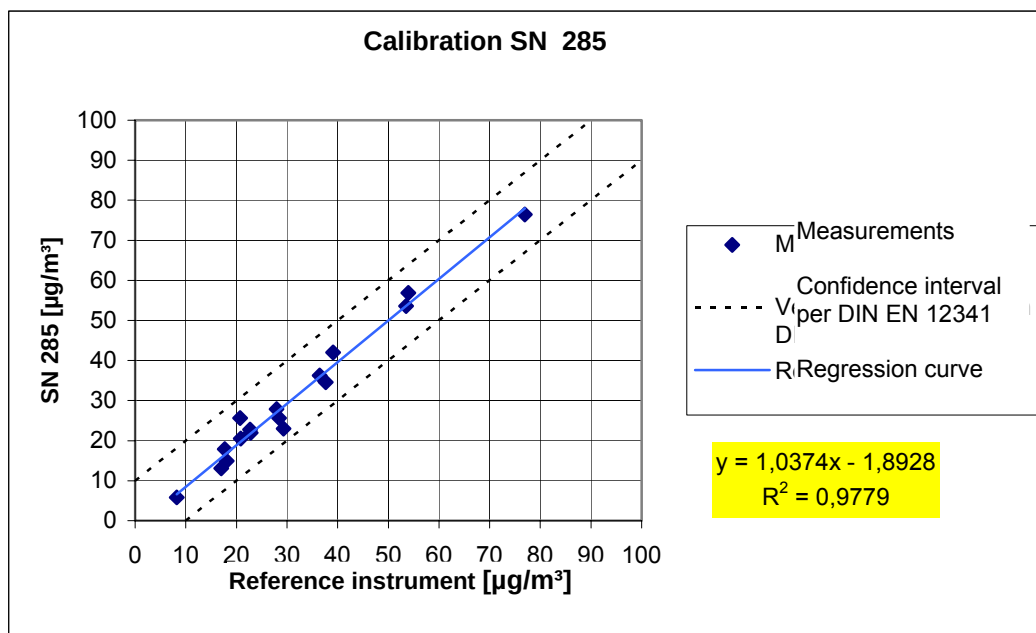


Fig. 18: Calibration curve for the tested instrument SN 285

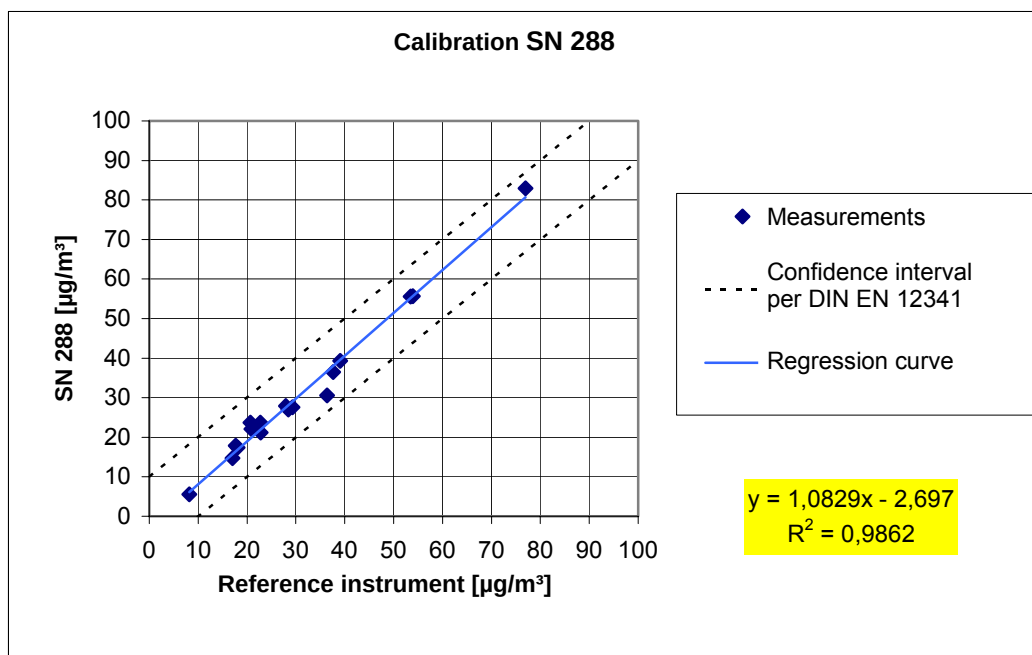


Fig. 19: Calibration curve for the tested instrument SN 288

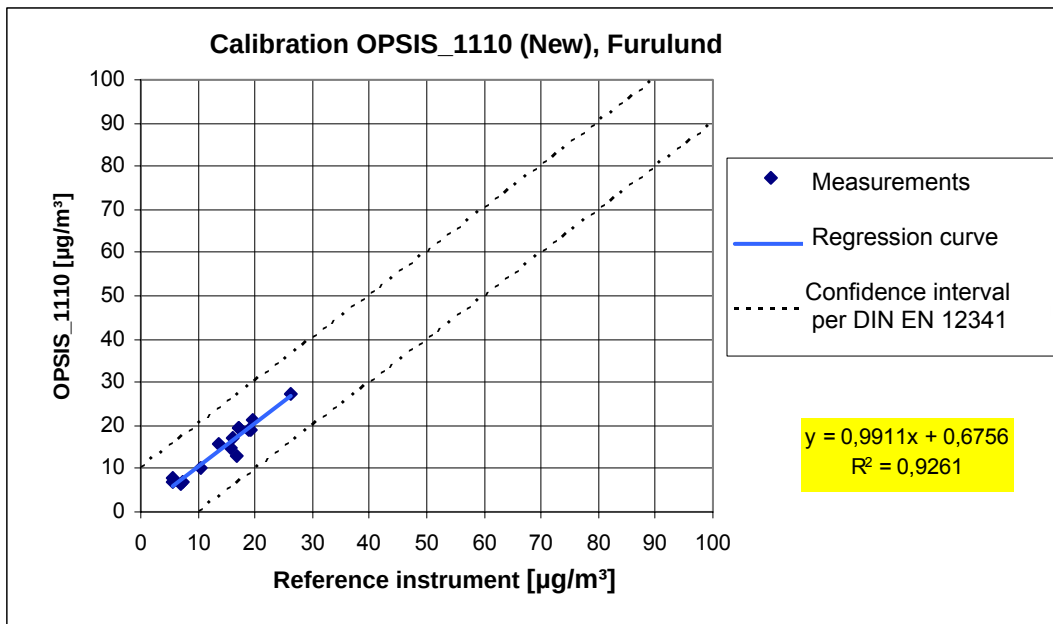


Fig. 18: Calibration curve for the tested instrument SN 1110

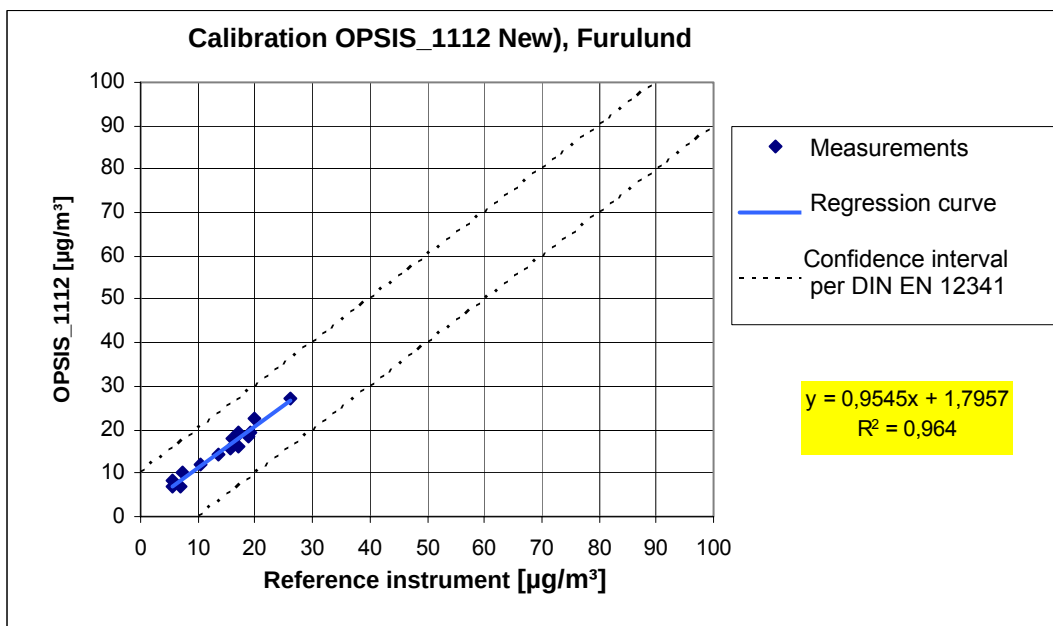


Fig. 18: Calibration curve for the tested instrument SN 285

The individual readings taken to determine the calibration curves are listed in the enclosed Appendix 2 and Appendix 11, respectively.

6.1 5.2.4 Linearity

Linearity is deemed to be ensured if the group averages of the readings taken deviate from the calibration function (determined in accordance with section 5.2.1) by not more than 5% of B_1 in the "0 ... B_1 " range, and by not more than 1% of B_2 in the "0 ... B_2 " range.

6.2 Equipment

Refer to module 5.3.1.

6.3 Implementation of the test

For measuring systems designed to monitor suspended particulates this test must be performed pursuant to minimum requirement 5.3.1, "Equivalence of the sampling system".

6.4 Evaluation

Refer to module 5.3.1.

6.5 Findings

For measuring systems designed to monitor suspended particulates this test must be performed pursuant to minimum requirement 5.3.1, "Equivalence of the sampling system".

Refer to module 5.3.1.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Refer to module 5.3.1.

6.1 5.2.5 Limit of detection

*The measuring system's limit of detection shall not exceed the B_0 reference value.
The limit of detection shall be determined by field testing.*

6.2 Equipment

For this test, "air containing no suspended particulates" was provided.

6.3 Implementation of the test

The limit of detection of the SN285 and SN288 instruments was determined by supplying each unit with particle-free air for a period of 24 hours each. The limit of detection was determined in the laboratory since it proved impossible to provide dust-free air under field testing conditions. The evaluation was conducted on the basis of readings appearing in the instrument's display and of the measured values calculated according to section 3.1 from the beta radiation intensity before and after sampling (F_{blank} , F_{collect} , stored in memory as "blank" [cpm] and "collect" [cpm], respectively).

6.4 Evaluation

The limit of detection $\underline{x}$ is determined from the standard deviation s_{x0} of the measurements obtained with both instruments drawing particle-free sample air. It corresponds to three times the standard deviation of the average $\bar{x}_0$ of the readings x_{0i} obtained when the respective instrument was taking in particle-free sample air.

$$\underline{x} = 3 \cdot s_{x0} \quad \text{where } s_{x0} = \sqrt{\frac{1}{n-1} \cdot \sum_{i=1,n} (x_{0i} - \bar{x}_0)^2}$$

Reference value: VDI: $B_0 = 2 \mu\text{g}/\text{m}^3$

6.5 Findings

The investigations revealed a detection limit of $0.41 \mu\text{g}/\text{m}^3$ for instrument No. 1 (SN 285) and of $1.08 \mu\text{g}/\text{m}^3$ for instrument No. 2 (SN 288) (instrument's measurement output), and of $0.96 \mu\text{g}/\text{cm}^3$ for instrument 1 (SN 285) and $1.29 \mu\text{g}/\text{m}^3$ for instrument 2 (SN 288) (values calculated from cpm).

A detection limit of $0.41 \mu\text{g}/\text{m}^3$ for instrument 1 (SN 285) and of $1.08 \mu\text{g}/\text{m}^3$ for instrument 2 (SN 288) was determined in the tests on the basis of the display readings (subject to a statistical confidence level of 99 %). From the calculated measuring results, a detection limit of $0.96 \mu\text{g}/\text{m}^3$ for instrument 1 (SN 285) and of $1.29 \mu\text{g}/\text{m}^3$ for instrument 2 (SN 288) were obtained with a statistical confidence of 99%. Thus, the limit of detection was below $< 2 \mu\text{g}/\text{m}^3$ in both cases.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 13: *Limit of detection (based on display readings)*

		Instrument SN 285	Instrument SN 288
Number of measurements n		15	15
Average of blank values x	$\mu\text{g}/\text{m}^3$	0.05	0.09
Standard deviation of values s	$\mu\text{g}/\text{m}^3$	0.14	0.36
Limit of detection 3 s	$\mu\text{g}/\text{m}^3$	0.41	1.08

Table 14: *Limit of detection (calculated measurements)*

		Instrument SN 285	Instrument SN 288
Number of measurements n		15	15
Average of blank values x	$\mu\text{g}/\text{m}^3$	-0.08	0.01
Standard deviation of values s	$\mu\text{g}/\text{m}^3$	0.32	0.43
Limit of detection 3 s	$\mu\text{g}/\text{m}^3$	0.96	1.29

The individual measurement values underlying the limit-of-detection determination are listed in the enclosed Appendix 1 hereto.

6.1 5.2.6 Response time

The measuring system's response time (90% time) shall not exceed 5% of the averaging time (24 h).

According to item 4.4.1 of VDI 4203 / Sheet 3, this test criterion is not relevant for dust measuring instruments with pre-separation using a physical measuring method.

6.2 Equipment

Not applicable.

6.3 Implementation of the test

Not applicable.

6.4 Evaluation

Not applicable.

6.5 Findings

Not applicable.

Minimum requirement fulfilled? -

6.6 Comprehensive presentation of the test result

Not applicable.

6.1 5.2.7 Zero point dependence on ambient temperature

The temperature dependence of the zero point must not exceed the B_0 reference value when the ambient temperature changes by 15 K in the +5°C to +20°C temperature range, and by 20 K in the +20°C to +40°C temperature range.

6.2 Equipment

Climatic chamber for the +5 to +40 °C temperature range, particle-free air.

6.3 Implementation of the test

To test the dependence of the instrument's zero point on ambient temperature, the entire measuring systems were operated in a climatic chamber. The two instruments, SN 285 and SN 248, were supplied with particle-free air. The temperature condition in the climatic chamber were varied in the sequence 20°C – 10°C – 5°C – 10°C – 20°C – 30°C – 40°C – 30°C – 20°C three times. The sampling time per temperature level was 24 hours. The relative humidity was kept constant.

6.4 Evaluation

The concentration readings returned by the instrument as well as the measurement values calculated from the pre-sampling and post-sampling beta radiation intensity (F_{blank} , F_{collect} , stored in the instrument's memory as "blank" [cpm] and "collect" [cpm]) in accordance with section 3.1 were recorded. The parameter considered was the absolute deviation in $\mu\text{g}/\text{m}^3$ per temperature step, related to the 20°C baseline level.

Reference value: VDI: $B_0 = 2 \mu\text{g}/\text{m}^3$.

6.5 Findings

An analysis of the readings supplied by the instrument showed no ambient temperature influence on its zero point.

An evaluation of the values calculated from the beta radiation intensities [cpm] showed maximum variations of $0.42 \mu\text{g}/\text{m}^3$ for instrument No. 1 (SN 285) and of $-0.48 \mu\text{g}/\text{m}^3$ for instrument No. 2 (SN 288), respectively.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 15: Zero point dependence on ambient temperature - Deviation in $\mu\text{g}/\text{m}^3$, average of three measurements (5°C and 40°C) and six measurements (10°C and 30 °C), respectively (display readings)

Temperature		Deviation	
Start temperature	End temperature	Instrument1 (SN 285)	Instrument 2 (SN 288)
°C	°C	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
20	10	0	0
20	5	0	0
20	30	0	0
20	40	0	0

Table 16: Zero point dependence on ambient temperature - Deviation in $\mu\text{g}/\text{m}^3$, average of three measurements (5°C and 40°C) and six measurements (10°C and 30 °C), respectively (calculated measurements)

Temperature		Deviation	
Start temperature	End temperature	Instrument1 (SN 285)	Instrument 2 (SN 288)
°C	°C	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
20	10	-0.05	0.17
20	5	-0.13	0.19
20	30	0.05	-0.02
20	40	-0.42	-0.48

In the individual steps, too, no deviations exceeding $-0,67 \mu\text{g}/\text{m}^3$ were found. The results of the three individual measurements are given in the enclosed Appendix 3.

6.1 5.2.8 Dependence of the measured value on ambient temperature

The temperature dependence of the reading in the region of the B_1 reference value must not exceed $\pm 5\%$ of the measured value when the temperature changes by 15 K in the $+5^\circ\text{C}$ to $+20^\circ\text{C}$ temperature range, and by 20 K in the $+20^\circ\text{C}$ to $+40^\circ\text{C}$ temperature range.

6.2 Equipment

Climatic chamber for the $+5 \dots +40^\circ\text{C}$ temperature range, reference films, evaluation software (Long).

6.3 Implementation of the test

To examine the dependence of the instrument readings on ambient temperature, the instruments were operated in a climatic chamber. Testing of their sensitivity was made possible by the use of a special software made available by the manufacturer and of four different reference films each corresponding to a defined mass. This method is also used by the instrument manufacturer to verify the linearity of the radiometric measurement.

Note: In the modified instrument version OPSIS SM 200 (New), the sensitivity verification software is directly implemented in a hidden service menu.

The temperature conditions in the climatic chamber were varied in the sequence $20^\circ\text{C} - 10^\circ\text{C} - 5^\circ\text{C} - 10^\circ\text{C} - 20^\circ\text{C} - 30^\circ\text{C} - 40^\circ\text{C} - 30^\circ\text{C} - 20^\circ\text{C}$ three times. The relative humidity was kept constant.

6.4 Evaluation

The parameter considered was the absolute change in the mass reading for each temperature step, related to the 20°C baseline level.

As an aside it should be mentioned here that the reference films can merely simulate mass values, not concentration levels. An examination in the range of B_1 ($= 40 \mu\text{g}/\text{m}^3$) was therefore not possible.

6.5 Findings

Deviations from the 20°C baseline value were found not to exceed -0.58% (instrument 1 - SN 285) and $-1,22\%$ (instrument 2 - SN 288), respectively.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 17: Dependence of instrument sensitivity on ambient temperature - Instrument SN 285, deviation in %, average of three measurements (5°C and 40°C) and six measurements (10°C and 30 °C), respectively

Temperature		Deviation (SN 285)			
Start temperature	End temperature	Film 1	Film 2	Film 3	Film 4
°C	°C	%	%	%	%
20	10	-0.51	-0.24	-0.08	-0.06
20	5	-0.34	-0.1	0.09	0.04
20	30	0.03	0.13	-0.11	-0.02
20	40	-0.58	-0.38	-0.56	-0.39

Table 18: Dependence of instrument sensitivity on ambient temperature - Instrument SN 288, deviation in %, average of three measurements (5°C and 40°C) and six measurements (10°C and 30 °C), respectively

Temperature		Deviation (SN 288)			
Start temperature	End temperature	Film 1	Film 2	Film 3	Film 4
°C	°C	%	%	%	%
20	10	0.45	0.39	0.36	0.24
20	5	0.52	0.41	0.40	0.31
20	30	-0.61	-0.38	-0.34	-0.24
20	40	-1.22	-0.89	-0.78	-0.52

In the individual steps, too, no deviations exceeding -1,98 µg/m³ were found. The results of the three individual measurements are given in the enclosed Appendix 4.

6.1 5.2.9 Zero point drift

The change in the zero point reading over time must not exceed the B_0 reference value over 24 hours or over the maintenance interval.

6.2 Equipment

Compliance with this requirement can be tested without any further equipment.

6.3 Implementation of the test

The test was conducted as part of the field testing campaign over a scheduled period of approx. 9 months. The daily zero point check stipulated in the catalogue of test requirements is not possible with this dust monitor. This being the case, the measuring instruments were run without their pump for a 24-hour sampling period once every month. The mass deposited on the filter was then determined as during the normal sampling cycle.

According to the equations in Fig. 3.1, the mass deposited on the filter can be calculated from the beta radiation intensity before and after the sampling cycle (F_{blank} , F_{collect} , stored in the instrument's memory as "blank" [cpm] and "collect" [cpm]) for a known impingement surface area and known $K(x)$. (Both of these values were supplied by the manufacturer).

As part of the supplementary tests, this test was also carried out at the "Furulund" site in accordance with the procedure outlined above.

6.4 Evaluation

The evaluation was conducted on the basis of the results for the mass values determined by running the measuring instrument without the pump, by comparing the respective readings with the "measured values" from the previous test and the "measured value" from the first (baseline) test.

6.5 Findings

The measurements were all within the permitted limits.

All measurements lay within the permitted deviation limits. Values outside the permitted tolerance range were found only on June 24 and June 25, 2002. No explanation of this untypical behaviour could be found, despite extensive investigations. Given the coincidence of the two events, an external influence is assumed. In summary, however, it must be stated the measured system fully meets the requirements concerning zero point drift.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 19 and Table 20 present the zero point measurements, as well as the calculated deviations from the previous and baseline values (in $\mu\text{g}/\text{m}^3$).

Fig. 22 and Fig. 23 show a graphic diagram of the instrument's zero drift behaviour over the investigated period.

The results from the supplementary test are summarized in Table 21 and Table 22, and in Fig. 24 and Fig. 25, respectively.

Table 19: Zero point drift - SN 285

Date	Measured value	Deviation from previous value	Deviation from baseline value
	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
17-04-2002	- 0.83	-	-
20-05-2002	1.08	1.91	1.91
25-06-2002	-3.23	-4.31	-2.40
21-07-2002	0.93	4.16	1.76
02-09-2002	1,06	0.13	1.89
27-09-2002	0,39	-0.67	1.22
18-11-2002	-0,36	-0.75	0.47
07-01-2003	0,39	0.75	1.22

Table 20: Zero point drift - SN 288

Date	Measured value	Deviation from previous value	Deviation from baseline value
	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
17-04-2002	- 0.91	-	-
21-05-2002	0.77	1.68	1.68
24-06-2002	-3.05	-3.96	-2.14
19-07-2002	-0.27	2.78	0.64
30-08-2002	0.55	0.82	1.46
26-09-2002	0.56	0.01	1.47
16-11-2002	1.19	0.63	2.1
06-01-2003	0.54	-0,65	1,45

Table 21: Zero point drift - SN 1110

Date	Measured value	Deviation from previous value	Deviation from baseline value
	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
17-01-2005	0.42	-	-
17-02-2005	0.56	0.14	0.14
23-03-2005	-0.59	-1.15	-1.01
18-04-2005	-0.25	0.34	-0.67

Table 22: Zero point drift - SN 1112

Date	Measured value	Deviation from previous value	Deviation from baseline value
	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
17-01-2005	1.41	-	-
18-02-2005	-0.14	-1.55	-1.55
23-03-2005	0.05	0.19	-1.36
15-04-2005	-0.24	-0.29	-1.65

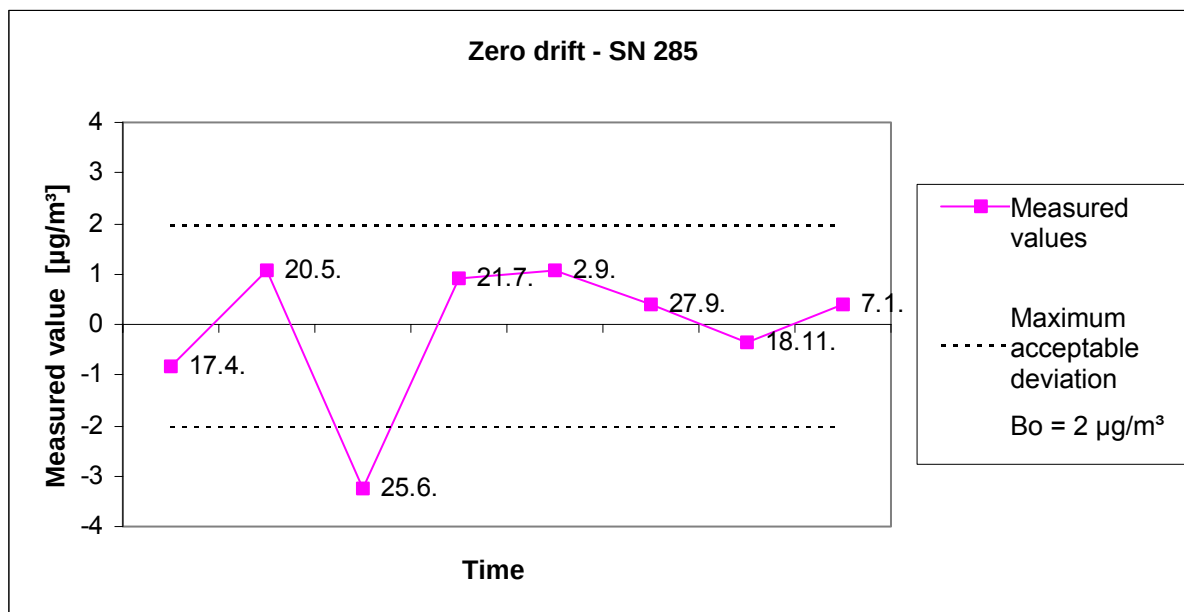


Fig. 22: Zero point drift - SN 285

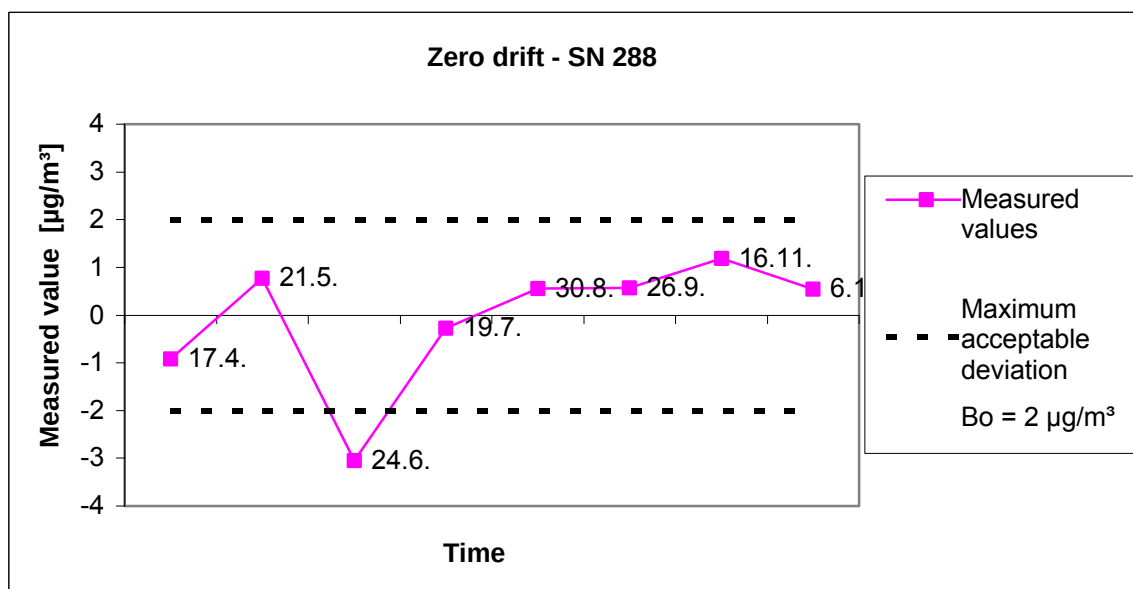


Fig. 23: Zero point drift - SN 288

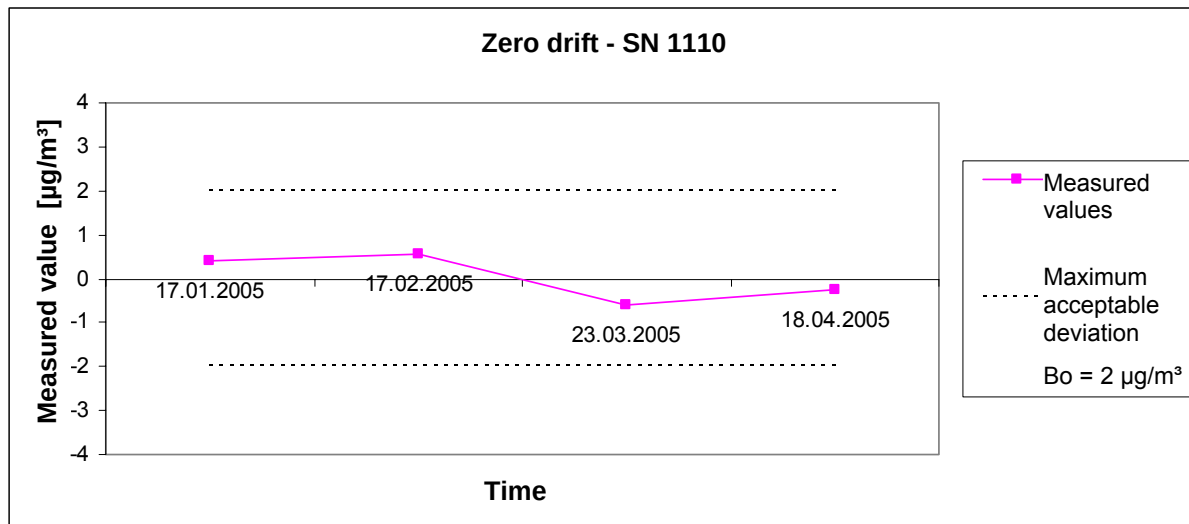


Fig. 24 Zero point drift - SN 1110

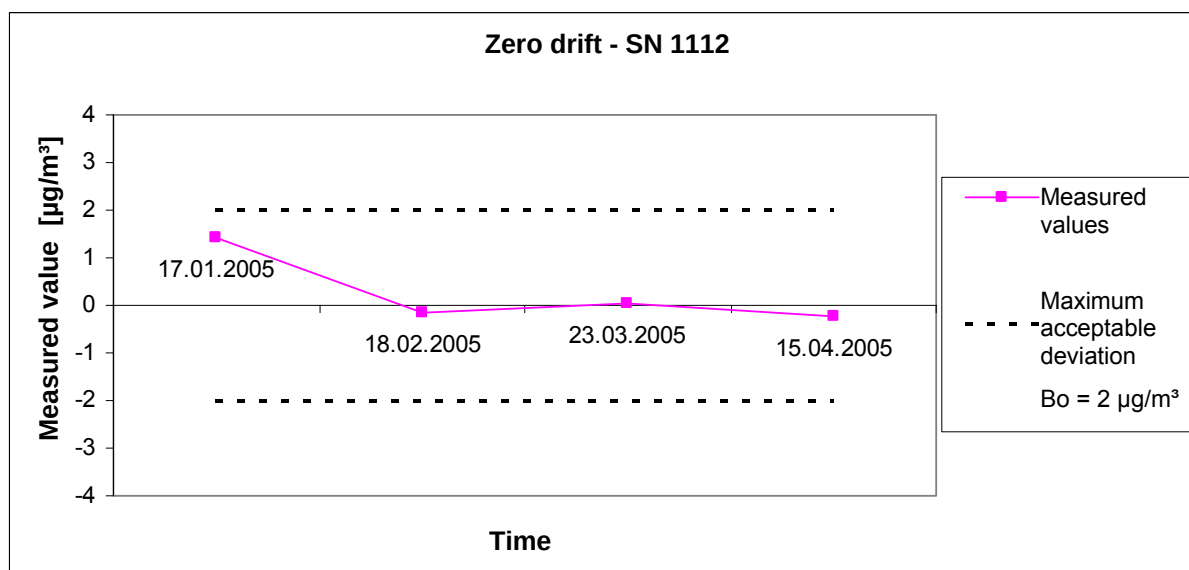


Fig. 25: Zero point drift - SN 1112

6.1 5.2.10 Measured value drift

The change of the measured value over time, in the region of the B_1 reference value, shall be within ± 5 % over 24 hours and over the maintenance interval.

6.2 Equipment

Reference films + evaluation software (Long).

6.3 Implementation of the test

This test was conducted at approximately monthly intervals as part of the field testing program over a period of about 9 months. For verification of their sensitivity, the instruments are operated with a special manufacturer-supplied software and four different reference films each corresponding to a defined mass. This method is used by the manufacturer to check the linearity of the radiometric measurement.

As part of the supplementary tests, this test was also carried out at the "Furulund" site in accordance with the procedure outlined above.

Note: In the modified instrument version OPSIS SM 200 (New), the sensitivity verification software is directly implemented in a hidden service menu.

6.4 Evaluation

For reasons of instrument design, an evaluation of the measured value drift over 24 hours is not possible.

The parameter examined, therefore, was the change (%) in the mass reading over the 1-month maintenance interval.

As an aside it should be noted here that the reference films can merely simulate mass values, not concentration levels. An examination in the region of B_1 ($= 40 \mu\text{g}/\text{m}^3$) was therefore not possible.

6.5 Findings

The measured value drift did not exceed 3.14% (SN 285) and 2.4% (SN 288), respectively, over the maintenance interval. In the supplementary tests, the measured values drifted by no more than 0.34% (SN 1110) and -0.80 (SN 1112), respectively.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Tables 23 and 24 present the percentage deviations of the measured values from the respective previous value. Fig. 26 and Fig. 27 illustrate the measured value drift determined with all four reference films in the form of graphic diagrams.

The results from the supplementary test are summarized in Table 25 and Table 26, and in Fig. 28 and Fig. 29, respectively.

Table 23: *Measured value drift - SN 285*

Date	Deviation (SN 285)			
	Film 1 %	Film 2 %	Film 3 %	Film 4 %
10-04-2002	0	0	0	0
21-05-2002	0.30	0.63	0.24	0.25
24-06-2002	0.56	-0.01	0.28	0.18
19-07-2002	-3.41	-2.40	-0.75	-1.18
30-08-2002	-0.19	0.30	-0.07	0.13
26-09-2002	-0.15	-0.50	-0.17	-0.13
16-11-2002	1.77	1.24	0.67	0.54
06-01-2003	0.82	0.55	0.44	0.20

Table 24: *Measured value drift - SN 288*

Date	Deviation (SN 288)			
	Film 1 %	Film 2 %	Film 3 %	Film 4 %
11-04-2002	0	0	0	0
20-05-2002	-0.15	-0.03	0.15	0.07
25-06-2002	-0.16	-0.09	0.07	0.18
21-07-2002	-2.28	-1.78	-0.53	-0.96
01-09-2002	0.73	0.55	0.29	0.25
27-09-2002	-0.45	-0.43	-0.14	-0.19
18-11-2002	2.44	1.94	1.13	0.80
07-01-2003	-0.16	-0.12	-0.19	0.00

Table 25: Measured value drift - SN 1110

Date	Deviation (SN 1110)			
	Film 1 %	Film 2 %	Film 3 %	Film 4 %
16-12-2004	-	-	-	-
17-01-2005	0.12	0.16	0.01	0.05
17-02-2005	-0.34	-0.02	0.01	0.02
24-03-2005	0.16	0.04	0.28	0.06
19-04-2005	0.20	0.15	-0.05	0.01

Table 26: Measured value drift - SN 1112

Date	Deviation (SN 1112)			
	Film 1 %	Film 2 %	Film 3 %	Film 4 %
16-12-2004	-	-	-	-
17-01-2005	0.06	0.27	0.01	0.07
17-02-2005	0.56	-0.24	-0.07	-0.06
24-03-2005	-0.80	-0.02	0.27	0.09
19-04-2005	0.17	0.04	0.05	0.03

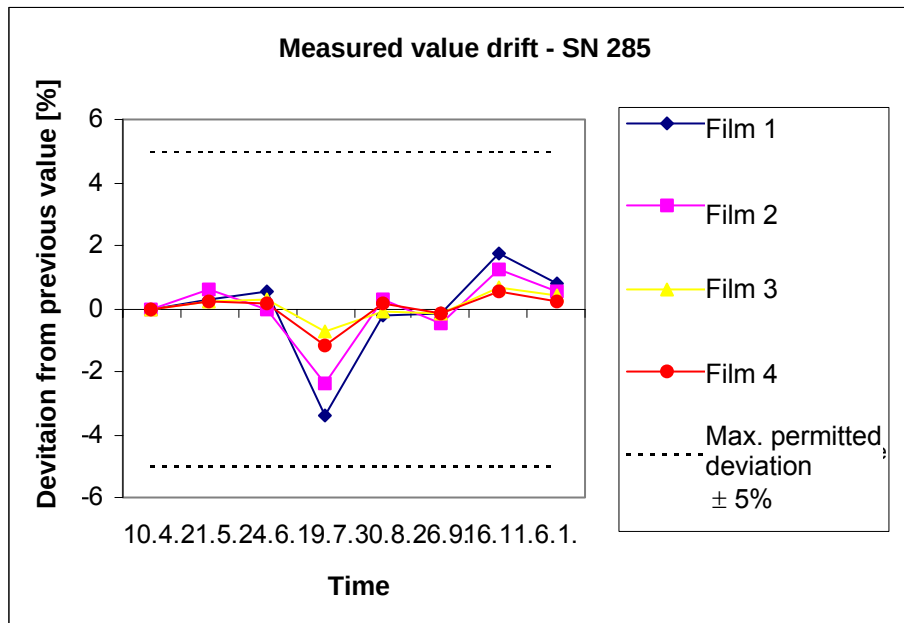


Fig. 26: Measured value drift - SN 285

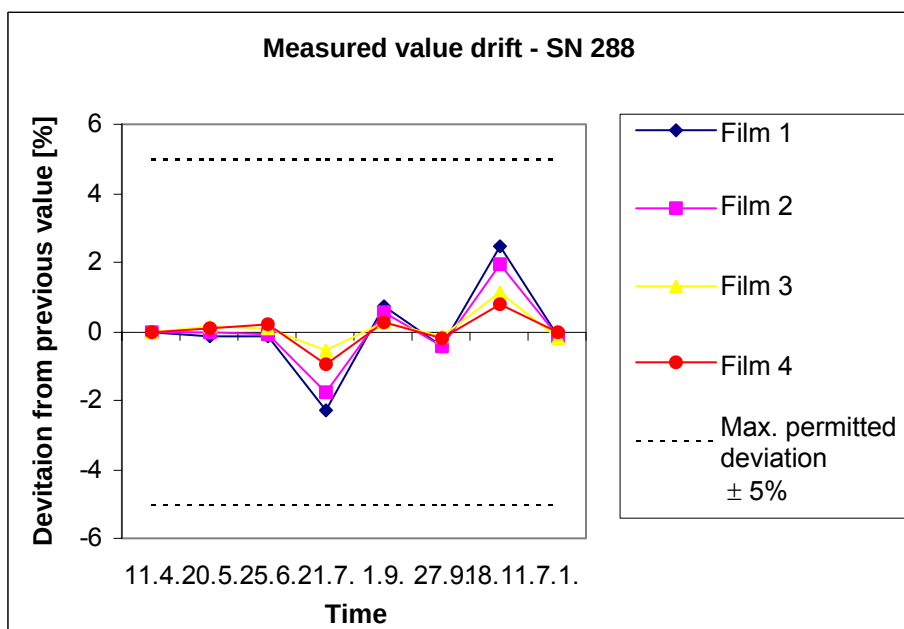


Fig. 27: Measured value drift - SN 288

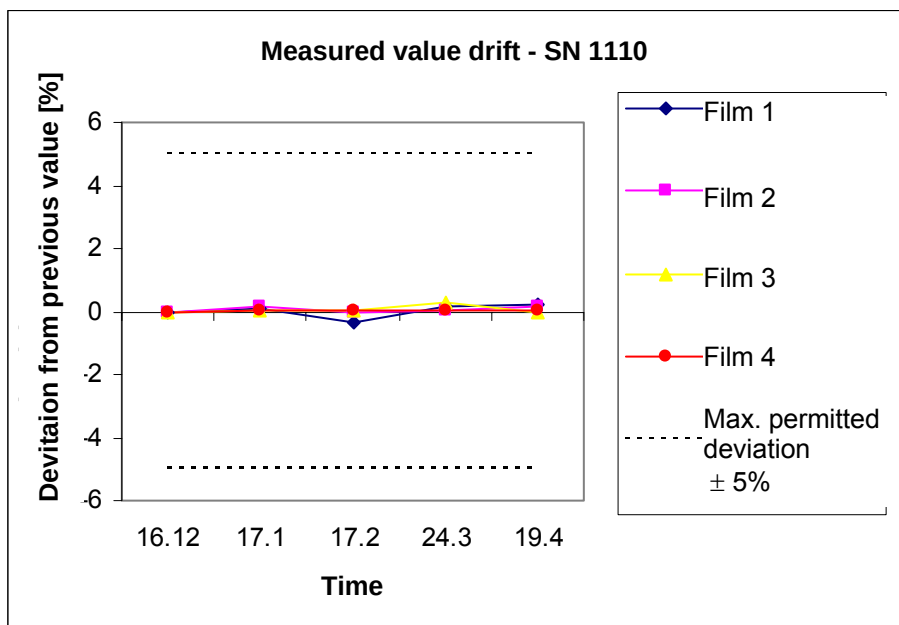


Fig. 28: Measured value drift - SN 1110

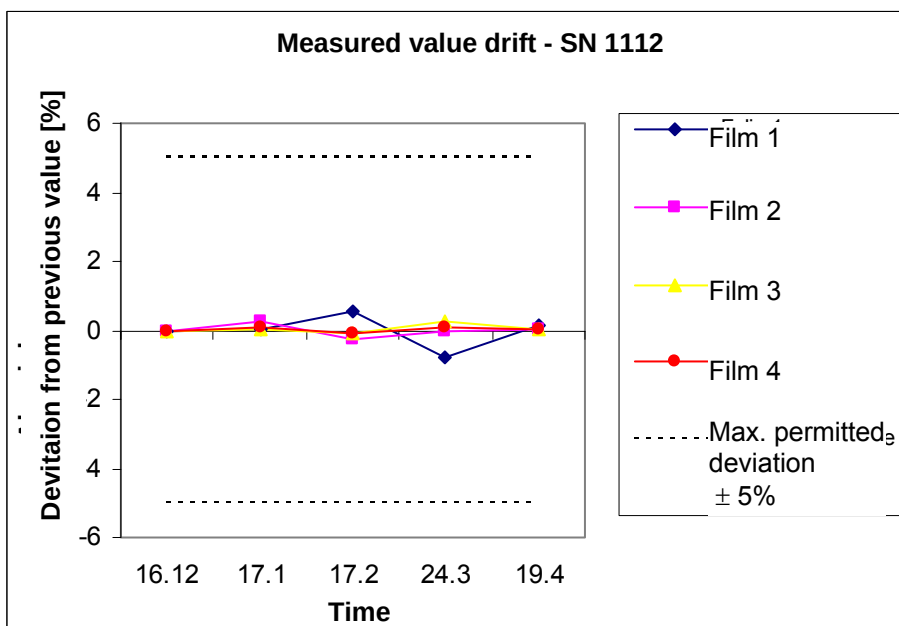


Fig. 29: Measured value drift - SN 1112

6.1 5.2.11 Cross-sensitivity

The absolute values of the sums of the positive and negative deviations, respectively, caused by cross-sensitivity to accompanying substances in the measured medium shall not exceed B_0 in the region of the zero point, nor 3% of B_2 in the region of B_2 . Accompanying substances shall be used in concentrations in the region of the B_2 level of the respective substance. Where no appropriate reference value is known, a suitable reference value shall be defined and stated by the Test Institute by agreement with the other test institutes.

This item is not relevant for dust measuring instruments. The minimum requirement 5.3.4 applies here. The results of these tests are therefore presented under Module 5.3.4.

6.2 Equipment

Not applicable.

6.3 Implementation of the test

Not applicable.

6.4 Evaluation

Not applicable.

6.5 Findings

Not applicable.

Minimum requirement fulfilled? -

6.6 Comprehensive presentation of the test result

Not applicable.

6.1 5.2.12 Repeatability

The repeatability R_D of the measurements is to be determined via double tests using two instruments of identical design and should not be less than 10. For reference, the value of B_1 shall be used.

6.2 Equipment

For the repeatability tests we additionally used the instruments listed in Chapter 5.

6.3 Implementation of the test

Repeatability is defined as the maximum difference between two randomly selected individual values obtained under reference conditions. The level of repeatability was determined by means of two identical instruments run concurrently during the field test. Measuring data from the entire field testing campaign are adduced for this purpose.

Since the supplementary tests involved comparative measurements between SM 200 (New) and SM 200 (Old) units, the data gained in these measurements were likewise evaluated for this purpose.

6.4 Evaluation

The repeatability is calculated as follows:

$$R = \frac{B_1}{U} \geq 10 \quad \text{wherein} \quad U = \pm s_D \cdot t_{(n;0,95)} \quad \text{and} \quad s_D = \sqrt{\frac{1}{2n} \cdot \sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

- R = Repeatability at B_1
- U = Uncertainty
- B_1 = $40 \mu\text{g}/\text{m}^3$ (VDI)
- s_D = Standard deviation from double determinations
- n = Number of double determinations
- $t_{(n;0,95)}$ = Student's factor for 95% confidence
- x_{1i} = Measuring signal from system 1 (e.g., SN 285) at i^{th} concentration
- x_{2i} = Measuring signal from system 2 (e.g., SN 288) at i^{th} concentration

6.5 Findings

The repeatability value was at least 10 - 16 in the field tests. In the supplementary tests, a repeatability level of at least 12 was determined.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

The results of the investigation are summarized in Table 27 and Table 28. Graphic diagrams plotting these findings are presented in Figs. 19 through 26.

Note: The uncertainties determined are related to the reference value B_1 for each site.

Table 27: Concentration averages, standard deviation, uncertainty range and repeatability under field conditions

Site	Number	$\bar{c}$ (SN 285)	$\bar{c}$ (SN 288)	$\bar{c}_{ges}$	s_D	t	U	R
		$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$		$\mu\text{g}/\text{m}^3$	
Cologne parking lot	39	27.15	27.51	27.33	1.956	2.023	3.957	10
Wesseling	62	29.89	30.12	30.01	1.445	1.999	2.889	14
Mechernich	114	19.46	19.32	19.39	1.281	1.981	2.538	16
Brühl	45	19.54	18.86	19.20	1.380	2.014	2.779	14

- $\bar{c}$ (SN 285): Average of the concentrations obtained with SN 285
- $\bar{c}$ (SN 288): Average of the concentrations obtained with SN 288
- $\bar{c}_{ges}$: Average of the concentrations obtained with SN 285 & SN 288 (total)

Table 27: Concentration averages, standard deviation, uncertainty range and repeatability under field conditions, Furulund site

Site	Number	$\bar{c}$ (G1)	$\bar{c}$ (G2)	$\bar{c}_{ges}$	s_D	t	U	R
Furulund		$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$		$\mu\text{g}/\text{m}^3$	
SN 1110 vs. SN 1112	104	18.12	18.63	18.38	1.142	1.983	2.265	18
SN 1110 vs. SN 276	80	18.14	18.71	18.43	1.654	1.990	3.291	12
SN 1112 vs. SN 276	80	18.53	18.71	18.62	1.412	1.990	2.810	14

Individual values are given in the enclosed Appendix 9 and Appendix 14.

6.1 5.2.13 Hourly values

The measuring method must permit the formation of hourly average values.

6.2 Equipment

Clock.

6.3 Implementation of the test

The measuring system's ability to form hourly values was verified.

6.4 Evaluation

Under the applicable Directive [7], the limits for PM10 fine particulates are related to a minimum averaging period of 24 hours. The formation of hourly averages is therefore not necessary for instruments designed to monitor these limits. As a result, the instruments will supply one reading every 24 hours in normal operating mode. However, averaging periods of at least 8 hours may be used alternatively.

6.5 Findings

The formation of hourly values of the PM10 fine particulates fraction is not necessary for monitoring the relevant threshold values.

Minimum requirement fulfilled? not applicable

6.6 Comprehensive presentation of the test result

Not required here.

T

6.1 5.2.14 Mains voltage and mains frequency

Changes in instrument readings occurring at the B_1 reference value as a result of typical mains voltage fluctuations in the (230^{+15}_{-20}) range shall not exceed B_0 . In mobile use, moreover, the changes in instrument readings caused by mains frequency fluctuations in the 50 ± 2 Hz interval shall not exceed B_0 .

6.2 Equipment

Isolation variable-ratio transformer, reference films + evaluation software (Long)

6.3 Implementation of the test

To investigate the mains voltage dependence of the instrument reading, the mains voltage was reduced from 230 V to 210 V, then raised again to 245 V via a 230 V intermediate step. For the sensitivity check the instruments were run with a special manufacturer-supplied software and four different reference films corresponding to defined masses. This procedure is used by the instrument manufacturer to determine the linearity of the radiometric measurement.

The impact of mains frequency fluctuations on the instrument reading was tested by reducing the mains frequency from 50 to 48 Hz, then raising it to 52 Hz via a 50 Hz intermediate level. The instrument's sensitivity to these changes was determined in a manner similar to its sensitivity to mains voltage fluctuations.

6.4 Evaluation

The reference films can merely simulate mass values, not concentration levels. Direct testing in the B_1 concentration range ($= 40 \mu\text{g}/\text{m}^3$) was therefore not possible.

The evaluation was instead conducted as follows:

The percentage change in the mass reading per voltage step (or mains frequency step) was determined. From the ratio of B_0 ($2 \mu\text{g}/\text{m}^3$) and B_1 ($40 \mu\text{g}/\text{m}^3$) the maximum acceptable deviation of 5% was calculated ($2/40 \cdot 100$).

6.5 Findings

Compliance with the minimum requirements was rated as outlined above.

The change in instrument reading caused by mains voltage variations did not exceed 0.56 % in the case of instrument No. 1 (SN 285) and 0.77% in the case of instrument No. 2 (SN 277), respectively (deviation from previous value in each case).

The change in instrument reading caused by mains frequency variations never exceeded -0.45 % in the case of instrument No.1 (SN 285) and -0.57% in the case of instrument No. 2 (SN 288), respectively (deviation related to previous value in each case).

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 29 and Table 30 summarize the test results.

Table 29: Impact of mains voltage changes on measurement signals (deviation in %)

Mains voltage		Deviation (SN 285)			
Initial voltage	End voltage	Film 1	Film 2	Film 3	Film 4
V	V	%	%	%	%
230	210	0.56	0.46	0.25	0.27
230	245	0.47	0.44	0.36	0.14
Mains voltage		Deviation (SN 288)			
Initial voltage	End voltage	Film 1	Film 2	Film 3	Film 4
V	V	%	%	%	%
230	210	0.77	0.47	0.42	0.24
230	245	0.52	0.35	0.27	0.47

Table 30: Impact of mains frequency changes on measurement signals (deviation in %)

Mains frequency		Deviation (SN 285)			
Initial frequency	End frequency	Film 1	Film 2	Film 3	Film 4
Hz	Hz	%	%	%	%
50	48	0.18	0.24	0.09	0.06
50	52	-0.39	-0.45	-0.16	0.11
Mans frequency		Deviation (SN 288)			
Initial frequency	End frequency	Film 1	Film 2	Film 3	Film 4
Hz	Hz	%	%	%	%
50	48	0.34	0.30	0.09	0.05
50	52	-0.57	-0.43	-0.24	0.11

For individual results refer to the enclosed Appendix 5.

T

6.1 5.2.15 Power failure

In the case of a system breakdown or electrical power failure, an uncontrolled release of operating gas and calibration gas must be effectively prevented. Instrument parameters must be protected by buffering against loss in the case of a power failure. Upon restoration of the power supply the instrument must return automatically to a condition in which it is ready for measurements, and must commence measurements consistent with operating specifications.

6.2 Equipment

No necessary with this minimum requirement..

6.3 Implementation of the test

An electrical power failure was simulated to find out whether the instrument remains undamaged and is ready to resume measurements upon restoration of the mains supply.

6.4 Evaluation

Since the measuring instruments need no operating or calibration gas to function, an uncontrolled release of such gases is not possible.

As regards the case of an electric power failure, a distinction must be made between two scenarios.

- 1) In the event of a power failure the instrument will re-start automatically and resume sampling with the same filter. The downtime is taken into account in the analysis of that filter, and is stored in the instrument's memory along with the data for that measurement.
- 2) A filter change will not be carried out after a power failure except if the routine filter change (cycle time 1) would have been due during the downtime. The fact that a filter has been changed is reliably evident from the data stored in memory.

6.5 Findings

All instrument parameters are protected by buffering against data loss.

Once re-energized, the measuring system returns to an error-free state and is ready to operate.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not required here.

6.1 5.2.16 Instrument functions

It must be possible to monitor key equipment functions by means of telemetrically transmittable status signals.

6.2 Equipment

Modem, ComVisioner software with SM 200 protocol.

6.3 Implementation of the test

A modem was connected to the measuring system. In addition, the OPSIS ComVisioner software and SM200 protocol are needed. The instrument's status signals, etc., were polled by remote data transmission.

6.4 Evaluation

The measuring system has full telemetric monitoring and control capabilities.

6.5 Findings

The tested instruments can be controlled and monitored from a remote computer just as if they were locally operated.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not required here.

6.1 5.2.17 Mode switching

The changeover between measuring and functional test mode and/or calibration mode must be achievable telemetrically by computer control, apart from being manually triggerable.

6.2 Equipment

Not necessary with this minimum requirement.

6.3 Implementation of the test

The measuring instrument can be controlled locally and by telemetric remote control in the same manner.

6.4 Evaluation

All test procedures can be initiated by an operator standing before the instrument or by telemetric remote control.

Where testing requires the insertion of reference films into the instrument, the on-site presence of an operator is indispensable.

6.5 Findings

As a rule, all functional testing and calibration steps can be performed locally (directly on the instrument) or by telemetric remote control.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not required here.

6.1 5.2.18 Availability

The instrument shall have an availability of at least 90 %.

6.2 Equipment

Not necessary with this minimum requirement.

6.3 Implementation of the test

The starting point and end point of the availability tests coincided with the start and end of field testing at each of the Wesseling, Mechernich and Brühl sites, respectively. All disruptions of the test, e.g., due to malfunctions or for maintenance purposes, were recorded.

In the supplementary tests, system availability was also determined for the modified instrument version OPSIS SM 200 (New) between December 18, 2004, and April 12, 2005, in a manner similar to the previous tests.

6.4 Evaluation

Table 31 and Table 32 summarize the operating, maintenance and malfunction times of the measuring systems. Equipment availability was assessed in two ways, i.e., with and without test-related downtimes. At the Mechernich site, two extended power failures accounted for a total of 48 hours of downtime. The high number of hours expended on maintenance reflects the monthly zero point drift and sensitivity drift tests, which implied major downtimes for each instrument. Note that these downtimes are not attributable to the equipment but entirely to test requirements.

In the supplementary tests, all recorded downtimes were evaluation-related. A list of the operating, maintenance and malfunction times is given in Table 33 and Table 34.

6.5 Findings

The availability rate was 97.8% excluding test-related downtimes and 90.6% including test-related downtimes for both instruments. In the supplementary tests, 100% availability was achieved if one disregards various test-induced downtimes. Including such test-related downtimes, an availability of 89.7% was observed.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 31: *Determination of availability rates (not including test-related downtimes)*

		Instrument 1 (SN 285)	Instrument 2 (SN 288)
Operating time	h	5424	5424
Downtime due to failures	h	48	48
Maintenance time	h	72	72
Net operating time	h	5304	5304
Availability	%	97.8	97.8

Table 32: *Determination of availability rates (including test-related downtimes)*

		Instrument 1 (SN 285)	Instrument 2 (SN 288)
Operating time	h	5856	5856
Downtime due to failures	h	48	48
Maintenance time	h	504	504
Net operating time	h	5304	5304
Availability	%	90.6	90.6

*Table 32: Determination of availability rates (not including test-related downtimes),
Furulund site*

		Instrument 1 (SN 1110)	Instrument 2 (SN 1112)
Operating time	h	2784	2784
Downtime due to failures	h	0	0
Maintenance time	h	0	0
Net operating time	h	2784	2784
Availability	%	100.0	100.0

*Table 32: Determination of availability rates (including test-related downtimes),
Furulund site*

		Instrument 1 (SN 1110)	Instrument 2 (SN 1112)
Operating time	h	2784	2784
Downtime due to failures	h	0	0
Maintenance time	h	288	288
Net operating time	h	2496	2496
Availability	%	89.7	89.7

6.1 5.2.19 Converter efficiency

In measuring instruments equipped with a converter, the latter must have an efficiency of at least 95 %.

According to the Guideline VDI 4203, sheet 3, item 4.4.1, this test item is not applicable to dust measuring instruments with pre-separator operating on a physical mass determination principle.

6.2 Equipment

Not applicable.

6.3 Implementation of the test

Not applicable.

6.4 Evaluation

Not applicable.

6.5 Findings

Not applicable.

Minimum requirement fulfilled? -

6.6 Comprehensive presentation of the test result

Not applicable.

6.1 5.2.20 Maintenance interval

The maintenance interval of the measuring instrument shall be determined and specified. It should be 28 days where possible, but in no case less than 14 days.

6.2 Equipment

Not necessary with this minimum requirement.

6.3 Implementation of the test

In testing for compliance with this minimum requirement, it was examined which maintenance steps are necessary at which intervals to ensure proper instrument operations. The results of the zero and reference point drift test performed in accordance with modules 5.2.9 and 5.2.10, respectively, were likewise taken into account.

6.4 Evaluation

No unacceptable drift was observed on the instruments over a 3-month period. The maintenance interval is therefore determined by the requisite maintenance interventions. The instrument has a filter stack comprising 40 filter elements, so theoretically a maintenance interval of 40 days should be achievable with the system. To insure proper instrument operation, measurements should be interrupted at least once a month for one day to check all instrument functions (refer also to module 4.1.2).

During operating periods, maintenance can be limited to checking for dirt, error messages, or any non-compliances identified in the quality assurance tests.

If non-compliances are detected by the built-in tests and the problem cannot be solved by the steps described in section 7 of the Operating Manual, an employee of the manufacturing company must be contacted. Likewise, a re-calibration of the mass determination process and volumetric flow rate can only be performed by manufacturer's staff.

6.5 Findings

The maintenance interval is determined by the required maintenance operations. It has a duration of 1 month.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

For requisite maintenance operations refer to module 4.1.2 herein. A description is also given in section 6 of the Operator Manual.

6.1 5.2.21 Overall uncertainty

The extended measuring uncertainty of the instrument shall be determined. The value thus obtained shall not exceed the requirements of the EC daughter directives concerning air quality.

6.2 Equipment

Not necessary with this measuring requirement.

6.3 Implementation of the test

The extended measuring uncertainty of the measuring instrument was determined for individual values in the range of the short-term immission threshold and for average values in the concentration range of the long-term immission threshold. The instrument process characteristics determined in the suitability tests were then compiled.

In addition, the overall uncertainty was calculated in the light of the supplementary test results. For this process, those process characteristics which were not re-tested during the supplementary tests were estimated using the least favourable value of the respective parameter from the previous tests.

6.4 Evaluation

The expanded overall uncertainty of the measuring system was determined on the basis of VDI Guideline 4202, Sheet 1, Appendix C [1].

6.5 Findings

To calculate the extended measuring uncertainties, the individual results for the respective test items were evaluated in aggregate. Where several independent values were available from the individual tests, the least favourable value was used.

The overall uncertainties were found to amount to 15.4% and 15.1%, respectively, for $U(c)$ and 10.6% for both systems for $U(\bar{c})$. In the supplementary tests, overall uncertainties of 8.6 % and 9.1 % for $U(c)$ and of 9.0 % for $U(\bar{c})$ (both systems) were determined.

The individual values are given in Table 35 through Table 42. All values obtained are below the specified 25% threshold for overall uncertainties.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 35: Extended measuring uncertainty $U(c)$ for instrument SN 285
Reference value: $50 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 285	Requirement	Result	Uncertainty	Uncertainty squared
			$\mu\text{g}/\text{m}^3$	$(\mu\text{g}/\text{m}^3)^2$
Repeatability	≥ 10	10	2.000	4.000
Confidence range CI_{95} per EN 12341	$\leq 5 \mu\text{g}/\text{m}^3$	$3.96 \mu\text{g}/\text{m}^3$	2.286	5.227
Temperature dependence of zero point	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.42 \mu\text{g}/\text{m}^3$	-0.242	0.059
Temperature dependence of measured value	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.23 \mu\text{g}/\text{m}^3$	-0.134	0.018
Zero point drift	$\leq 2 \mu\text{g}/\text{m}^3$	$-3.23 \mu\text{g}/\text{m}^3$	-1.865	3.478
Measured value drift	$\leq 2 \mu\text{g}/\text{m}^3$	$1.36 \mu\text{g}/\text{m}^3$	0.788	0.621
Mains voltage	$\leq 2 \mu\text{g}/\text{m}^3$	$0.22 \mu\text{g}/\text{m}^3$	0.129	0.017
Cross-sensitivities	$\leq 6 \mu\text{g}/\text{m}^3$	$0 \mu\text{g}/\text{m}^3$	0.000	0.000
Uncertainty of the test standard	$\leq 1 \mu\text{g}/\text{m}^3$	$2 \mu\text{g}/\text{m}^3$	1.155	1.333
			$\sum_K u^2(c_K)$	14.753
			$U(c) = 2 u(c)$	$\mu\text{g}/\text{m}^3$
				7.682
			$\frac{U(c)}{I_2}$	15.4%

Table 36: *Extended measuring uncertainty $U(c)$ for instrument SN 288*
Reference value: $50 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 288	Requirement	Result	Uncertainty	Uncertainty squared
			$\mu\text{g}/\text{m}^3$	$(\mu\text{g}/\text{m}^3)^2$
Repeatability	≥ 10	10	2.000	4.000
Confidence range CI ₉₅ per EN 12341	$\leq 5 \mu\text{g}/\text{m}^3$	$3.96 \mu\text{g}/\text{m}^3$	2.286	5.227
Temperature dependence of zero point	$\leq 2 \mu\text{g}/\text{m}^3$	$0.48 \mu\text{g}/\text{m}^3$	-0.277	0.077
Temperature dependence of measured value	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.49 \mu\text{g}/\text{m}^3$	-0.282	0.079
Zero point drift	$\leq 2 \mu\text{g}/\text{m}^3$	$3.05 \mu\text{g}/\text{m}^3$	-1.761	3.101
Measured value drift	$\leq 2 \mu\text{g}/\text{m}^3$	$0.97 \mu\text{g}/\text{m}^3$	0.562	0.316
Mains voltage	$\leq 2 \mu\text{g}/\text{m}^3$	$0.31 \mu\text{g}/\text{m}^3$	0.178	0.032
Cross-sensitivities	$\leq 6 \mu\text{g}/\text{m}^3$	$0.11 \mu\text{g}/\text{m}^3$	0.064	0.004
Uncertainty of the test standard	$\leq 1 \mu\text{g}/\text{m}^3$	$2 \mu\text{g}/\text{m}^3$	1.155	1.333
			$\sum_K u^2(c_K)$	14.169
			$U(c) = 2u(c)$	$\mu\text{g}/\text{m}^3$
				7.528
			$\frac{U(c)}{I_2}$	15.1%

Table 37: Extended measuring uncertainty $U(\bar{c})$ for instrument SN 285
Reference value: $40 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 285	Uncertainty (individual value)	Time base	Number n_K	Uncertainty squared (mean)
	$\mu\text{g}/\text{m}^3$			$(\mu\text{g}/\text{m}^3)^2$
Repeatability	2,000	24 hours	365	0.011
Confidence range CI_{95} per EN 12341	2,286	1 year	1	5.27
Temperature dependence of zero point	-0,242	1 year	1	0.059
Temperature dependence of measured value	-0,134	1 year	1	0.018
Zero point drift	-1,865	1 month	12	0.290
Measured value drift	0,788	1 month	12	0.052
Mains voltage	0,129	1 year	1	0.017
Cross-sensitivities	0,000	1 year	1	0.000
Uncertainty of the test standard	1,155	1 year	1	1.333
				$\sum_K u_M^2(c_K)$
				7.007
				$U(c) = 2\bar{u}(c)$
				$\mu\text{g}/\text{m}^3$
				5.294
				$\frac{U(c)}{I_1}$
				10.6%

Table 38: Extended measuring uncertainty $U(\bar{c})$ for instrument SN 288
Reference value: $40 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 288	Uncertainty (individual value)	Time base	Number n_K	Uncertainty squared (mean)
	$\mu\text{g}/\text{m}^3$			$(\mu\text{g}/\text{m}^3)^2$
Repeatability	2.000	24 hours	365	0.011
Confidence range CI_{95} per EN 12341	2.286	1 year	1	5.227
Temperature dependence of zero point	-0.277	1 year	1	0.077
Temperature dependence of measured value	-0.282	1 year	1	0.079
Zero point drift	-1.761	1 month	12	0.258
Measured value drift	0.562	1 month	12	0.026
Mains voltage	0.178	1 year	1	0.032
Cross-sensitivities	0.064	1 year	1	0.004
Uncertainty of the test standard	1.155	1 year	1	1.333
				$\sum_K u_M^2(c_K)$
				7.048
				$U(c) = 2\bar{u}(c)$
				$\mu\text{g}/\text{m}^3$
				5.310
				$\frac{U(c)}{I_1}$
				10.6%

Table 39: Extended measuring uncertainty $U(c)$ for instrument SN 1110
Reference value: $50 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 1110	Requirement	Result	Uncertainty	Uncertainty squared
			$\mu\text{g}/\text{m}^3$	$(\mu\text{g}/\text{m}^3)^2$
Repeatability	≥ 10	18	1.111	1.234
Confidence range CI_{95} per EN 12341	$\leq 5 \mu\text{g}/\text{m}^3$	$2.27 \mu\text{g}/\text{m}^3$	1.311	1.719
Temperature dependence of zero point*	$\leq 2 \mu\text{g}/\text{m}^3$	$0.48 \mu\text{g}/\text{m}^3$	0.277	0.077
Temperature dependence of measured value*	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.49 \mu\text{g}/\text{m}^3$	-0.283	0.080
Zero point drift	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.59 \mu\text{g}/\text{m}^3$	-0.341	0.116
Measured value drift	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.14\mu\text{g}/\text{m}^3$	-0.079	0.006
Mains voltage*	$\leq 2 \mu\text{g}/\text{m}^3$	$0.31 \mu\text{g}/\text{m}^3$	0.178	0.032
Cross-sensitivities*	$\leq 6 \mu\text{g}/\text{m}^3$	$0.11\mu\text{g}/\text{m}^3$	0.064	0.004
Uncertainty of the test standard*	$\leq 1 \mu\text{g}/\text{m}^3$	$2 \mu\text{g}/\text{m}^3$	1.155	1.333
			$\sum_K u^2(c_K)$	4.601
			$U(c) = 2 u(c)$	$\mu\text{g}/\text{m}^3$
				4.290
			$\frac{U(c)}{I_2}$	8.6%

* Least favourable result from the suitability tests of OPSIS SM 200 (Old)

Table 40: Extended measuring uncertainty $U(c)$ for instrument SN 1112
Reference value: $50 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 1112	Requirement	Result	Uncertainty	Uncertainty squared
			$\mu\text{g}/\text{m}^3$	$(\mu\text{g}/\text{m}^3)^2$
Repeatability	≥ 10	18	1.111	1.234
Confidence range CI_{95} per EN 12341	$\leq 5 \mu\text{g}/\text{m}^3$	$2.27 \mu\text{g}/\text{m}^3$	1.311	1.719
Temperature dependence of zero point*	$\leq 2 \mu\text{g}/\text{m}^3$	$0.8 \mu\text{g}/\text{m}^3$	0.277	0.077
Temperature dependence of measured value*	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.9 \mu\text{g}/\text{m}^3$	-0.282	0.079
Zero point drift	$\leq 2 \mu\text{g}/\text{m}^3$	$1.1 \mu\text{g}/\text{m}^3$	0.814	0.663
Measured value drift	$\leq 2 \mu\text{g}/\text{m}^3$	$-0.2 \mu\text{g}/\text{m}^3$	-0.185	0.034
Mains voltage*	$\leq 2 \mu\text{g}/\text{m}^3$	$0.1 \mu\text{g}/\text{m}^3$	0.178	0.032
Cross-sensitivities	$\leq 6 \mu\text{g}/\text{m}^3$	$0.1 \mu\text{g}/\text{m}^3$	0.064	0.004
Uncertainty of the test standard*	$\leq 1 \mu\text{g}/\text{m}^3$	$2 \mu\text{g}/\text{m}^3$	1.155	1.333
			$\sum_K u^2(c_K)$	5.175
			$U(c) = 2u(c)$	$\mu\text{g}/\text{m}^3$
				4.550
			$\frac{U(c)}{I_2}$	9.1%

* Least favourable result from the suitability tests of OPSIS SM 200 (Old).

Table 41: Extended measuring uncertainty $U(\bar{c})$ for instrument SN 1110
Reference value: $40 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 1110	Uncertainty (individual value)	Time base	Number n_K	Uncertainty squared (mean)
	$\mu\text{g}/\text{m}^3$			$(\mu\text{g}/\text{m}^3)^2$
Repeatability	1,111	24 hours	365	0.000
Confidence range CI_{95} per EN 12341	1,311	1 year	1	1.719
Temperature dependence of zero point*	0,277	1 year	1	0.077
Temperature dependence of measured value*	-0,283	1 year	1	0.080
Zero point drift	-0,341	1 month	12	0.000
Measured value drift	-0,079	1 month	12	0.000
Mains voltage*	0,178	1 year	1	0.032
Cross-sensitivities*	0,064	1 year	1	0.004
Uncertainty of the test standard*	1,155	1 year	1	1.333
			$\sum_K u_M^2(c_K)$	3.245
			$U(c) = 2\bar{u}(c)$	$\mu\text{g}/\text{m}^3$
				3.603
			$\frac{U(c)}{I_1}$	9.0%

* Least favourable result from the suitability tests of OPSIS SM 200 (Old)

Table 42: Extended measuring uncertainty $U(\bar{c})$ for instrument SN 1112
Reference value: $40 \mu\text{g}/\text{m}^3$

Characteristic process parameter Instrument SN 1112	Uncertainty (individual value)	Time base	Number n_K	Uncertainty squared (mean)
	$\mu\text{g}/\text{m}^3$			$(\mu\text{g}/\text{m}^3)^2$
Repeatability	1.111	24 hours	365	0.000
Confidence range CI ₉₅ per EN 12341	1.311	1 year	1	1.719
Temperature dependence of zero point*	0.277	1 year	1	0.077
Temperature dependence of measured value*	-0.282	1 year	1	0.079
Zero point drift	0.814	1 month	12	0.004
Measured value drift	-0.185	1 month	12	0.000
Mains voltage*	0.178	1 year	1	0.032
Cross-sensitivities*	0.064	1 year	1	0.004
Uncertainty of the test standard	1.155	1 year	1	1.333
				$\sum_K u_M^2(c_K)$
				3.248
				$U(c) = 2\bar{u}(c)$
				$\mu\text{g}/\text{m}^3$
				3.604
				$\frac{U(c)}{I_1}$
				9.0%

* Least favourable result from the suitability tests of OPSIS SM 200 (Old)

6.1 5.3.1 Equivalence of the sampling system

For PM10 sampling systems, equivalence to the DIN EN 12 341 [Part 5] reference method shall be determined.

6.2 Equipment

For this test item, the additional equipment listed in para. 3 of the present report is used.

6.3 Implementation of the test

The investigation was carried out by field testing at various sites as stated in para. 4.3.3.1 of the present report. Various seasons and different PM10 concentration levels were taken into account. At the Mechernich field test site, the readings of July 10 and July 13, 2002 and August 20, 2002 had to be discarded for implausibility due to intense rainfall on these days (> 20 mm in 2 hours).

The test was conducted in the same manner at the Furulund site. Equivalence to the reference method was examined for both the two SM 200 (New) specimen instruments (SN 1110 and SN 1112) and for the SM 200 (Old) predecessor model (SN 276).

At least 15 value pairs were determined for each site.

6.4 Evaluation

Requirements according to DIN EN 12341:

The calculated functional relationship $y = f(x)$ between the concentration readings taken with the test instrument and those determined with the reference units should be limited by a two-sided acceptance range. This acceptance range is defined as follows:

$y = (x \pm 10) \mu\text{g}/\text{m}^3$ for concentration averages $\leq 100 \mu\text{g}/\text{m}^3$ and

$y = 0.9x \mu\text{g}/\text{m}^3$ or $1.1x \mu\text{g}/\text{m}^3$, respectively, for concentration averages $> 100 \mu\text{g}/\text{m}^3$

Moreover, the variation coefficient R^2 of the calculated reference equivalence function should be at least equal to 0.95.

This test is aimed at establishing the functional relationship between the concentrations identified by the double readings taken with the tested instrument and the reference device, respectively. Ideally both systems will detect the same suspended dust fraction, i.e., $y = x$. The evaluation process is as follows:

A linear regression analysis was performed on the measured data from each of the four test sites individually and – following compilation of all readings – from all four test sites taken together.

In the supplementary tests, a linear regression analysis was carried out for each specimen instrument from the readings recorded at the "Furulund" site.

The foregoing method yields a reference equivalence function corresponding to the general equation

$$y_i = m \cdot x + b \quad \text{with } i = \text{SN 200 specimen instrument}$$

for each measured value y_i of the respective test instrument as well as for the concentration measured with the reference unit.

6.5 Findings

The following observation applies to all test sites examined:

The reference equivalence functions lie within the limits of the respective acceptance range. Also, the variation coefficient R^2 of the calculated reference equivalence functions is $\geq 0,95$ in the relevant concentration range, in compliance with the requirements. The foregoing holds true for the OPSIS SM 200 (Old) and OPSIS SM 200 (New) units alike.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

The results of the regression analyses are shown in Tables 43 through Table 45. The graphic renderings have been included as Fig. 30 through Fig. 42. Plotted in these diagrams are the fitted lines for the two tested instruments, the curve $y=x$ representing the "ideal" case, and the two-sided acceptance range. An overview of all individual values obtained with the test instruments and the reference devices by test site is given in the enclosed Appendix 12 hereto.

Table 43: *Result of the linear regression analysis of the readings taken with the two test instruments SN 285 and SN 288 at the four sites*

SN 285	Number of values N	Gradient m	Ordinate intercept	R^2
Cologne parking lot	17	1.0374	-1.8928	0.978
Wesseling	40	1.0043	2.0421	0.958
Mechernich	66	1.0345	-0.4712	0.950
Brühl	18	1.0062	-1.3519	0.956

SN 288	Number of values N	Gradient m	Ordinate intercept	R²
Cologne parking lot	17	1.829	-2.697	0.986
Wesseling	40	1.0193	1.8582	0.976
Mechernich	66	1.0243	-0.5352	0.951
Brühl	18	0.9209	0.0913	0.962

Table 44: Results of the linear regression analysis of the readings taken with the test instruments SN 285 and SN 288 (aggregate)

Tested instrument	Number of values N	Gradient m	Ordinate intercept b	R²
SN 285	141	1.0177	0.0478	0.956
SN 288	141	1.0211	-0.1718	0.959

Table 45: Results of the linear regression analysis of the readings taken with the test instruments SN 1110, SN 1112 and SN 276, Furulund site

Tested instrument	Number of values N	Gradient m	Ordinate intercept b	R²
SN 1110	43	1.0946	-1.0318	0.973
SN 1112	43	1.0490	0.3288	0.977
SN 276	34	1.0271	0.5794	0.954

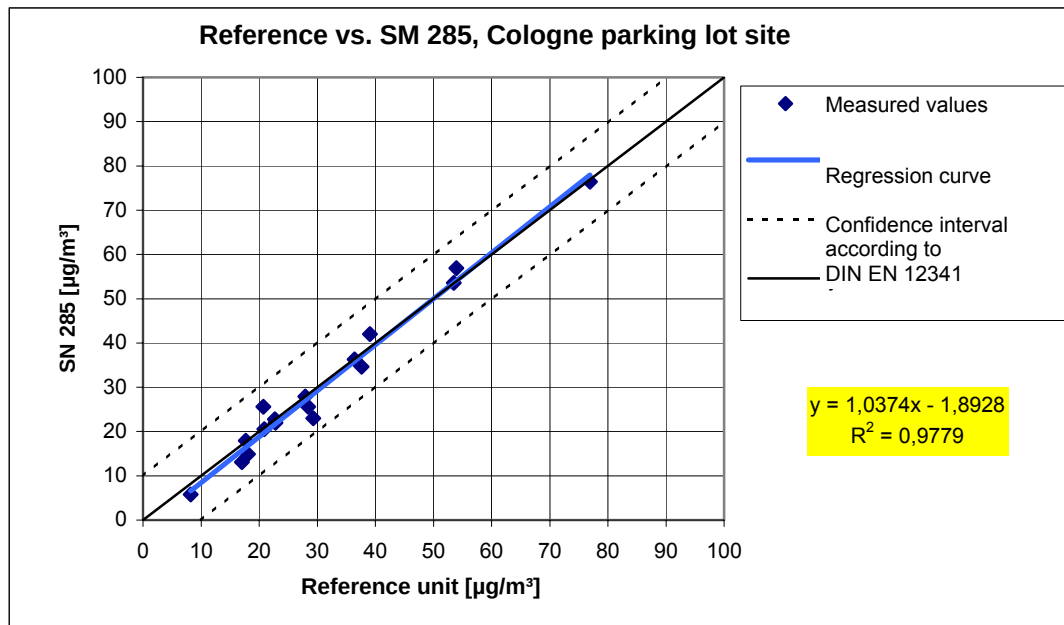


Fig. 30: Reference equivalence function for SN 285, Cologne parking lot site

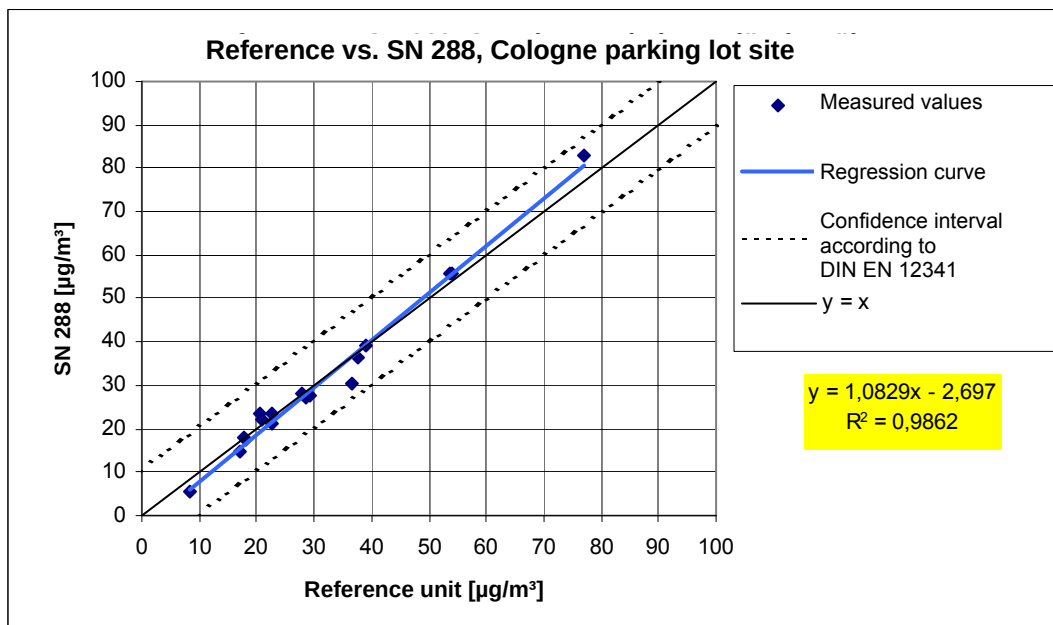


Fig. 31: Reference equivalence function SN 288, Cologne parking lot site

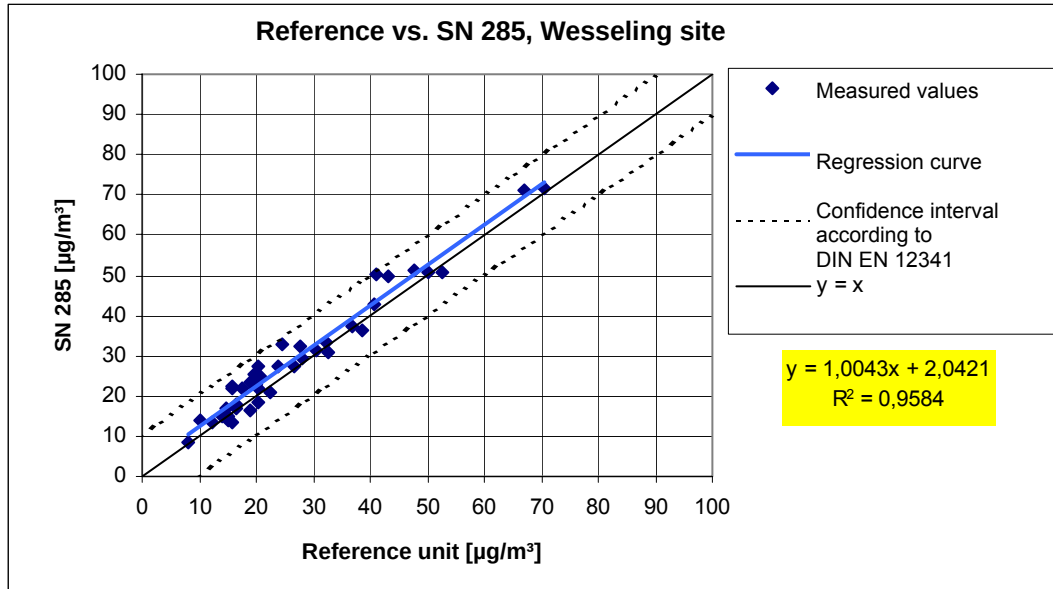


Fig. 32: Reference equivalence function for SN 285, Wesseling site

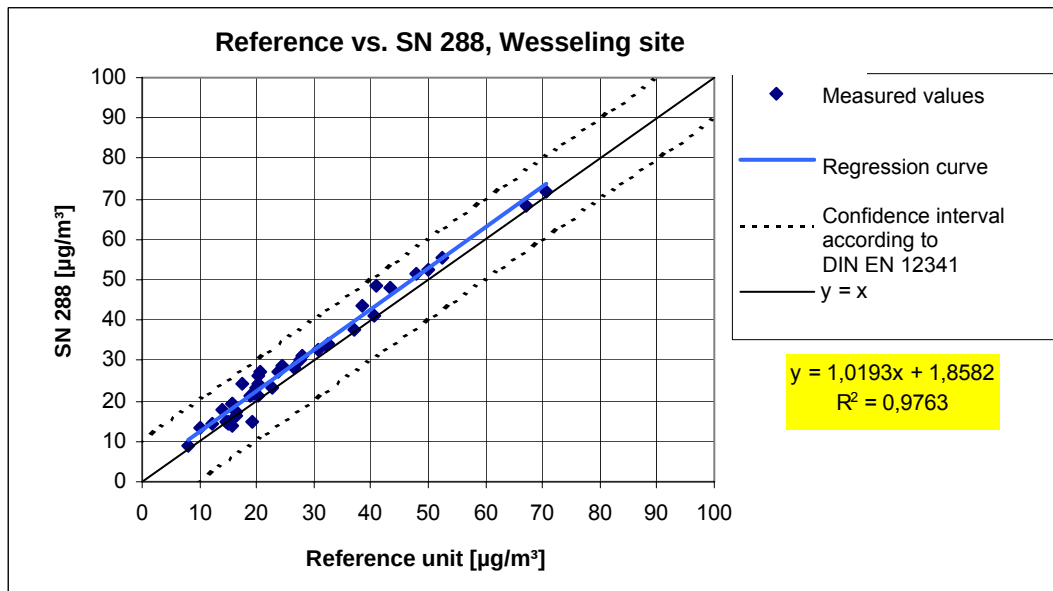


Fig. 33: Reference equivalence function for SN 288, Wesseling site

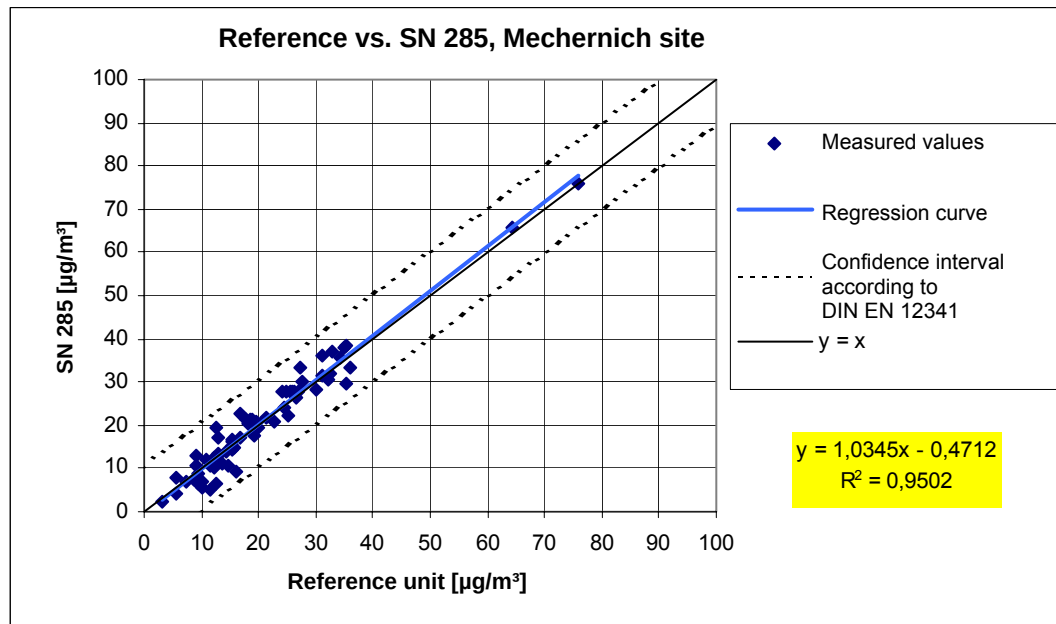


Fig. 34: Reference equivalence function for SN 285, Mechernich site

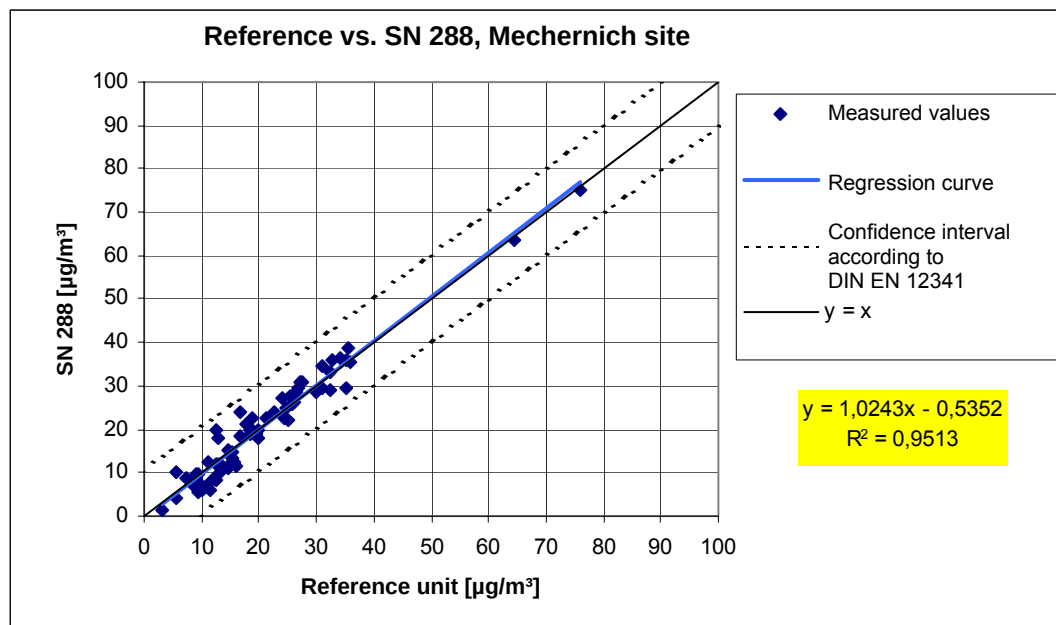


Fig. 35: Reference equivalence function for SN 288, Mechernich site

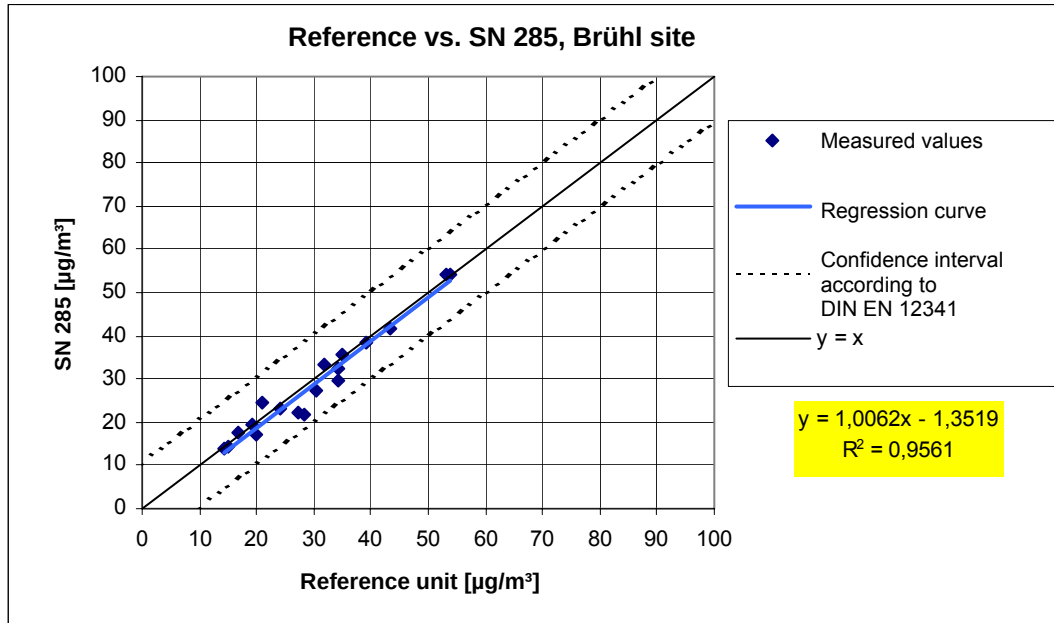


Fig. 36: Reference equivalence function for SN 285, Brühl site

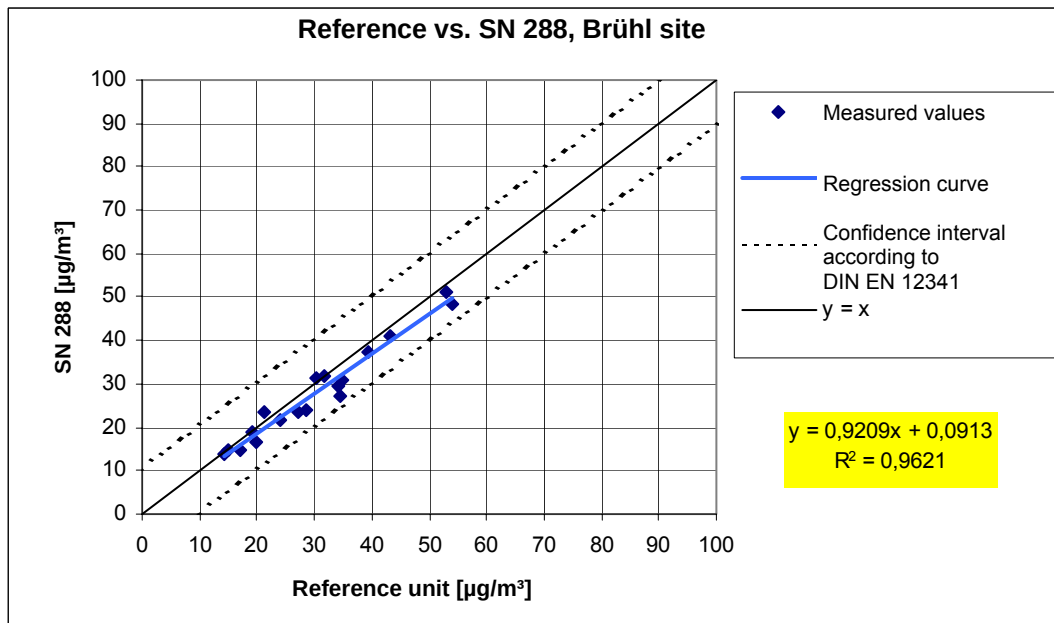


Fig. 37: Reference equivalence function for SN 288, Brühl site

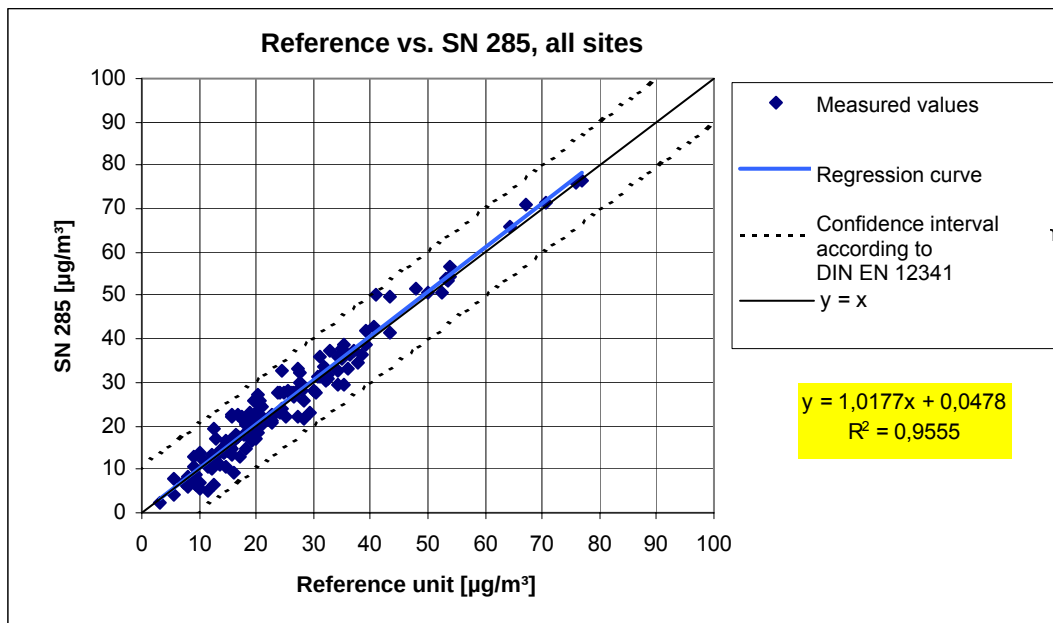


Fig. 38: Reference equivalence function for SN 285, all sites

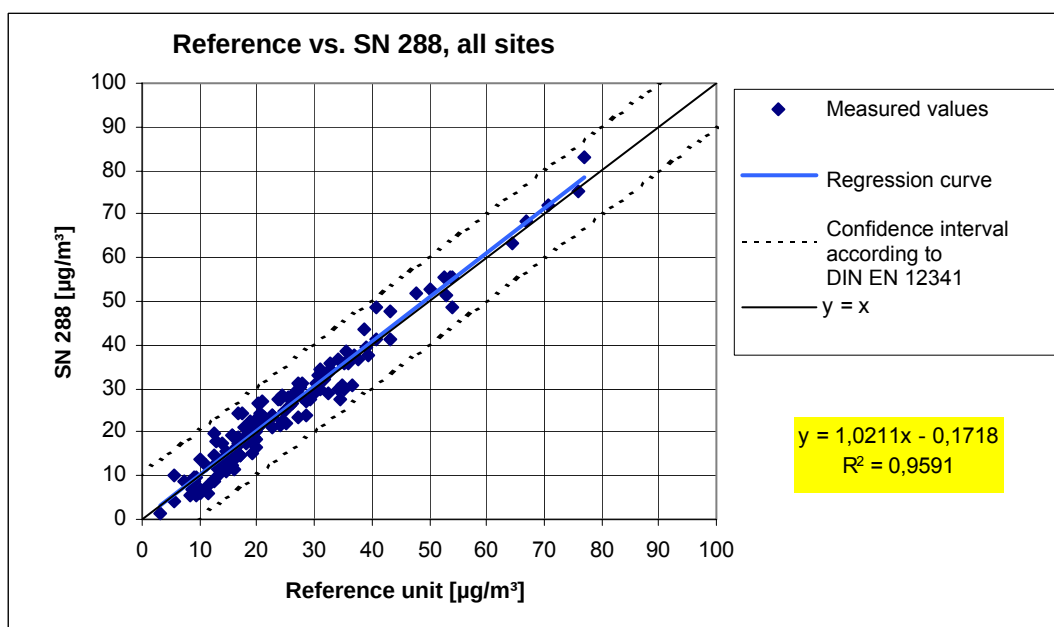


Fig. 39: Reference equivalence function for SN 288, all sites

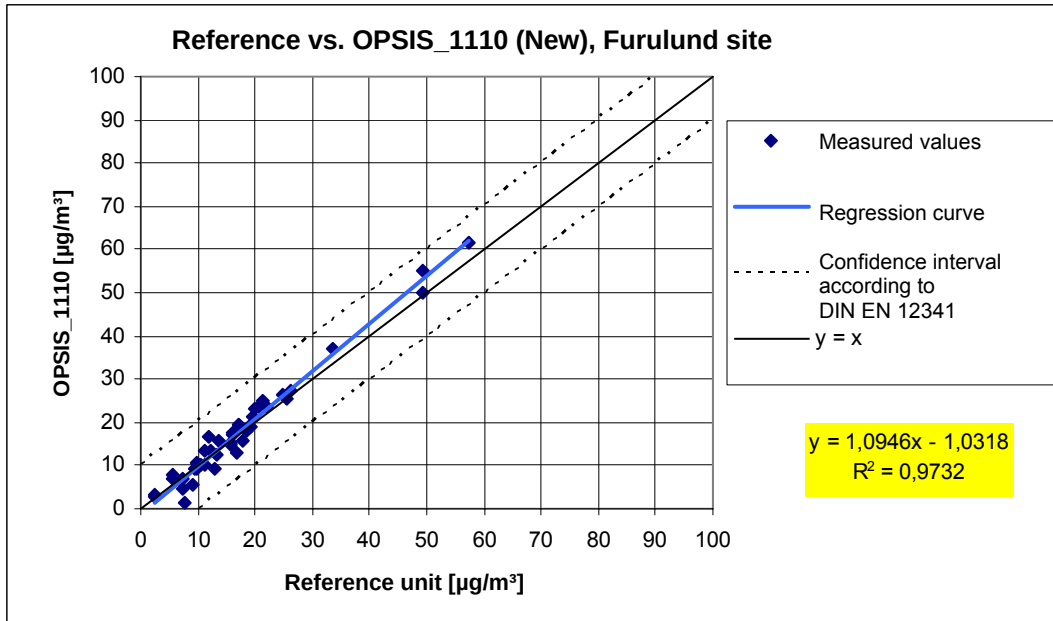


Fig. 40: Reference equivalence function for SN 1110, Furulund site

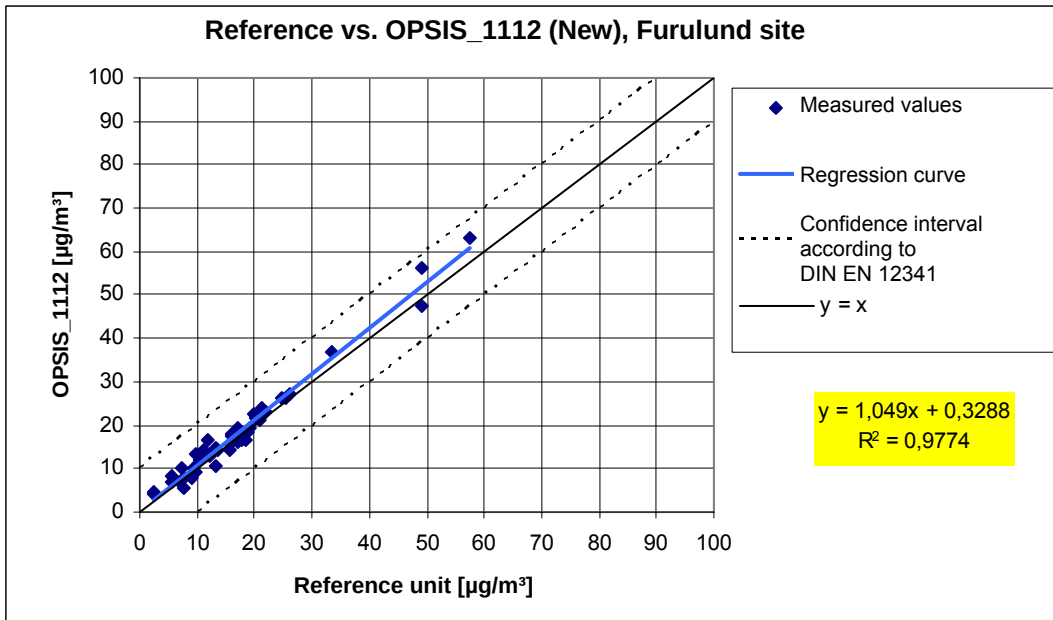


Fig. 41: Reference equivalence function for SN 1112, Furulund site

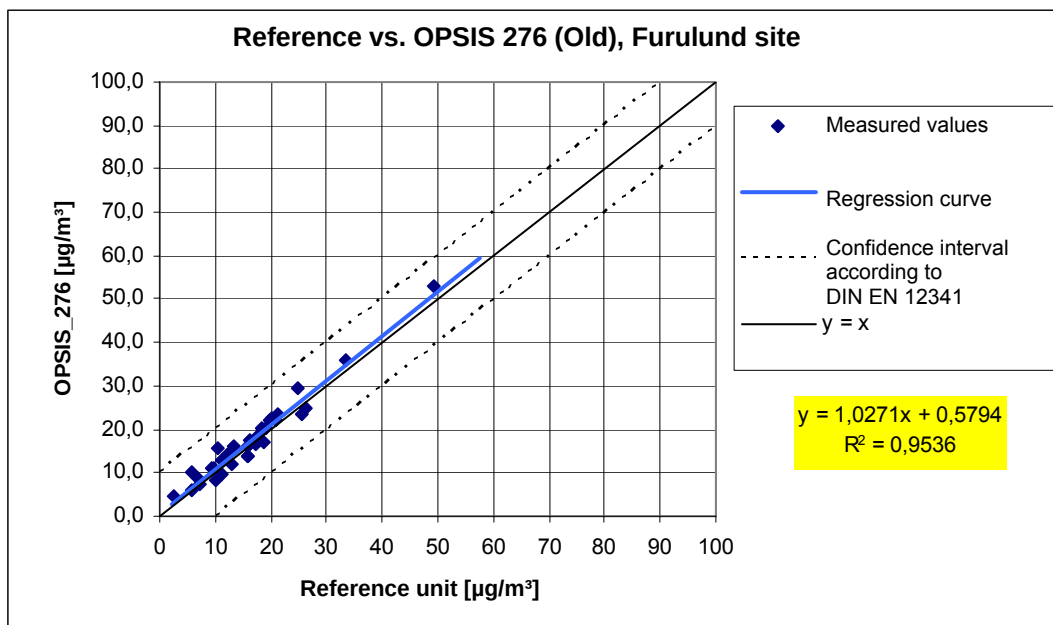


Fig. 42: Reference equivalence function for SN 276, Furulund site

T

6.1 5.3.2 Sampling system comparability

The PM10 sampling systems of two specimen instruments of identical design must be mutually comparable within the meaning of DIN EN 12341, Pt 5. Compliance with this requirement shall be demonstrated in the field tests

6.2 Equipment

Not necessary with this minimum requirement.

6.3 Implementation of the test

The investigation was conducted by field testing at multiple sites, taking into account different annual seasons and varying PM10 concentrations as well as different TSP/PM10 fraction ratios. At the Mechernich field test site, the readings of July 10 and July 13, 2002 and August 20, 2002 had to be discarded for implausibility due to intense rainfall on these days (> 20 mm in 2 hours).

The test was conducted in the same manner at the Furulund site. The comparability of the sampling systems was examined for both the two SM 200 (New) specimen instruments (SN 1110 and SN 1112) and for the SM 200 (Old) predecessor model (SN 276).

At least 15 value pairs were determined for each site.

6.4 Evaluation

The two-sided confidence interval CI_{95} calculated from the average concentration readings supplied by the instruments under test shall not exceed a value of $5 \mu\text{g}/\text{m}^3$ for average concentrations $\leq 100 \mu\text{g}/\text{m}^3$ and of 0,05 for average concentrations $> 100 \mu\text{g}/\text{m}^3$.

The comparability determination of the tested instruments was focused on the differences D_i between the concentration values Y_i returned by these instruments. Ideally, both instruments are identical and will therefore detect the same suspended particles fraction, meaning that $D_i = 0$. In analyzing the test data, the following procedure was adopted:

First, the concentration averages Y_i were calculated from the concentration readings obtained concurrently with the two tested instruments. These concentration averages Y_i were then split into two different data sets:

- Data set with $Y_i \leq 100 \mu\text{g}/\text{m}^3$ and the number of value pairs $n_{\leq}$
- Data set with $Y_i > 100 \mu\text{g}/\text{m}^3$ and the number of value pairs $n_{>}$

Re a):

From the value pairs in the data set with $Y_i \leq 100 \mu\text{g}/\text{m}^3$ the absolute standard deviation s_a is calculated thus

$$s_a = \sqrt{(\sum D_i^2 / 2n_{\leq})} ,$$

using the Student factor $t_{f_{\leq}; 0.975}$, defined as the 0.975 quantile of the two-sided confidence interval of the Student's t-distribution with $f_{\leq} = n_{\leq} - 2$ degrees of freedom.

The two-sided 95% confidence interval Cl_{95} for concentration averages $\leq 100 \mu\text{g}/\text{m}^3$ thus amounts to:

$$Cl_{95} = s_a \cdot t_{f_{\leq}; 0,975}$$

Re b):

From the value pairs in the data set with $Y_i > 100 \mu\text{g}/\text{m}^3$ the absolute standard deviation s_r is calculated thus

$$s_r = \sqrt{\left(\sum (D_i / Y_i)^2 / 2n_{>} \right)},$$

again using the Student factor $t_{f_{>}; 0,975}$, defined as the 0.975 quantile of the two-sided confidence interval of the Student's t-distribution with $f_{>} = n_{>} - 2$ degrees of freedom.

The two-sided confidence interval Cl_{95} for concentration averages $> 100 \mu\text{g}/\text{m}^3$ thus amounts to:

$$Cl_{95} = s_r \cdot t_{f_{>}; 0,975}$$

No concentration values $> 100 \mu\text{g}/\text{m}^3$ were measured during the field tests to date. The calculation under b) must therefore be considered not applicable.

6.5 Findings

The following observation has been confirmed for all sites examined to date:

The two-sided confidence interval Cl_{95} , at a maximum of $3.96 \mu\text{g}/\text{m}^3$, remains below the specified level of $5 \mu\text{g}/\text{m}^3$. In the supplementary tests, too, the two-sided confidence intervals Cl_{95} remained below the specified $5 \mu\text{g}/\text{m}^3$ threshold, ranging between a minimum of $2.27 \mu\text{g}/\text{m}^3$ and a maximum of $3.29 \mu\text{g}/\text{m}^3$ for all possible instrument combinations.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 46 and Table 47 show the values calculated for the standard deviation s_a and the two-sided confidence interval Cl_{95} . A graphic rendering is presented in Fig. 43 through to Fig. 50. These diagrams plot the fitted lines for the two tested instruments (determined by linear regression analysis), the curve $y = x$ representing the "ideal" case, and the two-sided acceptance range. An overview of all individual values obtained with the tested instruments and with the reference devices, arranged by test site, is given in the enclosed Appendices 9 and 14, respectively.

Table 46: *Two-sided 95% confidence interval CI_{95} for the specimen instruments
SN 285 and SN 288*

Instruments under test	Site	Number of values	Standard deviation s_a	Student factor t_f	Confidence interval CI_{95}
SN			$\mu\text{g}/\text{m}^3$		$\mu\text{g}/\text{m}^3$
285 / 288	Cologne parking lot	39	1.96	2.026	3.96
	Wesseling	62	1.46	2.000	2.91
	Mechernich	114	1.28	1.981	2.54
	Brühl	45	1.38	2.017	2.78
	TOTAL	260	1.46	1.969	2.87

Table 47: *Two-sided 95% confidence interval CI_{95} for the specimen instruments
tested at Furulund*

Instruments under test	Site	Number of values	Standard deviation s_a	Student factor t_f	Confidence interval CI_{95}
SN	Furulund		$\mu\text{g}/\text{m}^3$		$\mu\text{g}/\text{m}^3$
1110 / 1112		104	1.14	1.983	2.27
1110 / 276		80	1.65	1.991	3.29
1112 / 276		80	1.41	1.991	2.81

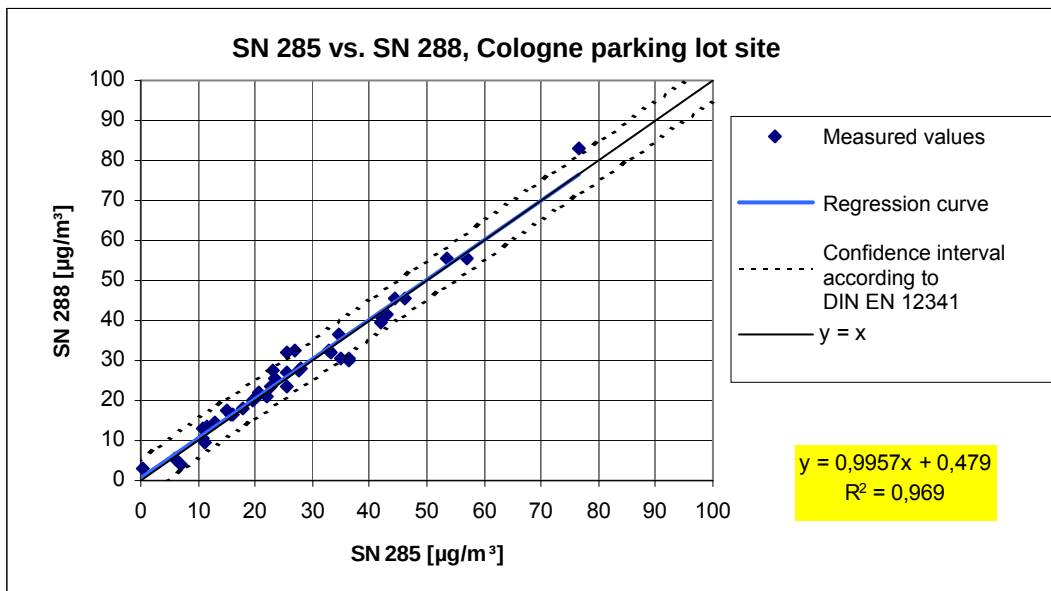


Fig. 43: Result of parallel measurements conducted with specimen instruments SN 285 / SN 288, Cologne parking lot site

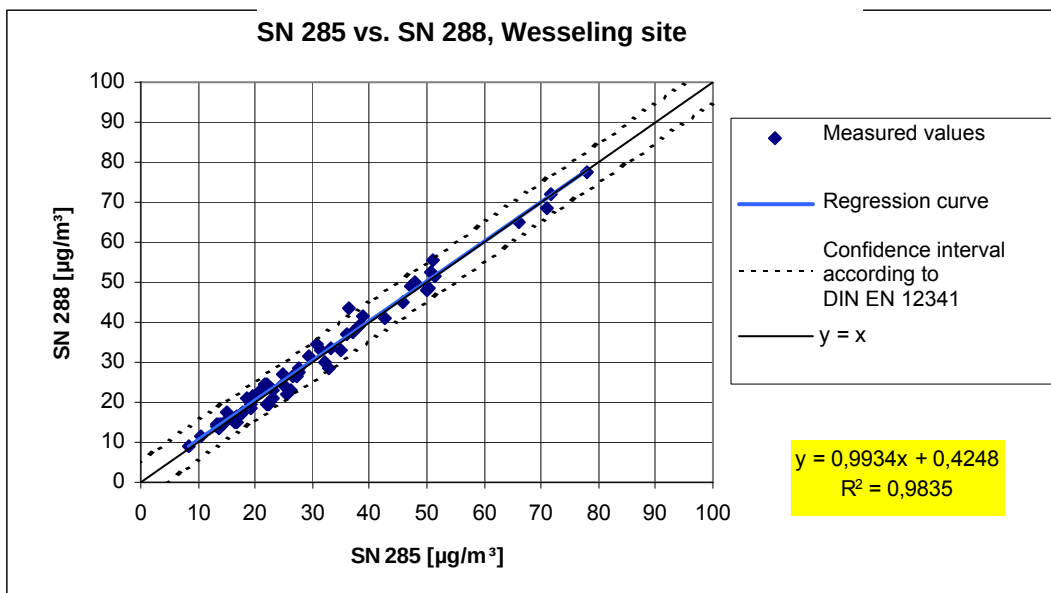


Fig. 44: Result of parallel measurements conducted with specimen instruments SN 285 / SN 288, Wesseling site

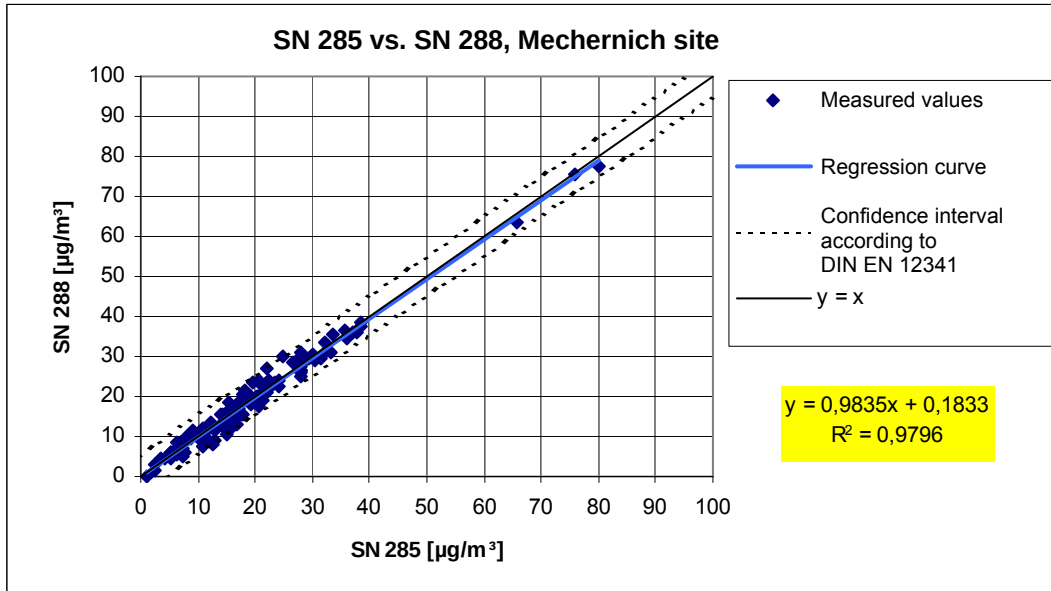


Fig. 45: Result of parallel measurements conducted with specimen instruments SN 285 / SN 288, Mechernich site

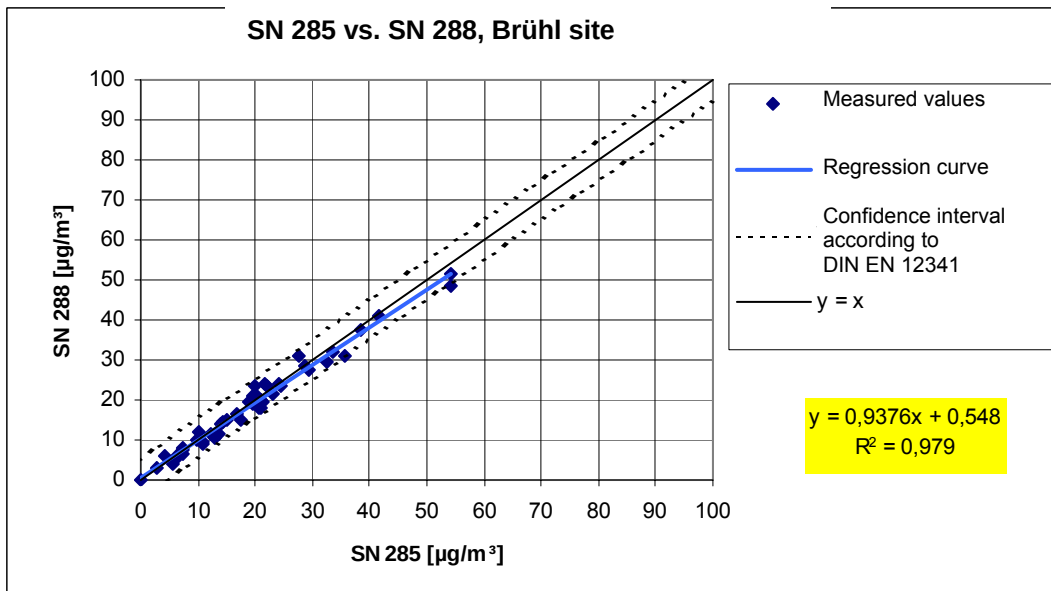


Fig. 46: Result of parallel measurements conducted with specimen instruments SN 285 / SN 288, Brühl site

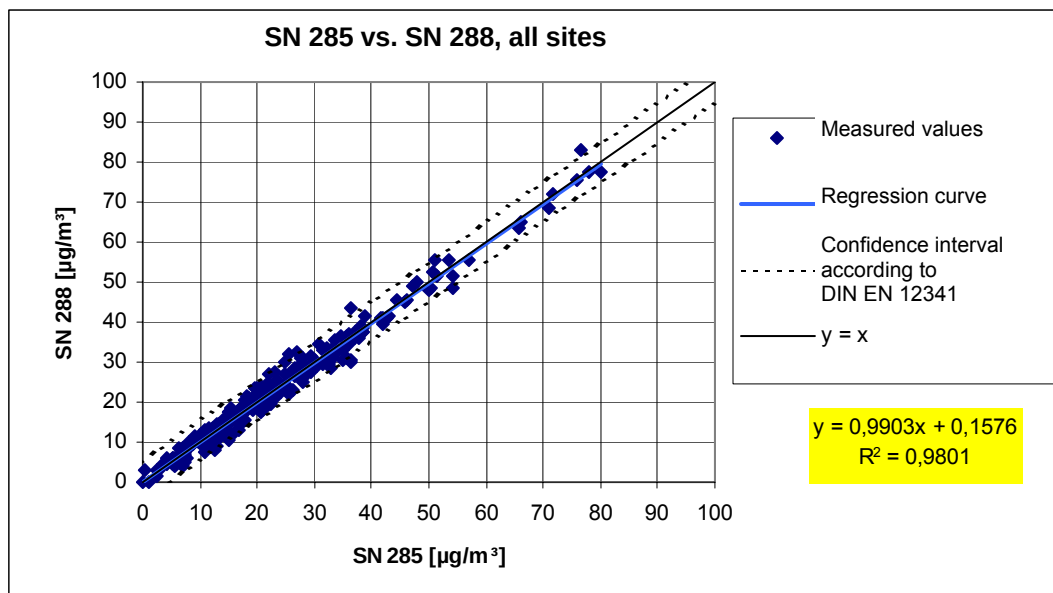


Fig. 47: Result of parallel measurements conducted with specimen instruments SN 285 / SN 288, all sites

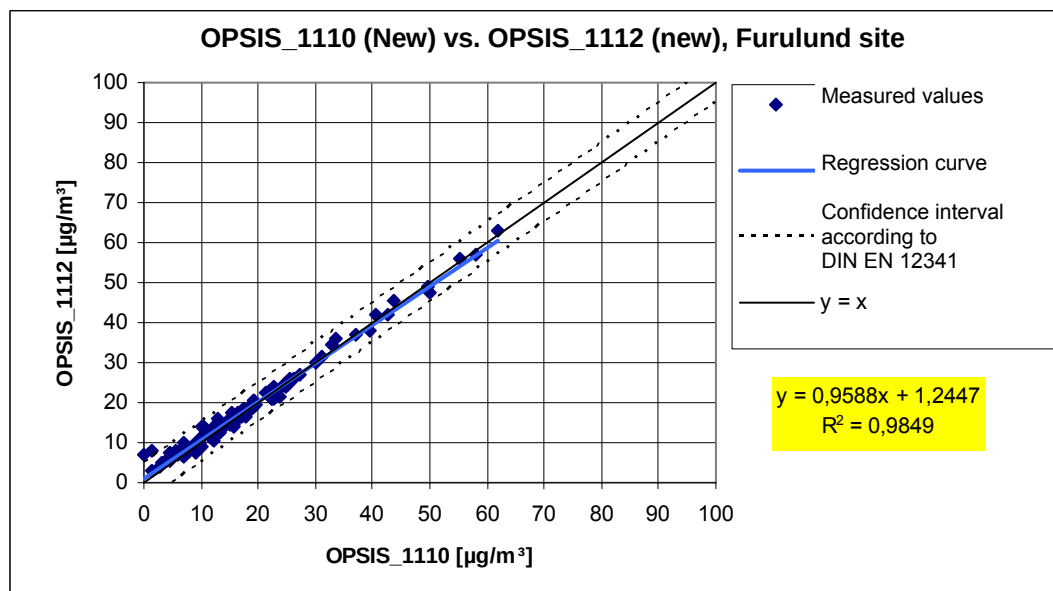


Fig. 48: Result of parallel measurements conducted with specimen instruments SN 1110 / SN 1112, Furulund site

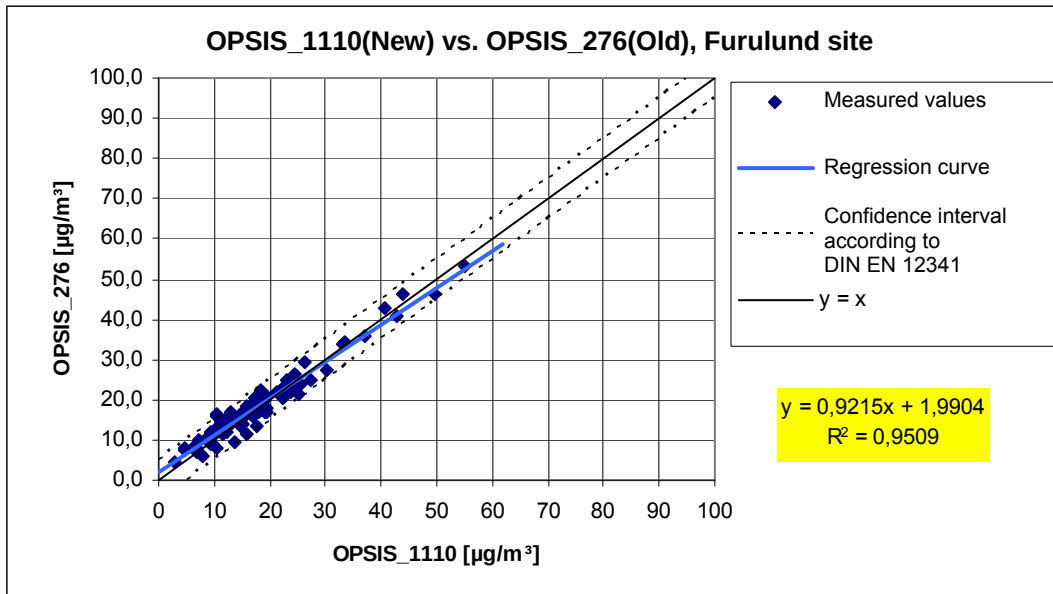


Fig. 49: Result of parallel measurements conducted with specimen instruments SN 1110 / SN 276, Furulund site

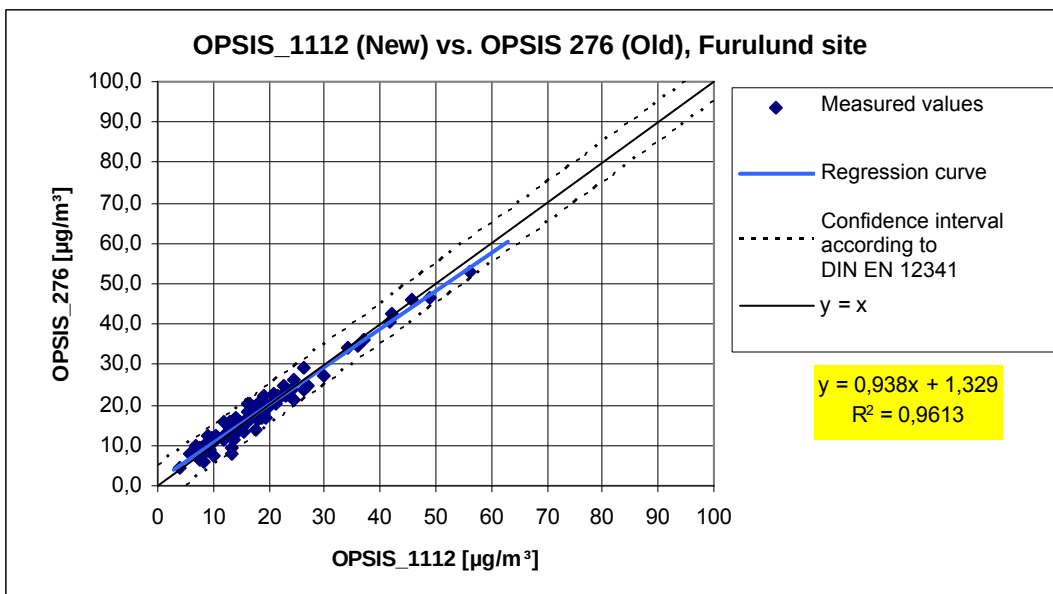


Fig. 50: Result of parallel measurements conducted with the specimen instruments SN 1112/ SN 276, Furulund site

6.1 5.3.3 Calibration

The PM10 specimens shall be calibrated in the field test via comparative measurements using a reference method consistent with DIN EN 12341, Part 5. As part of this process, the relationship between the measurement signal and the gravimetrically determined reference concentration shall be established as a continuous function.

For results of these tests refer to module 5.2.3.

6.2 Equipment

Refer to module 5.2.3.

6.3 Implementation of the test

Refer to module 5.2.3.

6.4 Evaluation

Refer to module 5.2.3.

6.5 Findings

Refer to module 5.2.3.

Minimum requirement fulfilled? -

6.6 Comprehensive presentation of the test result

Refer to module 5.2.3.

6.1 5.3.4 Cross-sensitivity

The interference caused by moisture in the measuring instrument shall, in the B_1 range, not exceed 10% of B_1 . If the sampling tube is heated, comparability with the gravimetric reference method shall be demonstrated for the stated temperature.

6.2 Equipment

Particle-free air, Hovacal steam generator.

6.3 Implementation of the test

To determine the interfering impact of air humidity contained in the measured medium, the instruments under test were supplied with particle-free air continuously loaded with a defined amount of water. A relative humidity >80 was thus established in the medium. The entire instrument system was operated in a climatic chamber at a temperature of 20°C . The test was repeated three times.

In addition, the reference equivalence functions for both specimen instruments were determined for those field testing days on which the relative humidity had exceeded 80%

6.4 Evaluation

Concentration readings were recorded from the instrument's display and calculated from the beta radiation intensity before and after the sampling cycle ($F_{\text{blank}}, F_{\text{collect}}$, stored in the instrument's memory as "blank" [cpm] and "collect" [cpm]). In both cases, the deviation from the target value was computed.

Reference value: VDI: $B_1 = 40 \mu\text{g}/\text{m}^3$

10 % of $B_1 = 4 \mu\text{g}/\text{m}^3$

In addition, it was examined whether the R^2 variation coefficient was also $\geq 0,95$ for measured values obtained on days with a relative humidity $> 80 \%$.

6.5 Findings

Moisture in the sampling medium was not observed to cause any interference with the zero point signal by more than $> 0.2 \mu\text{g}/\text{m}^3$ deviation from the target value (readings supplied by the instrument's display). Interference of sampling air moisture with the instrument's sensitivity could not be examined. However, changing relative air humidities recorded during the field tests were not found to have any impact on the measured results. The variation coefficient R^2 is $\geq 0,95$ for days with a relative moisture above 80% according to both reference equivalence functions.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

A summary is given in table 48 and Table 49 below.

Table 48: Zero point dependence on relative air humidity, deviation (in $\mu\text{g}/\text{m}^3$) from target value, average of three measurements - display readings

Relative air humidity	Deviation	
	Instrument 1 (SN 285)	Instrument 2 (SN 288)
%	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
> 80	0	0.2

Table 49: Zero point dependence on relative air humidity, deviation (in $\mu\text{g}/\text{m}^3$) from target value, average of three measurements - calculated measurements

Relative air humidity	Deviation	
	Instrument 1 (SN 285)	Instrument 2 (SN 288)
%	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
> 80	0	0.11

Individual values are listed in the enclosed Appendix 6.

Graphic diagrams of the measurements taken on days with > 80 % relative air humidity are presented in Fig. 10 and Fig. 11. Individual values are indicated in the enclosed Appendices 7 and 8.

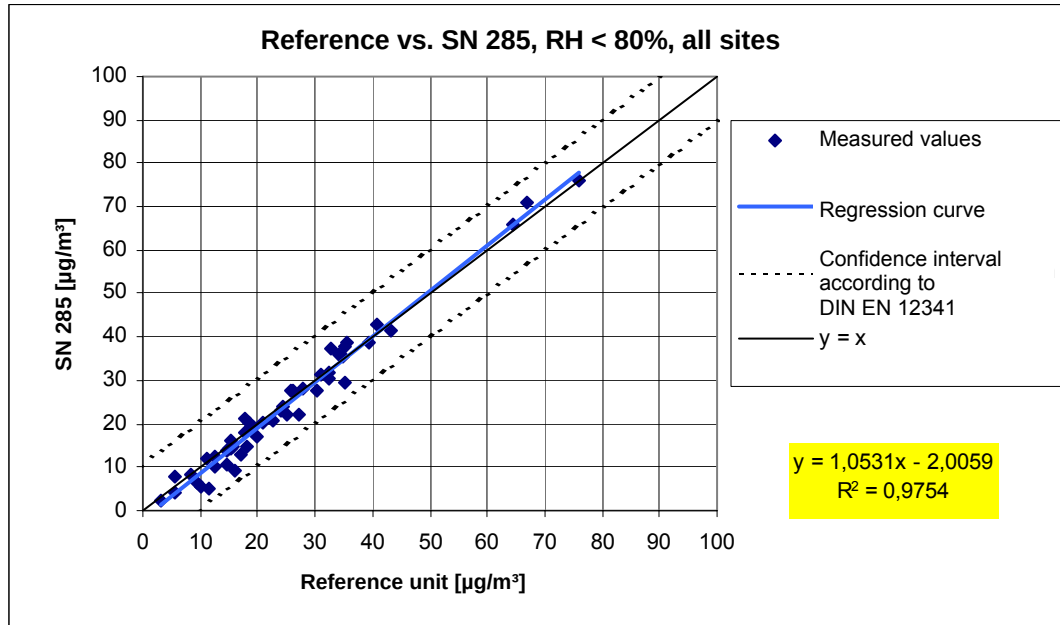


Fig. 51: Reference equivalence function for SN 285, > 80% air humidity, all sites

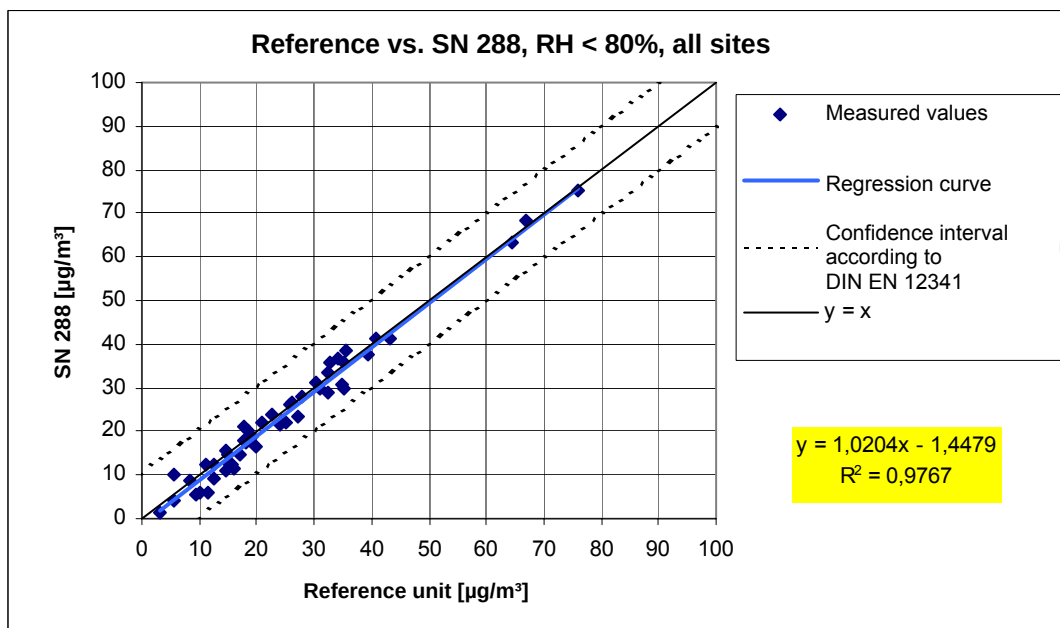


Fig. 52: Reference equivalence function for SN 288, > 80 % air humidity, all sites

6.1 5.3.5 Daily averages

The measuring system must permit the formation of 24-hour averages. Where filter changes are necessary, the total time needed for such time shall not exceed 1% of the averaging period.

6.2 Equipment

An additional clock was used for this test.

6.3 Implementation of the test

The instrument's capability to form daily averages was verified, and the time required for a filter change was measured.

6.4 Evaluation

The measuring instrument permits the formation of averages over sampling times ranging from 8 hours to 100 days. The filter change / filter transport within the system takes about 1 minute to complete, equivalent to approx. 0.07% idle time with a 24-hour sampling time setting.

The activation of a "pneumatic test" as described in para. 5.6.7 of the Operating Manual increases the total idle time of measurements for which this test is performed to about 3 – 4 minutes, corresponding to 0.28% idle time over a 24-hour sampling period

6.5 Findings

The formation of daily averages is possible. The maximum filter change time, with a 24-hour sampling interval and one "pneumatic test" per day, is about 0,28% of the sampling interval. At a maximum 0.28%, the filter change cycle is much shorter than the required 1% of the sampling time.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Not required here.

6.1 5.3.6 Constancy of the sampling flow rate

The sampling flow rate averaged over the sampling time must be constant to within $\pm 3\%$ of the target value. No transient value of the sampling flow rate shall exceed a fluctuation range of $\pm 5\%$ from the target value during the sampling time.

6.2 Equipment

An additional flow rate meter per para. 4.3. and some sampled filters were used for this test.

6.3 Implementation of the test

To determine the constancy of the sampling flow rate, a flow meter was connected to the measuring instruments during the field tests. For a period of 3 days, 3-minute flow rate values were recorded.

6.4 Evaluation

The average, standard deviation, maximum and minimum values of the flow rate readings were calculated.

6.5 Findings

From the diagrams of the flow rates recorded on the three measurement days it is evident that no measured value deviated from the 16.67 l/min target value by more than $\pm 5\%$. The deviation of the daily averages was likewise smaller than the specified $\pm 3\%$ of the target.

All daily averages determined were less than $\pm 3\%$ of the target value; no transient value deviated from the target by more than $\pm 5\%$.

The pump is automatically de-energized when the pressure drop across the filter exceeds 70 kPa. In the field tests the pressure loss was found to be in the region of 0.5 kPa per milligram of dust deposited (depending on dust properties and moisture). Assuming an average pressure loss of about 6 kPa across the filter, a mass of 128 mg would have to be collected theoretically before the pump is tripped. With the instrument set to a 24-hour sampling cycle, this would correspond to a concentration of $> 5000 \mu\text{g}/\text{m}^3$. Concentrations of this magnitude do not occur in practice.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 50 and Table 51 list the flow rate parameters determined. Fig. 53 and Fig. 54 show diagrams of the flow rate measurements conducted on the two specimen instruments SN 285 and SN 288

Table 50: *Characteristic parameters for the flow rate measurement - SN 285*

Parameter	Unit	12-07-2002	13-07-2002	14-07-2002
Average	l/min	16.83	16.93	16.86
Deviation of average	% of target value	0.94	1.56	1.15
Standard deviation	l/min	0.19	0.05	0.16
Maximum	l/min	17.13	17.17	17.01
Minimum	l/min	16.31	16.74	16.46

Table 51: *Characteristic parameters for the flow rate measurement - SN 288*

Parameter	Unit	07-06-2002	08-06-2002	09-06-2002
Average	l/min	16.93	16.90	16.90
Deviation of average	% of target value	1.59	1.35	1.37
Standard deviation	l/min	0.17	0.26	0.28
Maximum	l/min	17.19	17.30	17.30
Minimum	l/min	16.28	16.21	16.17

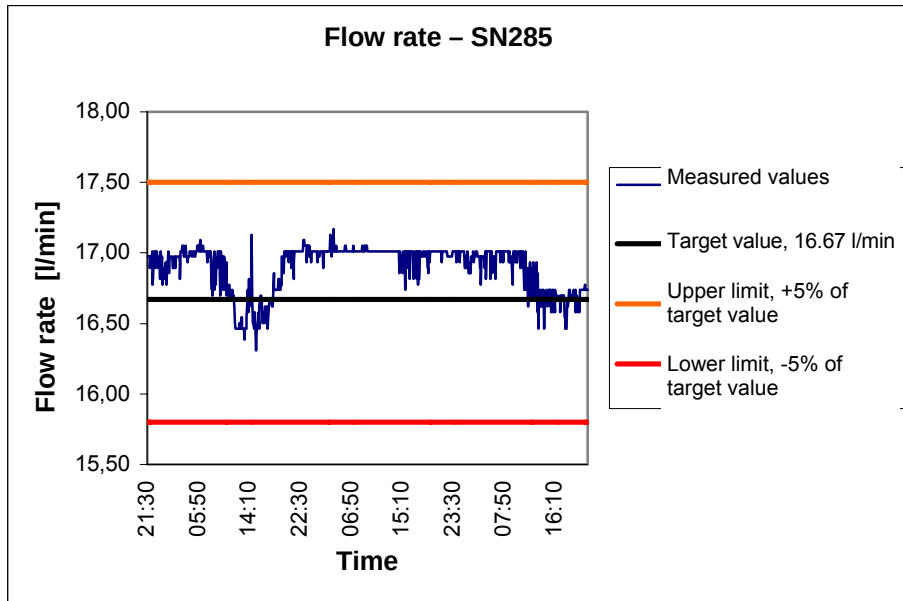


Fig. 53: Flow rate on specimen instrument SN 285, 12-07-2002 through 14-07-2002

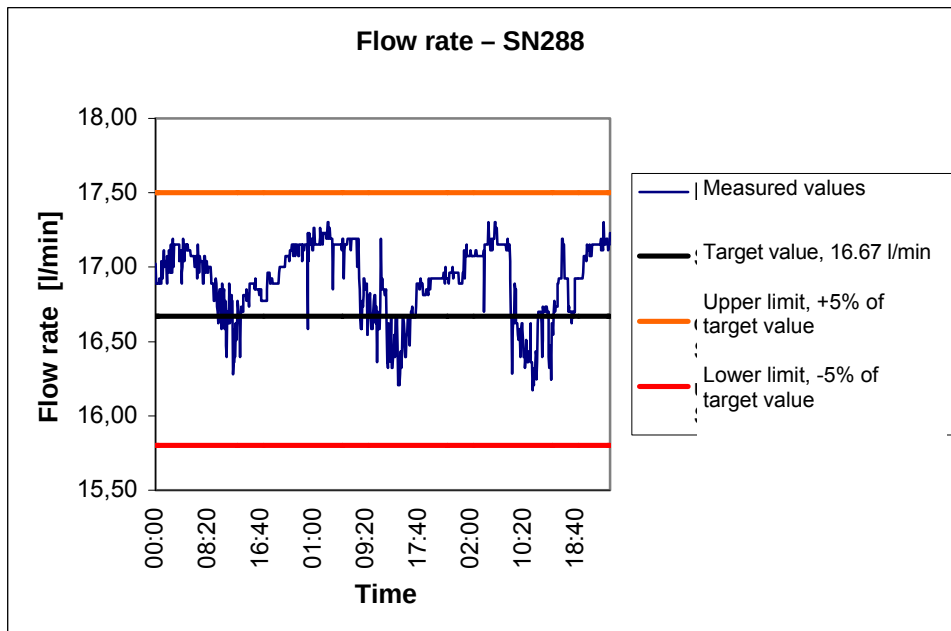


Fig. 54: Flow rate on specimen instrument SN 288 from 07-06-2002 through 09-06-2002

6.1 5.3.7 Tightness of the sampling system

The entire sampling system shall be tested for tightness. Any leaks in the circuit must not account for more than 1% of the sampling volume throughput

6.2 Equipment

Software (Long), rubber plugs, clock.

6.3 Implementation of the test

Testing the tightness of the sampling system carries a risk of causing lasting damage to the measuring system through excessive underpressures. The leakage rate was determined using a special test procedure based on the equipment manufacturer's instructions. With the aid of the "Long" software it is possible to deactivate the alarm messages and automatic flow rate control functions so that a controlled increase in negative pressure can be produced in the instrument with the pump running and the intake orifice of the sampling pipe closed off (e.g., with a rubber plug). The pump is de-energized when an underpressure of about 80 kPa is reached. The pressure loss over time can then be determined. The estimated overall system volume was 0.6 liters in this case.

This procedure was repeated three times.

On the OPSIS SM 200 (New) instrument, the tightness of the system can be checked by manually selecting the Leak Test valve while the pump is in operation. The pressure loss upon deactivation of the pump can be determined using the analog sensor. The estimated overall system volume is 0.3 liters in this case because the intake tube is not taken into account. This procedure was checked for its operability as part of the supplementary tests.

The procedure was repeated three times.

6.4 Evaluation

The leakage rate $\dot{V}_L$ is to be determined by means of the following equation:

$$\dot{V}_L = \frac{\Delta p}{p_0} \frac{V_{ges}}{\Delta t}$$

Where

Δp : pressure difference observed over the time difference Δt

p_0 : pressure at time t_0

V_{ges} : estimated overall system volume (dead volume), i.e., 0,6 l and 0,3 l, respectively, in this case

Δt : time difference required to complete a pressure difference Δp

The maximum value of the three leakage rates was determined.

6.5 Findings

The maximum leakage rates were 0.79 % for instrument 1 (SN 285) and 0.82 % for instrument 2. The OPSIS SM 200 (New) system was found to have a maximum leakage rate of 0.98 % in the supplementary tests.

Minimum requirement fulfilled? yes

6.6 Comprehensive presentation of the test result

Table 52 lists the results of the leakage tests.

Table 52: *Determination of the leakage rate*

SN 285							
No.	Date	p_0 [KPa]	Δp [KPa]	Δt [min]	V_{ges} [m ³]	Leakage rate [l/min]	% of target
1	10-01-03	84.1	18.5	1	0.0006	0.132	0.79
2	10-01-03	81.7	13.7	1	0.0006	0.101	0.60
3	10-01-03	79.9	15.3	1	0.0006	0.115	0.69
SN 288							
No.	Date	p_0 [KPa]	Δp [KPa]	Δt [min]	V_{ges} [m ³]	Leakage rate [l/min]	% of target
1	10-01-03	80.0	15.6	1	0.0006	0.117	0.70
2	10-01-03	78.6	17.8	1	0.0006	0.136	0.82
3	10-01-03	82.3	16.2	1	0.0006	0.118	0.71

Table 53: *Determination of the leakage rate, OPSIS SM 200 (New)*

No.	Date	p_0 [KPa]	Δp [KPa]	Δt [min]	V_{ges} [m ³]	Leakage rate [l/min]	% of target
1	21-04-04	92.64	47.87	1	0.0003	0.155	0.93
2	21-04-04	92.54	50.28	1	0.0003	0.163	0.98
3	21-04-04	92.52	49.84	1	0.0003	0.162	0.97

6.1 5.4 Requirements on multi-constituent measuring systems

*Multi-constituent measuring systems must meet the requirement for each individual constituents, even when all measuring channels are in use simultaneously.
In a sequential operating mode, the formation of hourly average values must be ensured.*

6.2 Equipment

Not applicable.

6.3 Implementation of the test

Not applicable.

6.4 Evaluation

Not applicable.

6.5 Findings

Not applicable.

Minimum requirement fulfilled? -

6.6 Comprehensive presentation of the test result

Not applicable.

7 Recommendations on practical use

7.1 Work in the maintenance interval

The instrument under test requires the following periodic steps: :

- Regular visual inspections / telemetric monitoring
- - Instrument status o.k.
- - No error messages
- - Internal quality assurance tests o.k.
- - No accumulation of dirt
- Verification of instrument functions per manufacturer's instructions
- Monthly cleaning/degreasing of the impaction plates
To prevent excessive disruptions of the sampling process by sampling head maintenance (degreasing of the impaction plates), it is recommended to replace the entire sampling head locally with another one prepared in the laboratory.

Apart from the foregoing, the manufacturer's instructions shall be observed.

7.2 Functional test and calibration

The following functional testing and pre-calibration procedure is recommended:

- Visual inspection of the instrument and its sampling system
- Inspection for leaks
- Flow rate check
- Zero point check as indicated
- Reference point check using the reference films
- Testing of the data transmission (analog and status signals) to the evaluation system

Further details concerning functional testing and calibration are given in the Operating Instructions.

Air Quality Management



Karsten Pletscher



Dr. Peter Wilbring

Cologne, 5 July 2005
936/21201592/A

8 Literature

- [1] VDI Guideline 4202, Sheet 1, "Minimum requirements for suitability tests of automated ambient air quality measuring systems – Point-related measurement methods of gaseous and particulate pollutants, June 2002
- [2] VDI Guideline 4203, Sheet 3, "Testing of automatic measuring systems – Test procedures for point-related ambient air measuring systems of gaseous and particulate pollutants", June 2002
- [3] European standard EN 12341, „Air quality – Determination of the PM10 fraction of suspended particulate matter – Reference method and field test procedure to determine reference equivalence of measurement methods, German version EN 12341: 1998
- [4] VDI Guideline 2463, Sheet 7, „Particulate matter measurement; measurement of mass concentration in ambient air (immission); filter method; small filter device GS 050, 1982
- [5] Operating Manual for OPSIS SM200 Series, last revised 01/2000
- [6] Operating Manual for Low Volume Sampler LVS3, last revised 2000
- [7] Directive 1999/30/EC of the Council of 22 April 1999 relating to limit values for sulphur dioxide, nitrogen dioxide and oxides of nitrogen, particulate matter and lead in ambient air
- [8] Uniform Federal Immission Monitoring Practice:
Guidelines for the design and suitability testing of measuring systems for continuous air quality monitoring, Circular Communication by the Minister of the Interior dated August 19, 1981 - U II 8 – 556 134/4
- [9] Operating Manual for OPSIS SM200 (New) series, last revised 06/2005 (4th edition)
- [10] Report on the Suitability Tests of the OPSIS SM200 Dust Monitor with PM10 Pre-Separator marketed by OPSIS AB, Sweden, TÜV Report No 936/801013/A dated January 29, 2003.

9 Appendices

Annex 1 Measured and calculated values

- Appendix 1: Limits of detection
- Appendix 2: Calibration and analysis function
- Appendix 3: Temperature dependence of the zero point
- Appendix 4: Temperature dependence of the instrument's sensitivity
- Appendix 5: Mains voltage and mains frequency dependence
- Appendix 6: Cross-sensitivity of the zero points
- Appendix 7: Measurements from the field test sites
- Appendix 8: Ambient conditions at the field test sites
- Appendix 9: Comparability of the tested instruments in the field tests
- Appendix 10: Startup screen of the SM 200 (Old) instrument
- Appendix 11: Calibration and analysis function, Furulund site
- Appendix 12: Measured values from the Furulund field test site
- Appendix 13: Ambient conditions at the Furulund field test site
- Appendix 14: Comparability of the tested instruments in the field tests, Furulund site
- Appendix 15: Startup screen of the SM 200 (New) instrument

Annex 2 Manuals

Appendix 1

Limit of Detection (Instrument Readings)

Sheet 1 of 2

Manufacturer	: OPSIS AB		
Measuring range	: 0 ... 1000 µg/m³	Climatic chamber temperature:	20°C
Instrument type	: SM 200	Climatic chamber relative air humidity:	60%
Serial Nos.	: SN 285 & SN 288	Zero gas:	Dust-free zero air

No.	Date	24-hr. value [µg/m³]	
		SN 285	SN 288
1	28.11.00	0	1,4
2	29.11.00	0	0
3	30.11.00	0	0
4	01.12.00	0	0
5	02.12.00	0	0
6	03.12.00	0	0
7	04.12.00	0	0
8	05.12.00	0	0
9	06.12.00	0	0
10	07.12.00	0	0
11	08.12.00	0,2	0
12	09.12.00	0	0
13	10.12.00	0	0
14	11.12.00	0,5	0
15	12.12.00	0	0
	Number of values	15	15
	Mean	0,05	0,09
	Standard deviation s _{x0}	0,14	0,36
	Limit of detection x	0,41	1,08

$$s_{x0} = \sqrt{\left(\frac{1}{n-1}\right) \cdot \sum_{i=1,n} (x_{0i} - \bar{x}_0)^2}$$

NWG = 3 * s_{x0}

Appendix 1

Limit of Detection (Instrument Readings)

Sheet 2 of 2

Manufacturer : OPSIS AB Measuring range : 0 ... 1000 µg/m³ Instrument type : SM 200 Serial Nos. : SN 285 & SN 288				Climatic chamber temperature: 20°C Climatic chamber relative air humidity: 60% Zero gas: Dust-free zero air
No.	Date	24-hr. reading [µg/m³]		$s_{x0} = \sqrt{\left(\frac{1}{n-1}\right) \cdot \sum_{i=1,n} (x_{0i} - \bar{x}_0)^2}$
		SN 285	SN 288	
1	28.11.00	0,04	1,44	
2	29.11.00	0,03	0,04	
3	30.11.00	-0,12	-0,09	
4	01.12.00	-0,04	-0,22	
5	02.12.00	-0,88	0,03	
6	03.12.00	0,05	0,01	
7	04.12.00	-0,15	-0,42	
8	05.12.00	-0,33	0,03	
9	06.12.00	0,01	0,00	
10	07.12.00	-0,05	-0,06	
11	08.12.00	0,04	0,01	
12	09.12.00	0,22	-0,51	
13	10.12.00	-0,54	0,03	
14	11.12.00	0,03	-0,14	
15	12.12.00	0,51	0,00	
Number of values		15	15	
Mean		-0,08	0,01	
Standard deviation s_{x0} (Δcpm)		0,32	0,43	
Limit of detection x (Δcpm)		0,96	1,29	
				NWG = 3 * s_{x0}

Appendix 2

Calibration and Analysis Function

Sheet 1 of 1

Manufacturer : OPSIS AB Measuring range : 0 ... 1000 µg/m³ Instrument type : SM 200 Serial Nos. : SN 285 & SN 288					
Measured variable: Suspended particulates, PM 10, outdoor ambient air Site : Cologne parking lot					
No.	Date	24-h reading [µg/m³]		Ref. method readings [µg/m³]	Calibration & analysis function
		SN 285	SN 288		
1	15.12.00	13,1	14,7	17,0	Calibration function $y = m \cdot x + b$ SN 285 $y = 1.0374 \cdot x - 1.8928$ $R^2 = 0.9779$ SN 288 $y = 1.0829 \cdot x - 2.697$ $R^2 = 0.9862$ Analysis function $x = 1/m \cdot y - b/m$ SN 285 $x = 0.9639 \cdot y + 1.8246$ SN 288 $x = 0.9234 \cdot y + 2.4905$
2	21.12.00	22	21,2	22,8	
3	23.12.00	42	39,3	39,1	
4	28.12.00	20,5	22,1	20,9	
5	29.12.00	25,6	23,7	20,7	
6	04.01.01	5,8	5,6	8,2	
7	08.01.01	22,8	23,7	22,7	
8	09.01.01	23	27,6	29,3	
9	10.01.01	17,9	17,9	17,7	
10	11.01.01	27,9	27,9	28,0	
11	12.01.01	25,6	27	28,4	
12	13.01.01	34,6	36,5	37,6	
13	16.01.01	53,6	55,6	53,5	
14	17.01.01	76,4	82,9	76,9	
15	18.01.01	56,9	55,7	53,9	
16	22.01.01	14,9	17,4	18,1	
17	28.01.01	36,3	30,6	36,4	

Appendix 3 Temperature dependence of the zero point (instrument readings)

Sheet 1 of 2

Manufacturer	OPSIS AB		Climatic chamber	Temperature program
Measuring range	0 to 1000	µg/m³	Climatic chamber relative air humidity	60%
Instrument type	SM 200		Zero gas	zero air without particulates
Serial No.	SN 285		Reference value	B ₂ = 2 µg/m³
Absolute deviations from the starting point were determined at 20 °C.				
No.	Date	Temperature [°C]	24-h readings [µg/m³] SN 285	Deviation [µg/m³]
1	03.02.01	20	0	-
2	04.02.01	10	0	0
3	05.02.01	5	0	0
4	06.02.01	10	0	0
5	07.02.01	20	0	-
6	08.02.01	30	0	0
7	09.02.01	40	0	0
8	10.02.01	30	0	0
9	11.02.01	20	0	-
10	12.02.01	20	0	-
11	13.02.01	10	0	0
12	14.02.01	5	0	0
13	15.02.01	10	0	0
14	16.02.01	20	0	-
15	17.02.01	30	0	0
16	18.02.01	40	0	0
17	19.02.01	30	0	0
18	20.02.01	20	0	-
19	21.02.01	20	0	-
20	22.02.01	10	0	0
21	23.02.01	5	0	0
22	24.02.01	10	0	0
23	25.02.01	20	0	-
24	26.02.01	30	0	0
25	27.02.01	40	0	0
26	28.02.01	30	0	0
27	01.03.01	20	0	-

Appendix 3 Temperature dependence of the zero point (instrument readings)

Sheet 2 of 4

Manufacturer	OP SIS AB		Climatic chamber	Temperature program
Measuring range	0 to 1000	$\mu\text{g}/\text{m}^3$	Climatic chamber relative air humidity	60%
Instrument type	SM 200		Zero gas	Dust-free zero air
Serial No.	SN 288		Reference value	$B_2 = 2 \mu\text{g}/\text{m}^3$
Absolute deviations from the starting point were determined at 20 °C.				
No.	Date	Temperature [°C]	24-hr. reading [$\mu\text{g}/\text{m}^3$] SN 285	Deviation per VDI [$\mu\text{g}/\text{m}^3$] [c(T2)-c(T1)]
1	03.02.01	20	0	-
2	04.02.01	10	0	0
3	05.02.01	5	0	0
4	06.02.01	10	0	0
5	07.02.01	20	0	-
6	08.02.01	30	0	0
7	09.02.01	40	0	0
8	10.02.01	30	0	0
9	11.02.01	20	0	-
10	12.02.01	20	0	-
11	13.02.01	10	0	0
12	14.02.01	5	0	0
13	15.02.01	10	0	0
14	16.02.01	20	0	-
15	17.02.01	30	0	0
16	18.02.01	40	0	0
17	19.02.01	30	0	0
18	20.02.01	20	0	-
19	21.02.01	20	0	-
20	22.02.01	10	0	0
21	23.02.01	5	0	0
22	24.02.01	10	0	0
23	25.02.01	20	0	-
24	26.02.01	30	0	0
25	27.02.01	40	0	0
26	28.02.01	30	0	0
27	01.03.01	20	0	-

Appendix 3 Temperature dependence of the zero point (values calculated from cpm) Sheet 3 of 4

Manufacturer OPSIS AB				
Measuring range	0 to 1000	$\mu\text{g}/\text{m}^3$	Climatic chamber	Temperature program
Instrument type	SM 200		Climatic chamber relative air humidity	60%
Serial No.	SN 285		Zero gas	Dust-free zero air
			Reference value	$B_2 = 2 \mu\text{g}/\text{m}^3$
Absolute deviations from the starting point were determined at 20 °C.				
No.	Date	Temperature [°C]	24-hr. readings [$\mu\text{g}/\text{m}^3$] SN 285	Deviation [$\mu\text{g}/\text{m}^3$]
1	03.02.01	20	-0,08	-
2	04.02.01	10	0,04	0,12
3	05.02.01	5	-0,16	-0,08
4	06.02.01	10	0,02	0,10
5	07.02.01	20	-0,29	-
6	08.02.01	30	0,04	0,33
7	09.02.01	40	-0,42	-0,13
8	10.02.01	30	0,00	0,29
9	11.02.01	20	0,04	-
10	12.02.01	20	-0,04	-
11	13.02.01	10	-0,21	-0,17
12	14.02.01	5	0,02	0,06
13	15.02.01	10	-0,23	-0,19
14	16.02.01	20	0,02	-
15	17.02.01	30	-0,05	-0,07
16	18.02.01	40	-0,58	-0,60
17	19.02.01	30	-0,15	-0,17
18	20.02.01	20	0,00	-
19	21.02.01	20	0,04	-
20	22.02.01	10	-0,05	-0,09
21	23.02.01	5	-0,34	-0,37
22	24.02.01	10	-0,01	-0,05
23	25.02.01	20	0,01	-
24	26.02.01	30	-0,12	-0,13
25	27.02.01	40	-0,53	-0,54
26	28.02.01	30	0,04	0,03
27	01.03.01	20	0,05	-

Appendix 3 Temperature dependence of the zero point (values calculated from cpm) Sheet 4 of 4

Manufacturer	OPSIS AB		Climatic chamber	Temperature program
Measuring range	0 to 1000	$\mu\text{g}/\text{m}^3$	Climatic chamber relative air humidity	60%
Instrument type	SM 200		Zero gas	Dust-free zero air
Serial No..	SN 288		Reference value	$B_2 = 2 \mu\text{g}/\text{m}^3$
Absolute deviations from the starting point were determined at 20 °C.				
No.	Date	Temperature [°C]	24-h readings [$\mu\text{g}/\text{m}^3$] SN 285	Deviation per VDI [$\mu\text{g}/\text{m}^3$] [c(T2)-c(T1)]
1	03.02.01	20	-0,14	-
2	04.02.01	10	-0,13	0,01
3	05.02.01	5	-0,01	0,13
4	06.02.01	10	0,02	0,17
5	07.02.01	20	0,04	-
6	08.02.01	30	0,03	-0,01
7	09.02.01	40	-0,63	-0,67
8	10.02.01	30	0,01	-0,02
9	11.02.01	20	0,03	-
10	12.02.01	20	-0,30	-
11	13.02.01	10	-0,07	0,22
12	14.02.01	5	-0,08	0,21
13	15.02.01	10	0,04	0,33
14	16.02.01	20	0,00	-
15	17.02.01	30	-0,06	-0,06
16	18.02.01	40	-0,45	-0,45
17	19.02.01	30	-0,01	-0,01
18	20.02.01	20	0,03	-
19	21.02.01	20	-0,20	-
20	22.02.01	10	0,01	0,21
21	23.02.01	5	0,03	0,23
22	24.02.01	10	-0,14	0,06
23	25.02.01	20	0,00	-
24	26.02.01	30	-0,04	-0,04
25	27.02.01	40	-0,33	-0,33
26	28.02.01	30	0,00	0,00
27	01.03.01	20	0,03	-

Appendix 4

Temperature dependence of the instrument's sensitivity

Sheet 1 of 2

Manufacturer	OPSIS AB			Climatic chamber				Temperature program	
Measuring range	0 tp 1000	µg/m³	Climatic chamber relative air humidity				60%		
Instrument type	SM 200	Standards				Reference films			
Serial No.	SN 285								
Relative deviations from the starting point were determined at 20°C.									
No.	Temperature [°C]	Film 1 reading [mg]	Dev. [%]	Film 2 reading [mg]	Dev. [%]	Film 3 reading [mg]	Dev. [%]	Film 4 reading [mg]	Dev. [%]
1	20	28,702	-	39,481	-	59,061	-	78,857	-
2	10	28,692	-0,04	39,556	0,19	59,196	0,23	78,984	0,16
3	5	28,683	-0,07	39,584	0,26	59,287	0,38	79,053	0,25
4	10	28,443	-0,90	39,482	0,00	59,074	0,02	78,867	0,01
5	20	28,219	-	39,244	-	58,884	-	78,759	-
6	30	28,077	-0,50	39,194	-0,13	58,655	-0,39	78,601	-0,20
7	40	28,018	-0,71	39,110	-0,34	58,586	-0,51	78,433	-0,41
8	30	28,307	0,31	39,472	0,58	58,849	-0,06	78,844	0,11
9	20	28,457	-	39,528	-	58,910	-	78,948	-
10	20	28,322	-	39,449	-	58,892	-	78,862	-
11	10	28,205	-0,41	39,303	-0,37	58,766	-0,22	78,771	-0,12
12	5	28,143	-0,63	39,307	-0,36	58,850	-0,07	78,824	-0,05
13	10	28,079	-0,86	39,204	-0,62	58,708	-0,31	78,717	-0,18
14	20	28,183	-	39,341	-	58,778	-	78,770	-
15	30	28,065	-0,42	39,166	-0,45	58,586	-0,33	78,562	-0,26
16	40	27,887	-1,05	39,060	-0,71	58,381	-0,68	78,367	-0,51
17	30	28,173	-0,04	39,362	0,05	58,814	0,06	78,708	-0,08
18	20	28,332	-	39,543	-	58,963	-	78,918	-
19	20	28,269	-	39,430	-	58,864	-	78,857	-
20	10	28,267	-0,01	39,414	-0,04	58,886	0,04	78,836	-0,03
21	5	28,181	-0,31	39,351	-0,20	58,848	-0,03	78,805	-0,07
22	10	28,028	-0,85	39,204	-0,57	58,708	-0,26	78,718	-0,18
23	20	28,012	-	39,217	-	58,743	-	78,627	-
24	30	28,127	0,41	39,358	0,36	58,741	0,00	78,771	0,18
25	40	28,020	0,03	39,180	-0,09	58,453	-0,49	78,430	-0,25
26	30	28,122	0,39	39,350	0,34	58,777	0,06	78,752	0,16
27	20	28,253	-	39,523	-	58,991	-	78,859	-

Manufacturer OPSIS AB Measuring range 0 to 1000 $\mu\text{g}/\text{m}^3$ Instrument type SM 200 Serial No. SN 288 Relative deviations from the starting point were determined at 20 °C.									
					Climatic chamber		Temperaturr program		
					Climatic chamber relative air humidity		60%		
					Standards		Reference films		
No..	Temperature [°C]	Film 1 readingg [$\mu\text{g}/\text{m}^3$]	Dev. [%]	Film 2 reading [$\mu\text{g}/\text{m}^3$]	Dev. [%]	Film 3 reading [$\mu\text{g}/\text{m}^3$]	Dev. [%]	Film 4 reading [$\mu\text{g}/\text{m}^3$]	Dev. [%]
1	20	28,856	-	40,101	-	59,395	-	78,808	-
2	10	29,042	0,64	40,325	0,56	59,645	0,42	78,996	0,24
3	5	28,976	0,42	40,264	0,41	59,625	0,39	79,024	0,27
4	10	28,899	0,15	40,296	0,49	59,624	0,39	79,014	0,26
5	20	28,806	-	40,224	-	59,528	-	78,847	-
6	30	28,785	-0,07	40,205	-0,05	59,435	-0,16	78,896	0,06
7	40	28,621	-0,64	40,023	-0,50	59,192	-0,57	78,649	-0,25
8	30	28,803	-0,01	40,177	-0,12	59,433	-0,16	78,826	-0,03
9	20	28,943	-	40,305	-	59,566	-	78,933	-
10	20	28,808	-	40,202	-	59,468	-	78,838	-
11	10	28,944	0,47	40,306	0,26	59,630	0,27	79,003	0,21
12	5	28,944	0,47	40,348	0,36	59,662	0,33	79,096	0,33
13	10	29,007	0,69	40,391	0,47	59,740	0,46	79,115	0,35
14	20	28,968	-	40,362	-	59,577	-	79,009	-
15	30	28,755	-0,74	40,121	-0,60	59,350	-0,38	78,781	-0,29
16	40	28,395	-1,98	39,755	-1,50	58,894	-1,15	78,395	-0,78
17	30	28,509	-1,58	39,999	-0,90	59,152	-0,71	78,687	-0,41
18	20	28,728	-	40,186	-	59,398	-	78,873	-
19	20	29,003	-	40,422	-	59,649	-	79,055	-
20	10	29,097	0,32	40,528	0,26	59,791	0,24	79,215	0,20
21	5	29,197	0,67	40,610	0,47	59,938	0,48	79,319	0,33
22	10	29,132	0,44	40,550	0,32	59,886	0,40	79,200	0,18
23	20	28,940	-	40,367	-	59,632	-	79,053	-
24	30	28,737	-0,70	40,240	-0,31	59,443	-0,32	78,847	-0,26
25	40	28,637	-1,05	40,096	-0,67	59,255	-0,63	78,636	-0,53
26	30	28,772	-0,58	40,239	-0,32	59,450	-0,30	78,676	-0,48
27	20	28,923	-	40,381	-	59,610	-	79,049	-

Appendix 5 Mains voltage / frequency dependence of the instrument's sensitivity

Sheet 1 of 1

Manufacturer OPSIS AB		Standards Reference							
Measuring	0 bis 1000 µg/m³								
Instrument type	SM 200								
Serial No.	SN 285 / SN 288								
Deviations were computed using the following equation:		Deviation = [(m(x2)-m(x1))/m(x1)]							
SN 285 Nr.	Mains voltage [V]	Film 1 reading [mg]	Dev. [%]	Film 2 reading [mg]	Dev. [%]	Film 3 reading [mg]	Dev. [%]	Film 4 reading [mg]	Dev. [%]
1	230	28,173	-	39,362	-	58,814	-	78,708	-
2	210	28,332	0,56	39,543	0,46	58,963	0,25	78,918	0,27
3	230	28,122	-0,74	39,350	-0,49	58,777	-0,32	78,753	-0,21
4	245	28,253	0,47	39,523	0,44	58,991	0,36	78,859	0,13
SN 288 Nr.	Mains voltage [V]	Film 1 reading [mg]	Dev. [%]	Film 2 reading [mg]	Dev. [%]	Film 3 reading [mg]	Dev. [%]	Film 4 reading [mg]	Dev. [%]
1	230	28,509	-	39,999	-	59,152	-	78,687	-
2	210	28,728	0,77	40,186	0,47	59,398	0,42	78,873	0,24
3	230	28,773	0,15	40,239	0,13	59,450	0,09	78,676	-0,25
4	245	28,923	0,52	40,381	0,35	59,610	0,27	79,049	0,47
SN 285 Nr.	Mains frequency [Hz]	Film 1 reading [mg]	Abw. [%]	Film 2 reading [mg]	Dev. [%]	Film 3 reading [mg]	Dev. [%]	Film 4 reading [mg]	Dev. [%]
1	50	28,067	-	39,302	-	59,599	-	78,959	-
2	48	28,118	0,18	39,395	0,24	59,655	0,09	79,003	0,06
3	50	28,100	-0,06	39,442	0,12	59,586	-0,12	78,776	-0,29
4	52	27,991	-0,39	39,265	-0,45	59,490	-0,16	78,859	0,11
SN 288 Nr.	Mains frequency [Hz]	Film 1 reading [mg]	Abw. [%]	Film 2 reading [mg]	Dev. [%]	Film 3 reading [mg]	Dev. [%]	Film 4 reading [mg]	Dev. [%]
1	50	28,280	-	39,648	-	59,512	-	79,022	-
2	48	28,376	0,34	39,766	0,30	59,567	0,09	79,058	0,05
3	50	28,333	-0,15	39,752	-0,04	59,576	0,01	78,841	-0,28
4	52	28,171	-0,57	39,580	-0,43	59,435	-0,24	78,923	0,11

Appendix 6

Zero point cross-sensitivity

Sheet 1 of 1

Manufacturer OPSIS			Climatic chamber 20			
Measuring range 0 to 1000 µg/m³			Rel. air humidity 60%			
Instrument type SM 200			Zero gas Dust-free zero air			
Serial No. SN 285 & SN 288			H2O			
			Target value 0 µg/m³			
No	Date	Rel. air humidity [%]	8-hr readings [µg/m³] (supplied by instrument) SN 285	Target deviation [µg/m³] SN 285 [c(actual)-c(target)]	8-hr readings [µg/m³] (supplied by instrument) SN 288	Target deviation [µg/m³] SN 288 [c(actual)-c(target)]
1	04-04-2001 (SN 275)	>80	0	0	0,5	0,5
2		>80	0	0	0	0
3	22-05-2001(SN 288)	>80	0	0	0	0
No.	Date	Rel. air humidity [%]	8-hr readings [µg/m³] (readings computed from cpm) SN	Target deviation [µg/m³] SN 285 [c(actual)-c(target)]	8-hr readings [µg/m³] (readings computed from cpm) SN 288	Target deviation [µg/m³] SN 288 [c(actual)-c(target)]
1	04-04-2001 (SN 285)	>80	-0,08	-0,08	0,49	0,49
2		>80	0,04	0,04	0,04	0,04
3	22.05.2001 (SN 288)	>80	0,04	0,04	-0,19	-0,19

Appendix 7

Readings from the field test sites

Sheet 1 of 12

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Cologne parking lot			
Instrument type		SM 200					
Serial No.		SN 285 & SN 288					
Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
1	15.12.00	17	13,1	14,7			Cologne
2	16.12.00		23,5	25,7			
3	17.12.00		41,8	40,8			
4	18.12.00		27,7	27,6			
5	19.12.00		35	30,7			
6	20.12.00	22,8	10,7	13,1			
7	21.12.00		22	21,2			
8	22.12.00		46,3	45,4			
9	23.12.00		42	39,3			
10	24.12.00		27	32,4			
11	25.12.00	39,1	19,7	20			
12	26.12.00		32,7	32,5			
13	27.12.00		42,9	41,7			
14	28.12.00		20,5	22,1			
15	29.12.00		25,6	23,7			
16	30.12.00	20,7	33,2	32,2			
17	31.12.00		25,6	31,8			
18	01.01.01		16	16,4			
19	02.01.01		20,3	20,9			
20	03.01.01		11,2	9,6			
21	04.01.01	8,2	5,8	5,6			
22	05.01.01		7	4,1			

Appendix 7

Readings from the field test sites

Sheet 2 of 12

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Cologne parking lot / Wesseling			
Instrument type		SM 200					
Serial No.		SN 285 & SN 288					
Average concentration levels ? 100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
23	06.01.01		11,7	13,4			Köln
24	07.01.01		15,8	16,3			
25	08.01.01	22,7	22,8	23,7			
26	09.01.01	29,3	23	27,6	38,7	75,6	
27	10.01.01	17,7	17,9	17,9	22,3	79,7	
28	11.01.01	28	27,9	27,9			
29	12.01.01	28,4	25,6	27	40,4	70,4	
30	13.01.01	37,6	34,6	36,5			
31	14.01.01		44,3	45,4			
32	15.01.01		36,5	30			
33	16.01.01	53,5	53,6	55,6			
34	17.01.01	76,9	76,4	82,9	91,9	83,7	
35	18.01.01	53,9	56,9	55,7	59,4	90,8	
36	22.01.01	18,1	14,9	17,4	21,9	82,5	
37	23.01.01		0,3	3,2			
38	27.01.01		10,7	10,3			
39	28.01.01	36,4	36,3	30,6	79,7	45,7	
40	20.04.02	40,9	50,3	48,7			Wesseling
41	21.04.02	43,2	49,9	47,8	45,5	94,9	
42	22.04.02		47,3	49,1			
43	23.04.02	47,8	51,4	51,7	61,3	78,0	
44	24.04.02	70,6	71,6	71,8			

Appendix 7

Readings from the field test sites

Sheet 3 of 12

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 285 & SN 288 Average concentration levels $\leq 100 \mu\text{g}/\text{m}^3$ Measured variable: Suspended particulates PM10, outdoor ambient air Site: Wesseling							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
45	25.04.02		78,1	77,5			Wesseling
46	26.04.02		36,1	36,8			
47	27.04.02	15,1	14,1	14,3			
48	28.04.02	15,9	13,3	13,8			
49	29.04.02	16,4	16,9	16,5			
50	30.04.02		26,7	26,5			
51	01.05.02	15,2	14,7	14,9	22,1	68,8	
52	02.05.02	14,6	16,8	14,8			
53	03.05.02		34,9	32,8			
54	04.05.02		19,6	21,4			
55	05.05.02	8,2	8,5	8,8	10	82,0	
56	06.05.02	19,9	23,1	23,1			
57	07.05.02	19,5	25,6	22	50,9	38,3	
58	08.05.02		45,9	45,2			
59	09.05.02	50,1	50,6	52,7	89,1	56,2	
60	10.05.02	40,6	42,8	41,1	49,2	82,5	
61	11.05.02		66	64,8			
62	12.05.02	67	71,1	68,5	83,7	80,0	
63	13.05.02	32,2	33,3	33,5	48,2	66,8	
64	14.05.02		17,5	16,9			
65	15.05.02	20,2	27,4	26,4	43	47,0	
66	16.05.02	30,6	31,4	32,8			

Appendix 7

Readings from the field test sites

Sheet 4 of 12

Manufacturer	OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air				
Measuring range	0 ... 1000 µg/m³		Site: Wesseling				
Instrument type	SM 200						
Serial No.	SN 285 & SN 288						
Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OP SIS SN 285 [µg/m³]	OP SIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
67	17.05.02		48	50,2			Wesseling
68	18.05.02	26,6	27,5	28,4	36,5	72,9	
69	19.05.02	16,6	17,9	17,3			
70	22.05.02	20,4	21,8	24,3	36	56,7	
71	23.05.02	15,7	22,5	19,3	24,7	63,6	
72	24.05.02		24,8	24,6			
73	25.05.02	14	14,9	17,6	19	73,7	
74	26.05.02	12,4	13,2	14,6	20	62,0	
75	27.05.02		13,6	14,7			
76	28.05.02	17,4	22	24,5	37,2	46,8	
77	29.05.02	15,8	21,9	19,3	20,3	77,8	
78	30.05.02		19,3	18,6			
79	31.05.02	20,8	24,7	27	36,9	56,4	
80	01.06.02	23,8	27,5	27,4	47,2	50,4	
81	02.06.02		10,6	11,7			
82	03.06.02	22,6	21,1	23,2	34,5	65,5	
83	04.06.02	32,6	30,7	34,4	44,9	72,6	
84	05.06.02		22,6	23,6			
85	06.06.02		19,6	21,7			
86	07.06.02	38,6	36,5	43,7	50,7	76,1	
87	08.06.02	20,2	18,6	21,2	21,6	93,5	
88	09.06.02		19,3	19,7			

Appendix 7

Readings from the field test sites

Sheet 5 of 12

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Wesseling / Mechernich			
Instrument type		SM 200					
Serial No.		SN 285 & SN 288					
Average concentration levels $\leq 100 \mu\text{g}/\text{m}^3$							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
89	11.06.02	19,1	16,6	15	23,7	80,6	Wesseling
90	12.06.02	10,1	13,7	13,6	11	91,8	
91	13.06.02		14	13,8			
92	14.06.02	18,9	23,2	21,2	31	61,0	
93	15.06.02	24,5	32,7	28,6	31,3	78,3	
94	16.06.02		26,3	22,8			
95	17.06.02		38,7	41,5			
96	18.06.02	52,5	50,9	55,4	60,7	86,5	
97	19.06.02	27,7	32,1	30,2	48,9	56,6	
98	20.06.02		38,2	38,9			
99	21.06.02	36,9	37,2	37,5	48,9	75,5	
100	22.06.02	27,9	29,4	31,4	32,8	85,1	
101	23.06.02		16,4	16			Mechernich
102	02.07.02	7,4	7	8,7	12,1	61,2	
103	03.07.02	5,6	8	10	5,8	96,6	
104	04.07.02		18,2	21,6			
105	05.07.02		17,2	16,2			
106	06.07.02	9,2	13,1	8,9	12,8	71,9	
107	07.07.02	18,3	20,8	19,6	19,5	93,8	
108	08.07.02		14,2	15,5			
109	09.07.02	18,6	21,2	19,1	24,8	75,0	
110	11.07.02	15,5	16,7	13,2	16,4	94,5	

Appendix 7

Readings from the field test sites

Sheet 6 of 12

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Mechernich			
Instrument type		SM 200					
Serial No.		SN 285 & SN 288					
Average concentration levels $\leq 100 \mu\text{g}/\text{m}^3$							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
111	12.07.02		15,8	16,8			Mechernich
112	14.07.02	35,1	37,9	35,8	40,3	87,1	
113	15.07.02		32,3	31,1			
114	16.07.02	32,4	32	33,4	43,8	74,0	
115	17.07.02	31,1	31,3	29,7	39,8	78,1	
116	18.07.02	20	19,3	18,2	26,6	75,2	
117	23.07.02	13,1	12,9	12	20,0	65,5	
118	24.07.02	11	12,1	12,4			
119	25.07.02	30,1	28,2	28,7	37,0	81,4	
120	26.07.02		24,9	29,9			
121	27.07.02	12,7	19,4	19,8	22,4	56,7	
122	28.07.02	12,9	16,9	17,9			
123	29.07.02		24	24			
124	30.07.02	26,8	27,7	29,3	38,9	68,9	
125	31.07.02		21,9	23,4			
126	01.08.02	18,3	20,4	20,2	26,1	70,1	
127	02.08.02		15	15,9			
128	03.08.02	12,4	12,5	12,2	14,1	87,9	
129	04.08.02	9,5	8,8	9,6	12,1	78,5	
130	05.08.02		11,2	11,7			
131	06.08.02	25,7	27,8	26	26,5	97,0	
132	07.08.02	24,1	27,8	27,3	27,2	88,6	

Appendix 7

Readings from the field test sites

Sheet 7 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Mechernich			
Instrument type		SM 200					
Serial No.		SN 285 & SN 288					
Average concentration levels ≤100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OP SIS SN 285 [µg/m³]	OP SIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
133	08.08.02	24,9	27,8	24,9	32,6	76,4	Mechernich
134	09.08.02		16,3	15,5			
135	10.08.02	17,8	21,2	21			
136	11.08.02	34,3	35,8	36,6	35,9	95,5	
137	12.08.02		23,4	23,4			
138	13.08.02	27,2	33,2	31	32,4	84,0	
139	14.08.02	31	35,9	34,6	44,8	69,2	
140	15.08.02		22,4	22,3			
141	16.08.02		22,5	22			
142	17.08.02	16,7	22,5	24,1			
143	18.08.02	26,6	26,6	28,7	33	80,6	
144	21.08.02	14,6	10,5	10,9	15,2	96,1	
145	22.08.02		17,1	17			
146	23.08.02		38,5	37,3			
147	24.08.02	32,9	37,2	36	35,5	92,7	
148	25.08.02	35,2	29,4	29,7	38,76	90,8	
149	27.08.02	64,5	65,8	63,5	78,9	81,7	
150	28.08.02	75,8	75,9	75,3	84,2	90,0	
151	29.08.02		79,9	77,6			
152	03.09.02	15,3	15,3	14,8	24	63,8	
153	04.09.02	16,7	17,3	18,6	17,5	95,4	
154	05.09.02	25,4	28	27,8	31	81,9	

Appendix 7

Readings from the field test sites

Sheet 8 of 12

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Mechernich			
Instrument type		SM 200					
Serial No.		SN 285 & SN 288					
Average concentration levels ≤100 µg/m³							
No.	Date	Reference PM10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
155	06.09.02		27,8	26,4			Mechernich
156	07.09.02	21,3	21,7	22,8	28,2	75,5	
157	08.09.02	19,7	20,7	20	20,1	98,0	
158	09.09.02		16,1	16,3			
159	10.09.02	12,8	13,4	12,2	13,6	94,1	
160	11.09.02	18,9	21,3	22,6	21,2	89,2	
161	12.09.02	27,5	30,1	30,7	35	78,6	
162	13.09.02		22,1	21,1			
163	14.09.02	35,9	33,4	35,6	41	87,6	
164	15.09.02	19,4	17,6	18,7	24,1	80,5	
165	16.09.02		20,7	19			
166	17.09.02	26,3	27,8	26,4	34,1	77,1	
167	18.09.02	35,4	38,5	38,7	43,2	81,9	
168	19.09.02		17,7	20,5			
169	20.09.02		19,7	23,5			
170	21.09.02	22,8	20,8	23,8	36,1	63,2	
171	22.09.02	16,2	9,1	11,5	19,6	82,7	
172	23.09.02		11,4	11,5			
173	24.09.02	9,1	10,7	9,8	15,9	57,2	
174	25.09.02	15,6	14,9	12,5	18,9	82,5	
175	30.09.02	10	6,8	7,1	15,1	66,2	
176	01.10.02	13,5	11	10,6	16,1	83,9	

Appendix 7

Readings from the field test sites

Sheet 9 of 12

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 285 & SN 288 Measured variable: Suspended particulates PM10, outdoor ambient air Site: Mechernich Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
177	02.10.02		20,5	17,6			Mechernich
178	03.10.02		22	27,2			
179	04.10.02		17	14,9			
180	05.10.02		12,6	8,2			
181	06.10.02		5,5	5,8			
182	07.10.02		15,1	10,6			
183	08.10.02	14,5	14	15,4	21,4	67,8	
184	09.10.02	8,7	7,5	7,1	14,2	61,3	
185	10.10.02		15,4	18,4			
186	11.10.02		28	31,1			
187	12.10.02	24,5	24	22,7	31,6	77,5	
188	13.10.02	15,3	14,5	12,6	16,2	94,4	
189	14.10.02		10,7	9,1			
190	15.10.02		12,1	13,4			
191	16.10.02	10	5,4	6,1	11,6	86,2	
192	17.10.02	9,5	6,4	5,6	15	63,3	
193	18.10.02		11,1	11,5			
194	19.10.02		11,4	9,6			
195	20.10.02	11,5	10,8	7,7	21,6	53,2	
196	21.10.02	5,6	4,1	4,3	12,7	44,1	
197	22.10.02	3,1	2,4	1,4	6,6	47,0	
198	23.10.02		7,6	6			

Appendix 7

Readings from the field test sites

Sheet 10 of 12

Manufacturer	OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air Site: Mechernich / Brühl				
Measuring range	0 ... 1000 µg/m³						
Instrument type	SM 200						
Serial No.	SN 285 & SN 288						
Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
199	24.10.02		7	8,4			Mechernich
200	25.10.02		0,9	0,1			
201	26.10.02		10,7	11,5			
202	27.10.02		11	12,1			
203	29.10.02	12,7	6,4	8,5	18,1	70,2	
204	30.10.02	12,4	10,1	9,4	14,8	83,8	
205	31.10.02		17,8	19,1			
206	01.11.02		17,8	15,6			
207	02.11.02		7,2	5			
208	03.11.02		3,5	4,4			
209	04.11.02		14,5	13,4			
210	05.11.02	32,3	30,4	29	38	85,0	
211	06.11.02	15,3	16	12,6	15,6	98,1	
212	07.11.02		2,4	3			
213	08.11.02		5,2	4,4			
214	09.11.02	11,4	5,1	5,8	12,2	93,4	
215	10.11.02	25,2	22	22,2	27,7	91,0	
216	20.11.02	30,4	27,5	31,2	33,6	90,5	Brühl
217	21.11.02		15	15,1			
218	22.11.02	15	14,5	14,6	23,3	64,4	
219	23.11.02	14,4	13,9	13,8	16,8	85,7	
220	24.11.02		13,5	11,6			

Appendix 7

Readings from the field test sites

Sheet 11 of 12

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 285 & SN 288 Measured variable: Suspended particulates PM10, outdoor ambient air Site: Brühl Average concentration levels ≤ 100 µg/m³							
Nr.	Datum	Referenz PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	Verhältnis PM10/TSP [%]	Standort
221	25.11.02		20,7	17,8			Brühl
222	26.11.02	35	35,5	30,9	35,3	99,2	
223	27.11.02	19,8	16,9	16,5	23,8	83,2	
224	28.11.02		21,1	19,9			
225	29.11.02		18,9	19,3			
226	30.11.02	39,3	38,6	37,4	44,5	88,3	
227	01.12.02	16,9	17,6	14,8	18,7	90,4	
228	02.12.02		5,5	4			
229	03.12.02	24,1	23,1	21,6	27	89,3	
230	04.12.02	27,2	22	23,4	32,6	83,4	
231	05.12.02		28,6	28,6			
232	06.12.02		20,4	21			
233	07.12.02	28,4	21,7	23,9	52,4	54,2	
234	08.12.02	31,8	33,5	32	66,4	47,9	
235	09.12.02		24,1	24,1			
236	10.12.02	34,4	29,5	27,4	36,5	94,2	
237	13.12.02	53	54,1	51,3	53,6	98,9	
238	14.12.02	53,9	54,2	48,4	57,3	94,1	
239	15.12.02		20,1	23,4			
240	16.12.02		13	10,5			
241	17.12.02	43,3	41,5	41,1	46,9	92,3	
242	18.12.02	34,2	32,5	29,7	36,9	92,7	

Appendix 7

Readings from the field test sites

Sheet 12 of 12

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 285 & SN 288 Average concentration levels ≤ 100 µg/m³ Measured variable: Suspended particulates PM10, outdoor ambient air Site: Brühl							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 285 [µg/m³]	OPSIS SN 288 [µg/m³]	TSP [µg/m³]	PM10/TSP ratio [%]	Site
243	19.12.02	21,1 19,3	19,7	21,2			Brühl
244	20.12.02		24,4	23,3			
245	21.12.02		19,5	19			
246	22.12.02		7,2	6,6			
247	23.12.02		7,5	8			
248	24.12.02		10,9	9,6			
249	25.12.02		12,2	11,4			
250	26.12.02		6,3	6			
251	27.12.02		4,2	5,8			
252	28.12.02		9,7	10			
253	29.12.02		6,2	5,6			
254	30.12.02		2,7	3,2			
255	31.12.02		10,9	8,9			
256	01.01.03		10,3	11,8			
257	02.01.03		0	0,1			
258	03.01.03		7,5	7,5			
259	04.01.03		20,9	17,9			
260	05.01.03		21,5	19,7			

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 1 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
1	15.12.00	Parkplatz Köln	6,0	997	82,2	4,2	266	4,7
2	16.12.00		4,0	1003	65,9	3,9	269	0,0
3	17.12.00		1,5	1008	82,2	0,3	242	0,0
4	18.12.00		2,9	1011	84,8	1,8	122	1,1
5	19.12.00		4,7	1014	84,2	2,4	106	0,0
6	20.12.00		2,4	1015	62,3	4,2	123	0,0
7	21.12.00		1,4	1016	42,1	5,1	133	0,0
8	22.12.00		-1,4	1013	55,3	4,8	127	0,0
9	23.12.00		0,3	1005	59,9	2,1	137	0,0
10	24.12.00		2,8	995	72,0	1,0	146	1,1
11	25.12.00		2,2	990	87,8	2,0	56	0,0
12	26.12.00		1,1	985	93,3	0,3	39	4,0
13	27.12.00		3,6	982	98,6	1,1	107	4,0
14	28.12.00		1,8	989	84,0	2,1	248	0,7
15	29.12.00		1,5	991	77,6	0,8	225	0,0
16	30.12.00		1,7	994	82,0	2,7	262	1,1
17	31.12.00		2,3	1011	81,1	1,8	228	0,0
18	01.01.01		3,1	995	75,1	5,1	143	10,8
19	02.01.01		9,3	990	75,5	3,6	160	2,5
20	03.01.01		7,9	997	69,3	3,0	213	1,1
21	04.01.01		7,7	993	72,2	4,1	170	4,3
22	05.01.01		9,2	985	78,7	3,7	186	12,6
23	06.01.01		8,2	992	81,1	1,7	239	0,4
24	07.01.01		5,1	1003	83,7	1,0	227	0,4
25	08.01.01		5,1	1006	78,3	1,0	229	0,7
26	09.01.01		5,7	1008	78,1	1,0	249	0,0
27	10.01.01		2,9	1005	87,3	2,2	105	13,0
28	11.01.01		3,7	1013	85,1	1,9	64	0,0
29	12.01.01		0,6	1020	79,8	1,0	64	0,4
30	13.01.01		-0,2	1026	55,0	0,9	75	0,0

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 2 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
31	14.01.01	Parkplatz Köln	-0,6	1026	72,9	0,2	103	0,0
32	15.01.01		-0,5	1020	39,7	1,2	88	0,0
33	16.01.01		-2,1	1017	51,7	0,7	144	0,0
34	17.01.01		-2,5	1013	60,6	0,9	109	0,0
35	18.01.01		1,9	1012	72,3	2,0	124	2,2
36	22.01.01		4,7	1002	94,8	2,5	131	7,9
37	23.01.01		8,4	995	78,7	3,3	167	3,2
38	27.01.01		4,8	989	75,5	4,2	201	7,6
39	28.01.01		3,8	996	75,7	0,5	242	0,4
40	20.04.02	Wesseling	12,1	1022	63,2	3,5	309	0,0
41	21.04.02		15,2	1027	52,2	1,3	198	0,0
42	22.04.02		15,5	1025	62,1	0,9	121	0,0
43	23.04.02		16,2	1022	58,3	3,6	292	0,0
44	24.04.02		16,7	1019	65,7	2,5	231	0,0
45	25.04.02		16,7	1013	68,3	2,9	226	0,0
46	26.04.02		12,5	1002	72,6	4,0	271	6,8
47	27.04.02		10,5	1002	64,9	5,6	275	1,1
48	28.04.02		11,6	996	71,1	5,0	215	10,4
49	29.04.02		12,2	997	64,7	5,8	254	1,4
50	30.04.02		13,8	1002	50,2	5,0	224	0,0
51	01.05.02		15,3	999	53,5	4,7	228	0,0
52	02.05.02		13,4	1001	66,4	3,5	218	0,0
53	03.05.02		10,7	1007	74,5	4,0	265	2,9
54	04.05.02		10,1	1007	92,5	4,7	315	38,9
55	05.05.02		9,3	1007	92,4	4,8	272	0,7
56	06.05.02		11,3	1015	77,9	1,3	154	1,4
57	07.05.02		16,0	1014	61,4	1,7	115	0,0
58	08.05.02		19,1	1007	62,7	1,4	100	0,0
59	09.05.02		20,5	1000	67,4	1,4	216	0,0
60	10.05.02		18,9	1000	81,5	2,3	189	1,4

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 3 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
61	11.05.02	Wesseling	17,8	1002	76,5	3,0	253	0,0
62	12.05.02		15,5	1006	85,6	1,4	176	13,3
63	13.05.02		17,4	1008	68,9	3,1	170	0,0
64	14.05.02		16,9	1009	56,7	5,5	233	0,0
65	15.05.02		17,8	1018	54,5	2,8	209	0,0
66	16.05.02		21,7	1014	56,7	1,5	157	0,0
67	17.05.02		23,5	1006	53,1	1,3	107	0,0
68	18.05.02		18,6	1007	76,8	3,3	252	2,9
69	19.05.02		18,2	1013	56,9	2,9	241	0,0
70	22.05.02		20,5	995	54,4	3,4	198	0,0
71	23.05.02		16,5	1001	69,2	2,7	218	1,4
72	24.05.02		17,3	1003	57,2	4,2	224	0,7
73	25.05.02		16,0	1005	58,1	3,6	246	1,4
74	26.05.02		16,0	1002	56,5	4,3	200	0,4
75	27.05.02		16,7	1001	51,9	3,4	195	0,0
76	28.05.02		18,0	1002	50,1	2,9	198	0,0
77	29.05.02		16,9	1008	49,5	4,5	272	0,0
78	30.05.02		16,6	1016	53,0	2,5	189	0,0
79	31.05.02		17,7	1018	51,9	3,6	264	0,0
80	01.06.02		19,2	1018	35,5	2,1	85	0,0
81	02.06.02		20,1	1014	33,9	2,9	145	0,0
82	03.06.02		20,1	1005	51,9	3,9	167	3,6
83	04.06.02		22,5	1000	57,4	2,7	165	0,0
84	05.06.02		20,9	992	70,4	2,8	160	0,0
85	06.06.02		21,0	995	58,5	3,0	175	0,0
86	07.06.02		18,5	997	65,8	4,9	326	0,0
87	08.06.02		19,7	1001	50,7	2,5	223	0,0
88	09.06.02		20,1	1002	59,9	1,4	204	1,8
89	11.06.02		16,3	1008	73,6	3,1	205	4,0
90	12.06.02		18,8	1008	76,0	3,6	228	4,3

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 4 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
91	13.06.02	Wesseling	17,5	1009	82,3	4,0	260	1,4
92	14.06.02		22,5	1009	62,3	2,6	232	7,9
93	15.06.02		21,7	1008	67,8	4,2	265	0,0
94	16.06.02		22,8	1011	63,4	1,6	180	0,0
95	17.06.02		27,9	1010	56,7	1,5	171	0,0
96	18.06.02		29,8	1008	51,9	4,4	218	0,0
97	19.06.02		22,3	1015	64,2	3,5	237	0,0
98	20.06.02		22,2	1011	76,0	3,8	235	21,6
99	21.06.02		21,2	1014	61,4	2,6	188	0,0
100	22.06.02		24,1	1013	52,3	2,9	227	0,0
101	23.06.02		20,9	1013	55,2	4,1	266	0,0
102	02.07.02	Mechernich	13,8	956	75,1	3,2	291	1,4
103	03.07.02		14,0	954	86,4	3,6	279	3,2
104	04.07.02		15,3	965	70,9	2,0	285	0,0
105	05.07.02		17,1	966	60,9	1,9	283	1,1
106	06.07.02		16,3	966	78,5	0,9	286	3,6
107	07.07.02		18,0	969	68,0	1,0	294	0,0
108	08.07.02		22,1	965	51,5	1,7	278	0,0
109	09.07.02		22,8	961	56,8	1,5	274	2,9
110	11.07.02		16,1	970	66,0	1,5	283	0,0
111	12.07.02		18,1	972	59,6	0,7	199	0,4
112	14.07.02		16,7	972	97,6	1,6	228	13,3
113	15.07.02		18,5	971	93,5	1,4	141	0,4
114	16.07.02		17,7	967	84,0	1,3	181	0,0
115	17.07.02		16,9	966	85,3	2,0	203	0,0
116	18.07.02		14,8	968	73,8	1,8	217	0,0
117	23.07.02		16,9	966	63,5	1,7	292	0,0
118	24.07.02		16,5	964	80,5	1,4	255	1,4
119	25.07.02		16,4	967	68,9	0,6	262	0,7
120	26.07.02		17,6	969	80,2	0,7	278	0,0

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 5 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
121	27.07.02	Mechernich	22,7	971	65,2	0,9	200	0,0
122	28.07.02		24,4	971	44,3	1,7	192	0,0
123	29.07.02		25,7	970	42,0	0,9	179	0,0
124	30.07.02		23,8	964	60,0	1,2	237	15,8
125	31.07.02		21,0	961	80,1	0,7	209	54,7
126	01.08.02		19,1	963	84,3	0,7	237	1,1
127	02.08.02		19,4	966	60,5	0,8	268	1,4
128	03.08.02		16,0	965	81,9	0,7	220	3,6
129	04.08.02		18,1	964	66,7	1,1	276	0,0
130	05.08.02		16,7	965	72,0	0,6	262	0,0
131	06.08.02		15,5	964	87,3	0,2	235	27,0
132	07.08.02		17,5	967	78,8	0,8	198	0,0
133	08.08.02		18,6	965	72,4	0,5	204	0,4
134	09.08.02		18,5	958	74,7	1,0	253	10,4
135	10.08.02		17,3	956	80,7	0,8	228	0,0
136	11.08.02		17,1	958	87,4	0,9	248	1,4
137	12.08.02		17,2	962	76,5	1,4	256	0,0
138	13.08.02		18,7	970	68,4	1,1	248	0,0
139	14.08.02		19,8	972	68,8	1,0	195	0,0
140	15.08.02		22,4	970	52,5	0,9	205	0,0
141	16.08.02		22,9	970	48,5	1,0	191	0,0
142	17.08.02		23,4	968	48,4	1,3	195	0,0
143	18.08.02		24,4	966	47,6	1,5	209	0,0
144	21.08.02		16,2	966	96,5	0,7	280	11,2
145	22.08.02		16,4	969	92,5	0,7	241	0,4
146	23.08.02		17,4	969	90,0	0,8	149	0,0
147	24.08.02		17,9	970	86,8	0,4	246	7,9
148	25.08.02		16,0	974	83,5	0,9	187	0,0
149	27.08.02		19,5	966	93,0	0,6	211	0,7
150	28.08.02		20,2	967	90,9	1,0	217	35,3

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 6 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
151	29.08.02	Mechernich	19,6	969	86,5	0,6	226	0,0
152	03.09.02		15,2	969	56,4	1,8	163	2,2
153	04.09.02		16,2	964	78,9	1,3	266	1,1
154	05.09.02		17,7	966	71,3	0,6	247	0,4
155	06.09.02		17,4	968	64,1	1,2	289	0,0
156	07.09.02		17,4	965	62,6	1,7	286	0,0
157	08.09.02		18,7	963	59,6	1,6	284	0,4
158	09.09.02		19,5	964	63,7	1,4	261	0,4
159	10.09.02		14,1	971	80,3	0,7	190	2,9
160	11.09.02		14,1	975	87,6	1,3	170	0,4
161	12.09.02		16,0	975	73,8	2,6	114	0,0
162	13.09.02		14,4	975	61,4	0,9	159	0,0
163	14.09.02		13,7	973	86,5	1,0	290	0,4
164	15.09.02		14,1	972	76,0	1,8	274	0,0
165	16.09.02		12,7	970	80,3	0,8	311	0,0
166	17.09.02		12,8	969	81,4	0,6	319	0,0
167	18.09.02		11,9	967	85,2	0,9	287	0,0
168	19.09.02		13,5	966	72,5	1,0	285	0,0
169	20.09.02		13,3	969	77,3	1,1	251	0,0
170	21.09.02		12,3	968	82,3	0,2	206	0,0
171	22.09.02		11,4	962	89,6	1,1	270	9,4
172	23.09.02		9,3	964	84,8	3,9	187	0,4
173	24.09.02		8,3	970	68,1	3,8	134	0,0
174	25.09.02		8,3	968	93,8	0,5	271	5,4
175	30.09.02		13,3	977	48,7	1,8	213	0,0
176	01.10.02		15,9	974	49,8	0,8	325	0,0
177	02.10.02		15,2	972	64,9	1,3	322	0,0
178	03.10.02		15,2	973	84,9	0,3	323	5,4
179	04.10.02		13,4	976	81,3	0,9	212	0,0
180	05.10.02		10,7	970	93,4	1,0	293	13,3

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 7 of 9

No.	Date	Site	Air temperature	Air press.	Relative humid.	Wind speed	Wind direction	Rainfall/Precipitation
			[°C]	[hPa]	[%]	[m/s]	[°]	[mm]
181	06.10.02	Mechernich	9,9	965	98,2	2,0	193	5,4
182	07.10.02		7,9	968	82,6	0,3	256	0,4
183	08.10.02		7,3	965	88,3	1,6	117	0,0
184	09.10.02		7,3	961	73,7	2,5	98	0,0
185	10.10.02		7,8	960	80,1	1,9	94	0,0
186	11.10.02		5,9	963	92,3	2,0	75	0,0
187	12.10.02		6,8	962	88,6	0,8	146	0,7
188	13.10.02		9,0	962	94,8	0,5	191	0,0
189	14.10.02		9,7	958	84,0	3,9	243	0,0
190	15.10.02		10,8	960	90,2	2,7	268	2,5
191	16.10.02		12,3	953	86,7	3,4	295	7,9
192	17.10.02		9,3	952	91,7	0,9	327	2,5
193	18.10.02		6,3	958	89,6	1,6	307	3,2
194	19.10.02		6,1	966	83,8	1,1	283	0,4
195	20.10.02		5,6	967	72,8	1,3	206	1,1
196	21.10.02		11,3	952	86,9	3,8	247	0,0
197	22.10.02		13,1	945	87,4	3,4	267	13,0
198	23.10.02		9,8	948	84,4	3,2	274	3,2
199	24.10.02		7,5	961	75,5	2,9	260	0,0
200	25.10.02		10,3	952	93,7	3,5	262	26,3
201	26.10.02		9,7	961	77,5	4,1	283	11,5
202	27.10.02		10,5	959	81,5	6,5	277	6,5
203	29.10.02		8,0	970	74,3	2,1	230	0,7
204	30.10.02		10,8	965	97,3	0,4	218	4,7
205	31.10.02		10,5	968	98,7	0,3	197	0,0
206	01.11.02		11,3	968	91,0	2,0	272	9,0
207	02.11.02		12,8	963	85,7	1,3	206	8,6
208	03.11.02		11,1	958	86,5	2,6	232	3,2
209	04.11.02		8,3	964	94,0	0,8	259	0,4
210	05.11.02		7,8	973	96,3	0,4	215	1,8

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 8 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
211	06.11.02	Mechernich	5,9	969	86,8	2,7	254	6,5
212	07.11.02		4,8	961	97,0	1,8	252	8,3
213	08.11.02		4,7	961	90,5	3,9	272	14,8
214	09.11.02		6,9	952	91,6	2,7	255	1,1
215	10.11.02		5,2	958	92,5	2,4	260	2,4
216	20.11.02	Brühl	7,9	1002	84,8	3,0	134	0,0
217	21.11.02		9,3	991	78,2	3,5	125	2,2
218	22.11.02		9,3	992	84,6	1,5	180	0,4
219	23.11.02		8,4	998	78,2	3,0	144	0,4
220	24.11.02		8,6	1000	87,6	1,2	151	2,9
221	25.11.02		8,2	995	95,0	1,7	183	4,7
222	26.11.02		9,1	1005	84,7	1,6	255	0,4
223	27.11.02		7,5	1008	85,6	2,2	169	0,0
224	28.11.02		9,5	1003	80,4	2,6	154	2,5
225	29.11.02		9,3	1005	89,6	0,5	198	0,7
226	30.11.02		9,1	1011	88,2	1,3	282	0,0
227	01.12.02		7,9	1003	79,9	3,5	161	11,2
228	02.12.02		7,0	994	86,3	2,1	170	1,1
229	03.12.02		7,5	1003	86,5	0,4	151	0,0
230	04.12.02		6,7	1007	83,4	1,0	162	0,0
231	05.12.02		6,5	1011	77,1	2,6	119	0,0
232	06.12.02		3,4	1018	71,3	5,0	50	0,0
233	07.12.02		2,4	1016	67,1	4,2	85	0,0
234	08.12.02		0,7	1018	53,5	5,6	109	0,0
235	09.12.02		-2,5	1019	28,1	4,7	111	0,0
236	10.12.02		-3,5	1017	35,8	4,8	117	0,0
237	13.12.02		0,8	1015	70,7	2,6	117	1,1
238	14.12.02		3,0	1012	77,6	2,8	121	1,1
239	15.12.02		3,4	1005	74,1	6,4	131	1,8
240	16.12.02		6,8	1004	75,3	3,3	225	1,1

Appendix 8

Ambient Conditions at the Field Test Sites

Sheet 9 of 9

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
241	17.12.02	Brühl	2,9	1013	89,1	3,7	254	0,4
242	18.12.02		2,0	1023	61,4	1,6	132	0,0
243	19.12.02		1,1	1022	74,4	2,4	121	0,0
244	20.12.02		1,6	1012	79,5	1,4	133	1,1
245	21.12.02		4,9	1010	98,9	0,3	223	2,2
246	22.12.02		9,2	1001	91,1	5,1	185	18,0
247	23.12.02		8,1	1005	89,3	2,9	147	1,4
248	24.12.02		10,8	1003	78,6	2,6	140	1,8
249	25.12.02		10,9	1006	84,4	2,9	135	0,7
250	26.12.02		9,4	999	83,6	4,0	142	1,8
251	27.12.02		11,0	990	80,9	3,4	131	5,8
252	28.12.02		10,4	991	80,3	2,3	234	0,7
253	29.12.02		8,8	996	79,0	3,1	220	18,4
254	30.12.02		13,1	991	73,6	6,6	273	14,4
255	31.12.02		2,9	1002	74,4	4,7	185	0,0
256	01.01.03		5,1	996	76,8	4,7	172	11,5
257	02.01.03		11,4	980	68,7	6,4	239	14,4
258	03.01.03		9,0	988	72,5	5,0	271	4,3
259	04.01.03		1,0	998	65,3	4,7	132	0,0
260	05.01.03		-1,3	1005	65,5	2,6	189	0,0

Appendix 9

Comparability of the instruments in the field test

Sheet 1 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Cologne parking lot	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
1	15.12.00	13,1	14,7	2,56	Cologne parking lot site
2	16.12.00	23,5	25,7	4,84	
3	17.12.00	41,8	40,8	1	
4	18.12.00	27,7	27,6	0,01	
5	19.12.00	35	30,7	18,49	
6	20.12.00	10,7	13,1	5,76	
7	21.12.00	22	21,2	0,64	
8	22.12.00	46,3	45,4	0,81	
9	23.12.00	42	39,3	7,29	
10	24.12.00	27	32,4	29,16	
11	25.12.00	19,7	20	0,09	
12	26.12.00	32,7	32,5	0,04	
13	27.12.00	42,9	41,7	1,44	
14	28.12.00	20,5	22,1	2,56	
15	29.12.00	25,6	23,7	3,61	
16	30.12.00	33,2	32,2	1	
17	31.12.00	25,6	31,8	38,44	
18	01.01.01	16	16,4	0,16	
19	02.01.01	20,3	20,9	0,36	
20	03.01.01	11,2	9,6	2,56	
21	04.01.01	5,8	5,6	0,04	
22	05.01.01	7	4,1	8,41	

Appendix 9

Comparability of the instruments in the field test

Sheet 2 of 12

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 285 & SN 288 Average concentration levels ≤ 100 µg/m³ Measured variable: Suspended particulates PM10, outdoor ambient air Site: Cologne parking lot / Wesseling					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)² [µg/m³]²	Site
		SN 285	SN 288		
23	06.01.01	11,7	13,4	2,89	Cologne parking lot
24	07.01.01	15,8	16,3	0,25	
25	08.01.01	22,8	23,7	0,81	
26	09.01.01	23	27,6	21,16	
27	10.01.01	17,9	17,9	0	
28	11.01.01	27,9	27,9	0	
29	12.01.01	25,6	27	1,96	
30	13.01.01	34,6	36,5	3,61	
31	14.01.01	44,3	45,4	1,21	
32	15.01.01	36,5	30	42,25	
33	16.01.01	53,6	55,6	4	
34	17.01.01	76,4	82,9	42,25	
35	18.01.01	56,9	55,7	1,44	
36	22.01.01	14,9	17,4	6,25	
37	23.01.01	0,3	3,2	8,41	
38	27.01.01	10,7	10,3	0,16	
39	28.01.01	36,3	30,6	32,49	
40	20.04.02	50,3	48,7	2,56	Wesseling
41	21.04.02	49,9	47,8	4,41	
42	22.04.02	47,3	49,1	3,24	
43	23.04.02	51,4	51,7	0,09	
44	24.04.02	71,6	71,8	0,04	

Appendix 9

Comparability of the instruments in the field test

Sheet 3 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Wesseling	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
45	25.04.02	78,1	77,5	0,36	Wesseling
46	26.04.02	36,1	36,8	0,49	
47	27.04.02	14,1	14,3	0,04	
48	28.04.02	13,3	13,8	0,25	
49	29.04.02	16,9	16,5	0,16	
50	30.04.02	26,7	26,5	0,04	
51	01.05.02	14,7	14,9	0,04	
52	02.05.02	16,8	14,8	4	
53	03.05.02	34,9	32,8	4,41	
54	04.05.02	19,6	21,4	3,24	
55	05.05.02	8,5	8,8	0,09	
56	06.05.02	23,1	23,1	0	
57	07.05.02	25,6	22	12,96	
58	08.05.02	45,9	45,2	0,49	
59	09.05.02	50,6	52,7	4,41	
60	10.05.02	42,8	41,1	2,89	
61	11.05.02	66	64,8	1,44	
62	12.05.02	71,1	68,5	6,76	
63	13.05.02	33,3	33,5	0,04	
64	14.05.02	17,5	16,9	0,36	
65	15.05.02	27,4	26,4	1	
66	16.05.02	31,4	32,8	1,96	

Appendix 9

Comparability of the instruments in the field test

Sheet 4 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Wesseling	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
67	17.05.02	48	50,2	4,84	Wesseling
68	18.05.02	27,5	28,4	0,81	
69	19.05.02	17,9	17,3	0,36	
70	22.05.02	21,8	24,3	6,25	
71	23.05.02	22,5	19,3	10,24	
72	24.05.02	24,8	24,6	0,04	
73	25.05.02	14,9	17,6	7,29	
74	26.05.02	13,2	14,6	1,96	
75	27.05.02	13,6	14,7	1,21	
76	28.05.02	22	24,5	6,25	
77	29.05.02	21,9	19,3	6,76	
78	30.05.02	19,3	18,6	0,49	
79	31.05.02	24,7	27	5,29	
80	01.06.02	27,5	27,4	0,01	
81	02.06.02	10,6	11,7	1,21	
82	03.06.02	21,1	23,2	4,41	
83	04.06.02	30,7	34,4	13,69	
84	05.06.02	22,6	23,6	1	
85	06.06.02	19,6	21,7	4,41	
86	07.06.02	36,5	43,7	51,84	
87	08.06.02	18,6	21,2	6,76	
88	09.06.02	19,3	19,7	0,16	

Appendix 9

Comparability of the instruments in the field test

Sheet 5 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Wesseling / Mechernich	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
89	11.06.02	16,6	15	2,56	Wesseling
90	12.06.02	13,7	13,6	0,01	
91	13.06.02	14	13,8	0,04	
92	14.06.02	23,2	21,2	4	
93	15.06.02	32,7	28,6	16,81	
94	16.06.02	26,3	22,8	12,25	
95	17.06.02	38,7	41,5	7,84	
96	18.06.02	50,9	55,4	20,25	
97	19.06.02	32,1	30,2	3,61	
98	20.06.02	38,2	38,9	0,49	
99	21.06.02	37,2	37,5	0,09	
100	22.06.02	29,4	31,4	4	Mechernich
101	23.06.02	16,4	16	0,16	
102	02.07.02	7	8,7	2,89	
103	03.07.02	8	10	4	
104	04.07.02	18,2	21,6	11,56	
105	05.07.02	17,2	16,2	1	
106	06.07.02	13,1	8,9	17,64	
107	07.07.02	20,8	19,6	1,44	
108	08.07.02	14,2	15,5	1,69	
109	09.07.02	21,2	19,1	4,41	
110	11.07.02	16,7	13,2	12,25	

Appendix 9

Comparability of the instruments in the field test

Sheet 6 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Mechernich	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
111	12.07.02	15,8	16,8	1	Mechernich
112	14.07.02	37,9	35,8	4,41	
113	15.07.02	32,3	31,1	1,44	
114	16.07.02	32	33,4	1,96	
115	17.07.02	31,3	29,7	2,56	
116	18.07.02	19,3	18,2	1,21	
117	23.07.02	12,9	12	0,81	
118	24.07.02	12,1	12,4	0,09	
119	25.07.02	28,2	28,7	0,25	
120	26.07.02	24,9	29,9	25	
121	27.07.02	19,4	19,8	0,16	
122	28.07.02	16,9	17,9	1	
123	29.07.02	24	24	0	
124	30.07.02	27,7	29,3	2,56	
125	31.07.02	21,9	23,4	2,25	
126	01.08.02	20,4	20,2	0,04	
127	02.08.02	15	15,9	0,81	
128	03.08.02	12,5	12,2	0,09	
129	04.08.02	8,8	9,6	0,64	
130	05.08.02	11,2	11,7	0,25	
131	06.08.02	27,8	26	3,24	
132	07.08.02	27,8	27,3	0,25	

Appendix 9

Comparability of the instruments in the field test

Sheet 7 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Mechernich	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
133	08.08.02	27,8	24,9	8,41	Mechernich
134	09.08.02	16,3	15,5	0,64	
135	10.08.02	21,2	21	0,04	
136	11.08.02	35,8	36,6	0,64	
137	12.08.02	23,4	23,4	0	
138	13.08.02	33,2	31	4,84	
139	14.08.02	35,9	34,6	1,69	
140	15.08.02	22,4	22,3	0,01	
141	16.08.02	22,5	22	0,25	
142	17.08.02	22,5	24,1	2,56	
143	18.08.02	26,6	28,7	4,41	
144	21.08.02	10,5	10,9	0,16	
145	22.08.02	17,1	17	0,01	
146	23.08.02	38,5	37,3	1,44	
147	24.08.02	37,2	36	1,44	
148	25.08.02	29,4	29,7	0,09	
149	27.08.02	65,8	63,5	5,29	
150	28.08.02	75,9	75,3	0,36	
151	29.08.02	79,9	77,6	5,29	
152	03.09.02	15,3	14,8	0,25	
153	04.09.02	17,3	18,6	1,69	
154	05.09.02	28	27,8	0,04	

Appendix 9

Comparability of the instruments in the field test

Sheet 8 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Mechernich	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
155	06.09.02	27,8	26,4	1,96	Mechernich
156	07.09.02	21,7	22,8	1,21	
157	08.09.02	20,7	20	0,49	
158	09.09.02	16,1	16,3	0,04	
159	10.09.02	13,4	12,2	1,44	
160	11.09.02	21,3	22,6	1,69	
161	12.09.02	30,1	30,7	0,36	
162	13.09.02	22,1	21,1	1	
163	14.09.02	33,4	35,6	4,84	
164	15.09.02	17,6	18,7	1,21	
165	16.09.02	20,7	19	2,89	
166	17.09.02	27,8	26,4	1,96	
167	18.09.02	38,5	38,7	0,04	
168	19.09.02	17,7	20,5	7,84	
169	20.09.02	19,7	23,5	14,44	
170	21.09.02	20,8	23,8	9	
171	22.09.02	9,1	11,5	5,76	
172	23.09.02	11,4	11,5	0,01	
173	24.09.02	10,7	9,8	0,81	
174	25.09.02	14,9	12,5	5,76	
175	30.09.02	6,8	7,1	0,09	
176	01.10.02	11	10,6	0,16	

Appendix 9

Comparability of the instruments in the field test

Sheet 9 of 12

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Mechernich	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)² [µg/m³]²	Site
		SN 285	SN 288		
177	02.10.02	20,5	17,6	8,41	Mechernich
178	03.10.02	22	27,2	27,04	
179	04.10.02	17	14,9	4,41	
180	05.10.02	12,6	8,2	19,36	
181	06.10.02	5,5	5,8	0,09	
182	07.10.02	15,1	10,6	20,25	
183	08.10.02	14	15,4	1,96	
184	09.10.02	7,5	7,1	0,16	
185	10.10.02	15,4	18,4	9	
186	11.10.02	28	31,1	9,61	
187	12.10.02	24	22,7	1,69	
188	13.10.02	14,5	12,6	3,61	
189	14.10.02	10,7	9,1	2,56	
190	15.10.02	12,1	13,4	1,69	
191	16.10.02	5,4	6,1	0,49	
192	17.10.02	6,4	5,6	0,64	
193	18.10.02	11,1	11,5	0,16	
194	19.10.02	11,4	9,6	3,24	
195	20.10.02	10,8	7,7	9,61	
196	21.10.02	4,1	4,3	0,04	
197	22.10.02	2,4	1,4	1	
198	23.10.02	7,6	6	2,56	

Appendix 9

Comparability of the instruments in the field test

Sheet 10 of 12

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 285 & SN 288 Average concentration levels ≤ 100 µg/m³					
Measured variable: Suspended particulates PM10, outdoor ambient air Site: Mechernich / Brühl					
No.	Date	24-hr.readings [µg/m³]		(SN 285 - SN 288)² [µg/m³]²	Site
		SN 285	SN 288		
199	24.10.02	7	8,4	1,96	Mechernich
200	25.10.02	0,9	0,1	0,64	
201	26.10.02	10,7	11,5	0,64	
202	27.10.02	11	12,1	1,21	
203	29.10.02	6,4	8,5	4,41	
204	30.10.02	10,1	9,4	0,49	
205	31.10.02	17,8	19,1	1,69	
206	01.11.02	17,8	15,6	4,84	
207	02.11.02	7,2	5	4,84	
208	03.11.02	3,5	4,4	0,81	
209	04.11.02	14,5	13,4	1,21	
210	05.11.02	30,4	29	1,96	
211	06.11.02	16	12,6	11,56	
212	07.11.02	2,4	3	0,36	
213	08.11.02	5,2	4,4	0,64	
214	09.11.02	5,1	5,8	0,49	
215	10.11.02	22	22,2	0,04	
216	20.11.02	27,5	31,2	13,69	Brühl
217	21.11.02	15	15,1	0,01	
218	22.11.02	14,5	14,6	0,01	
219	23.11.02	13,9	13,8	0,01	
220	24.11.02	13,5	11,6	3,61	

Appendix 9

Comparability of the instruments in the field test

Sheet 11 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Brühl	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
221	25.11.02	20,7	17,8	8,41	Brühl
222	26.11.02	35,5	30,9	21,16	
223	27.11.02	16,9	16,5	0,16	
224	28.11.02	21,1	19,9	1,44	
225	29.11.02	18,9	19,3	0,16	
226	30.11.02	38,6	37,4	1,44	
227	01.12.02	17,6	14,8	7,84	
228	02.12.02	5,5	4	2,25	
229	03.12.02	23,1	21,6	2,25	
230	04.12.02	22	23,4	1,96	
231	05.12.02	28,6	28,6	0	
232	06.12.02	20,4	21	0,36	
233	07.12.02	21,7	23,9	4,84	
234	08.12.02	33,5	32	2,25	
235	09.12.02	24,1	24,1	0	
236	10.12.02	29,5	27,4	4,41	
237	13.12.02	54,1	51,3	7,84	
238	14.12.02	54,2	48,4	33,64	
239	15.12.02	20,1	23,4	10,89	
240	16.12.02	13	10,5	6,25	
241	17.12.02	41,5	41,1	0,16	
242	18.12.02	32,5	29,7	7,84	

Appendix 9

Comparability of the instruments in the field test

Sheet 12 of 12

Manufacturer		OP SIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range		0 ... 1000 µg/m³		Site: Brühl	
Instrument type		SM 200			
Serial No.		SN 285 & SN 288			
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 285 - SN 288)²	Site
		SN 285	SN 288	[µg/m³]²	
243	19.12.02	19,7	21,2	2,25	Brühl
244	20.12.02	24,4	23,3	1,21	
245	21.12.02	19,5	19	0,25	
246	22.12.02	7,2	6,6	0,36	
247	23.12.02	7,5	8	0,25	
248	24.12.02	10,9	9,6	1,69	
249	25.12.02	12,2	11,4	0,64	
250	26.12.02	6,3	6	0,09	
251	27.12.02	4,2	5,8	2,56	
252	28.12.02	9,7	10	0,09	
253	29.12.02	6,2	5,6	0,36	
254	30.12.02	2,7	3,2	0,25	
255	31.12.02	10,9	8,9	4	
256	01.01.03	10,3	11,8	2,25	
257	02.01.03	0	0,1	0,01	
258	03.01.03	7,5	7,5	0	
259	04.01.03	20,9	17,9	9	
260	05.01.03	21,5	19,7	3,24	



Appendix 10: Start-up screen

Sheet 1 of 1

MN009n	10:31:20	27/03/00		
	UsePrg	ModPrg	Buffer	Tools
[F1]	[F2]	[F3]	[F4]	[F5]

Appendix 11

Calibration and analysis function

Sheet 1 of 1

Manufacturer OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air Site: Furulund		
Measuring range	0 ... 1000 µg/m³			
Instrument type	SM 200			
Serial No.	SN 1110 & SN 1112 (New)			

No.	Date	24-hr. readings [µg/m³]		Ref. method readings [µg/m³]	Calibration and analysis function
		SN 1110	SN 1112		
1	18.12.2004	6,5	7,1	6,9	Calibration fuinction $y = m \cdot x + b$
2	20.12.2004	7,1	6,7	5,6	
3	22.12.2004	19,6	19,3	17,2	
4	24.12.2004	7,9	8,1	5,7	
5	26.12.2004	10,4	11,9	10,4	
6	28.12.2004	14,7	15,5	15,7	
7	30.12.2004	19,4	18,9	17,1	
8	02.01.2005	6,9	10,1	7,3	SN 1110 $y = 0,9911 \cdot x + 0,6756$ $R^2 = 0,9261$
9	04.01.2005	17,0	18,0	16,0	
10	06.01.2005	21,2	22,4	19,7	
11	09.01.2005	19,2	19,3	19,1	
12	11.01.2005	27,2	27	26,2	
13	13.01.2005	18,9	18,6	18,8	
14	15.01.2005	12,8	16,1	16,9	Analysis function $x = 1/m \cdot y - b/m$
15	20.01.2005	15,6	14,3	13,6	
					SN 1110 $x = 1,0090 \cdot y - 0,6817$
					SN 1112 $x = 1,0477 \cdot y + 1,8813$

Appendix 12

Readings from the Furulund field test site

Sheet 1 of 5

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Furulund			
Instrument type		SM 200					
Serial No.		SN 1110 & SN 1112 (New)		SN 276 (Old)			
Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 1110 [µg/m³]	OPSIS SN 1112 [µg/m³]	OPSIS SN 276 [µg/m³]	Note	Site
1	18.12.2004	6,9	6,5	7,1	8,6		Furulund
2	19.12.2004		4,6	5,6	7,8		
3	20.12.2004	5,6	7,1	6,7	10,1		
4	21.12.2004		18,1	18,0	18,7		
5	22.12.2004	17,2	19,6	19,3	16,8		
6	23.12.2004		15,8	13,8	11,3		
7	24.12.2004	5,7	7,9	8,1	5,9		
8	25.12.2004		8,5	8,8	10,3		
9	26.12.2004	10,4	10,4	11,9	15,7		
10	27.12.2004		4,5	7,3	7,4		
11	28.12.2004	15,7	14,7	15,5	13,6		
12	29.12.2004		18,5	19,1	22,2		
13	30.12.2004	17,1	19,4	18,9	18,0		
14	31.12.2004		12,3	12,9	14,3		
15	01.01.2005		30,2	30,0	27,4		
16	02.01.2005	7,3	6,9	10,1	7,2		
17	03.01.2005		18,0	17,4	17,1		
18	04.01.2005	16,0	17,0	18,0	16,1		
19	05.01.2005		21,9	22,9	22,0		
20	06.01.2005	19,7	21,2	22,4	22,1		
21	07.01.2005		19,1	20,3	19,9		
22	08.01.2005		15,3	16,2	16,0		

Appendix 12

Readings from the Furulund field test site

Sheet 2 of 5

Manufacturer OPSIS AB

Measuring range 0 ... 1000 µg/m³

Instrument type SM 200

Serial No. SN 1110 & SN 1112 (New) SN 276 (Old)

Average concentration levels ≤ 100 µg/m³

Measured variable: Suspended particulates PM10, outdoor ambient air

Site: Furulund

No	Date	Reference PM 10 [µg/m³]	OPSIS SN 1110 [µg/m³]	OPSIS SN 1112 [µg/m³]	OPSIS SN 276 [µg/m³]	Note	Site
23	09.01.2005	19,1	19,2	19,3	20,3		Furulund
24	10.01.2005		22,7	21,1	21,1		
25	11.01.2005	26,2	27,2	27,0	24,8		
26	12.01.2005		25,0	24,5	21,2		
27	13.01.2005	18,8	18,9	18,6	17,0		
28	14.01.2005		15,4	17,4	17,5		
29	15.01.2005	16,9	12,8	16,1	17,0		
30	16.01.2005		10,3	14,2	16,6		
31	20.01.2005	13,6	15,6	14,3		SN 276 parametrization error	
32	21.01.2005		7,5	7,9		SN 276 parametrization error	
33	22.01.2005	7,5	4,6	5,6		SN 276 parametrization error	
34	23.01.2005		5,9	6,8		SN 276 parametrization error	
35	24.01.2005	2,3	3,1	4,8		SN 276 parametrization error	
36	25.01.2005		4,7	6,4		SN 276 parametrization error	
37	26.01.2005		17,0	17,3		SN 276 parametrization error	
38	27.01.2005	22,1	23,5	23,1		SN 276 parametrization error	
39	28.01.2005		16,8	16,6		SN 276 parametrization error	
40	29.01.2005		19,2	20,3		SN 276 parametrization error	
41	30.01.2005		31,1	31,7		SN 276 parametrization error	
42	31.01.2005		13,0	13,1		SN 276 parametrization error	
43	01.02.2005		10,0	8,9	12,5		
44	02.02.2005		16,4	16,1	18,4		

Appendix 12

Readings from the Furulund field test site

Sheet 3 of 5

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Furulund			
Instrument type		SM 200					
Serial No.		SN 1110 & SN 1112 (New)		SN 276 (Old)			
Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM [µg/m³]	OPSIS SN 1110 [µg/m³]	OPSIS SN 1112 [µg/m³]	OPSIS SN 276 [µg/m³]	Note	Site
45	03.02.2005	13,1	9,4	10,7	11,9		Furulund
46	04.02.2005		22,7	23,0	23,5		
47	05.02.2005		33,0	34,3	34,1		
48	06.02.2005		43,8	45,7	46,1		
49	07.02.2005	49,3	50,1	47,3		SN 276 error after power outage	
50	08.02.2005		58,0	57,1		SN 276 error after power outage	
51	09.02.2005	57,5	61,8	63,0		SN 276 error after power outage	
52	10.02.2005		39,4	38,1		SN 276 error after power outage	
53	11.02.2005		13,4	12,7		SN 276 error after power outage	
54	12.02.2005	11,9	16,7	16,5		SN 276 error after power outage	
55	13.02.2005		1,5	2,8		SN 276 error after power outage	
56	14.02.2005	7,6	1,5	7,9		SN 276 error after power outage	
57	19.02.2005	21,4	24,8	24,0	23,6		
58	20.02.2005		24,4	24,6	26,4		
59	21.02.2005	17,8	15,9	16,7	18,4		
60	22.02.2005		12,3	12,1	13,1		
61	23.02.2005	16,1	17,8	17,7	17,7		
62	24.02.2005		12,4	13,6	14,4		
63	25.02.2005	20,1	23,1	21,7	22,4		
64	26.02.2005		11,5	11,7	11,3		
65	27.02.2005	2,5	2,8	4,0	4,6		
66	28.02.2005		17,8	17,5	13,6		

Appendix 12

Readings from the Furulund field test site

Sheet 4 of 5

Manufacturer		OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range		0 ... 1000 µg/m³		Site: Furulund			
Instrument type		SM 200					
Serial No.		SN 1110 & SN 1112 (New)		SN 276 (Old)			
Average concentration levels ≤ 100 µg/m³							
No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 1110 [µg/m³]	OPSIS SN 1112 [µg/m³]	OPSIS SN 276 [µg/m³]	Note	Site
67	01.03.2005	9,9	10,6	13,3	8,1	Fatal Error 2 on SN 276 Fatal Error 2 on SN 276 <	

Appendix 12

Readings from the Furulund field test site

Sheet 5 of 5

Manufacturer

OPSIS AB

Measuring range

0 ... 1000 µg/m³

Instrument type

SM 200

Serial No.

SN 1110 & SN 1112 (New) SN 276 (Old)

Average concentration levels ≤ 100 µg/m³

Measured variable: Suspended particulates PM10, outdoor ambient air

Site: Furulund

No.	Date	Reference PM 10 [µg/m³]	OPSIS SN 1110 [µg/m³]	OPSIS SN 1112 [µg/m³]	OPSIS SN 276 [µg/m³]	Note	Site
89	28.03.2005	9,3	9,3	10	11,0		Furulund
90	29.03.2005		9,9	9,8	11,7		
91	30.03.2005	11,3	10,4	14,1	13,1		
92	31.03.2005		22,9	22,6	24,7		
93	01.04.2005	33,6	37,2	36,9	36,0		
94	02.04.2005		42,8	41,8	40,8		
95	03.04.2005	49,2	55,1	56,1	53,2		
96	04.04.2005		49,6	49	46,5		
97	05.04.2005	18,5	17,9	16,7	20,3		
98	06.04.2005		33,6	36	34,6		
99	07.04.2005	25,4	25,6	26,2	23,6		
100	08.04.2005		17,5	18,3	20,7		
101	09.04.2005		12,2	10,6	12,2		
102	10.04.2005		14,3	14,3	15,7		
103	11.04.2005		19,4	20,7	20,9		
104	12.04.2005		22,6	23,9	22,1		

Appendix 13

Ambient conditions at the Furulund field test site

Sheet 1 of 4

No.	Date	Site	Air temperature	Air press.	Relative humid.	Wind speed	Wind direction	Rainfall/Precipitation
			°C	hPa	%	m/s	°	mm
1	18.12.2004	Furulund	3,1	987	79,9	3,6	122	4,8
2	19.12.2004		-0,2	996	70,2	1,2	180	0,0
3	20.12.2004		0,8	1008	76,8	1,2	134	3,3
4	21.12.2004		1,1	1016	75,8	2,1	107	0,4
5	22.12.2004		2,6	1002	79,1	7,9	104	2,2
6	23.12.2004		6,4	986	77,2	5,6	117	2,2
7	24.12.2004		1,4	986	85,7	0,6	216	15,2
8	25.12.2004		3,7	984	81,0	4,8	105	22,2
9	26.12.2004		1,0	1003	86,3	0,8	164	0,0
10	27.12.2004		-2,2	1011	90,4	0,2	278	0,0
11	28.12.2004		-0,2	1002	78,2	1,6	56	0,0
12	29.12.2004		0,9	1012	85,9	0,4	78	0,0
13	30.12.2004		4,5	1014	85,5	6,7	104	1,9
14	31.12.2004		4,5	1012	91,6	0,4	132	1,1
15	01.01.2005		3,2	1011	88,5	2,2	107	0,0
16	02.01.2005		4,0	995	73,2	6,8	112	5,9
17	03.01.2005		4,2	1005	67,6	5,8	138	2,6
18	04.01.2005		7,3	1004	80,9	5,3	128	11,5
19	05.01.2005		5,4	1009	73,1	4,7	121	0,0
20	06.01.2005		5,6	1009	72,3	5,5	121	1,1
21	07.01.2005		8,0	1005	82,3	7,0	107	3,3
22	08.01.2005		8,1	994	67,6	10,6	97	3,0
23	09.01.2005		6,3	1003	65,0	7,2	64	0,0
24	10.01.2005		8,6	1005	75,1	6,8	161	0,0
25	11.01.2005		7,1	1012	77,6	4,1	187	0,0
26	12.01.2005		6,8	1007	73,0	6,5	195	0,4
27	13.01.2005		5,2	1006	66,0	6,0	65	0,0
28	14.01.2005		4,2	1017	77,6	1,5	67	0,0
29	15.01.2005		-0,6	1026	81,1	0,1	201	0,0
30	16.01.2005		2,5	1020	74,9	2,7	293	0,0

Appendix 13

Ambient conditions at the Furulund field test site

Sheet 2 of 4

No.	Date	Site	Air temperature [°C]	Air press. [hPa]	Relative humid. [%]	Wind speed [m/s]	Wind direction [°]	Rainfall/Precipitation [mm]
31	20.01.2005	Furulund	4,0	977	74,2	5,9	105	6,7
32	21.01.2005		1,9	981	68,6	2,4	66	0,0
33	22.01.2005		-0,8	999	74,6	0,3	91	0,0
34	23.01.2005		-1,0	1003	72,5	1,4	99	0,4
35	24.01.2005		-3,5	1013	66,3	0,5	155	0,0
36	25.01.2005		-4,4	1022	66,5	1,0	161	0,0
37	26.01.2005		-5,1	1025	70,3	0,1	125	0,0
38	27.01.2005		-2,7	1014	78,2	0,0	97	0,0
39	28.01.2005		-3,5	1012	72,5	0,2	173	0,0
40	29.01.2005		-4,5	1017	68,3	0,3	232	0,0
41	30.01.2005		2,7	1012	85,0	2,9	120	0,4
42	31.01.2005		4,5	1003	64,4	3,3	67	0,0
43	01.02.2005		2,2	1010	61,4	1,2	98	0,0
44	02.02.2005		0,7	1014	85,3	0,2	72	0,0
45	03.02.2005		0,6	1024	81,3	0,0	253	0,4
46	04.02.2005		2,7	1026	85,8	0,5	313	0,0
47	05.02.2005		2,4	1026	81,2	1,1	304	0,0
48	06.02.2005		-1,8	1030	74,3	1,9	266	0,0
49	07.02.2005		-1,8	1031	70,5	1,1	240	0,0
50	08.02.2005		-2,4	1028	65,1	0,9	268	0,0
51	09.02.2005		-0,8	1024	69,6	0,8	294	0,0
52	10.02.2005		3,2	1008	82,9	4,5	212	8,1
53	11.02.2005		2,4	1008	71,4	2,2	88	0,0
54	12.02.2005		0,7	988	84,8	1,5	269	11,5
55	13.02.2005		0,5	970	84,2	0,8	126	7,8
56	14.02.2005		0,1	989	72,4	1,6	136	0,0
57	19.02.2005		1,1	999	84,3	1,7	237	2,6
58	20.02.2005		2,0	1004	76,2	0,2	228	0,0
59	21.02.2005		0,8	1014	70,0	1,9	202	0,0
60	22.02.2005		-1,2	1016	64,1	3,6	182	0,0

Appendix 13

Ambient conditions at the Furulund field test site

Sheet 3 of 4

No.	Date	Site	Air temperature	Air press.	Relative humid.	Wind speed	Wind direction	Rainfall/Precipitation
			°C	hPa	%	m/s	°	mm
61	23.02.2005	Furulund	-0,3	1014	65,9	7,3	194	0,0
62	24.02.2005		1,1	1015	76,4	4,8	196	0,4
63	25.02.2005		-0,1	1015	70,7	1,2	133	0,0
64	26.02.2005		-2,0	1014	55,7	1,9	158	0,4
65	27.02.2005		-5,4	1018	44,9	1,4	159	0,0
66	28.02.2005		-3,3	1011	61,3	1,4	99	0,0
67	01.03.2005		-1,3	998	74,2	4,6	207	0,0
68	02.03.2005		-3,4	1000	68,0	5,3	205	0,0
69	03.03.2005		-6,7	1010	52,2	1,4	125	0,7
70	04.03.2005		-7,8	1010	56,1	0,4	163	0,0
71	05.03.2005		-5,2	1010	52,6	1,6	188	0,0
72	06.03.2005		-4,3	1013	57,5	0,5	124	0,4
73	07.03.2005		0,5	1008	70,5	0,5	85	2,2
74	08.03.2005		2,1	1009	60,6	0,5	129	0,0
75	09.03.2005		-0,9	1020	51,5	0,7	163	0,0
76	10.03.2005		-0,7	1018	63,4	2,7	41	0,0
77	11.03.2005		2,0	987	75,6	5,1	65	5,9
78	12.03.2005		0,0	984	70,7	1,7	168	0,4
79	13.03.2005		-3,0	999	63,6	0,8	176	0,0
80	14.03.2005		2,2	1002	65,9	4,0	49	1,5
81	15.03.2005		2,2	1010	69,8	2,2	148	4,8
82	16.03.2005		5,9	1007	83,9	4,5	136	5,2
83	17.03.2005		8,3	1003	80,0	4,6	80	0,7
84	18.03.2005		4,0	1011	79,2	1,1	134	4,4
85	19.03.2005		-0,8	1026	45,9	1,6	164	0,0
86	25.03.2005		8,4	1014	71,4	0,5	255	0,4
87	26.03.2005		6,0	1011	68,5	4,7	218	0,0
88	27.03.2005		4,7	1013	60,9	5,7	218	0,0
89	28.03.2005		4,6	1016	55,5	2,2	229	0,0
90	29.03.2005		2,6	1020	52,9	4,4	226	0,0

Appendix 13

Ambient conditions at the Furulund field test site

Sheet 4 of 4

No.	Date	Site	Air temperature	Air press.	Relative humid.	Wind speed	Wind direction	Rainfall/Precipitation
			[°C]	[hPa]	[%]	[m/s]	[°]	[mm]
91	30.03.2005	Furulund	2,3	1025	59,0	3,8	222	0,0
92	31.03.2005		3,8	1029	59,4	0,8	181	0,0
93	01.04.2005		4,2	1029	58,7	0,8	82	0,0
94	02.04.2005		6,1	1027	51,6	0,9	190	0,0
95	03.04.2005		8,8	1023	45,8	1,1	282	0,0
96	04.04.2005		11,4	1018	44,8	0,9	292	0,0
97	05.04.2005		8,9	1013	58,7	3,2	107	1,5
98	06.04.2005		8,5	1012	60,6	3,2	192	0,0
99	07.04.2005		8,6	995	67,8	3,9	262	3,7
100	08.04.2005		7,0	994	70,7	2,9	216	0,4
101	09.04.2005		5,7	1005	57,2	1,8	83	0,0
102	10.04.2005		4,3	1014	68,6	2,7	198	0,0
103	11.04.2005		10,1	1017	70,2	3,3	76	0,0
104	12.04.2005		10,0	1015	64,6	3,6	71	0,0

Appendix 14

Comparability of the instruments in the field test

Sheet 1 of 13

Manufacturer OPSIS AB

Measuring range 0 ... 1000 µg/m³

Instrument type SM 200

Serial No. SN 1110 & SN 1112 (New)

Average concentration levels ≤ 100 µg/m³

Measured variable: Suspended particulates PM10, outdoor ambient air

Site: Furulund

No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 1112)² [µg/m³]²	Site
		SN 1110	SN 1112		
1	18.12.2004	6,5	7,1	0,36	Furulund
2	19.12.2004	4,6	5,6	1,00	
3	20.12.2004	7,1	6,7	0,16	
4	21.12.2004	18,1	18	0,01	
5	22.12.2004	19,6	19,3	0,09	
6	23.12.2004	15,8	13,8	4,00	
7	24.12.2004	7,9	8,1	0,04	
8	25.12.2004	8,5	8,8	0,09	
9	26.12.2004	10,4	11,9	2,25	
10	27.12.2004	4,5	7,3	7,84	
11	28.12.2004	14,7	15,5	0,64	
12	29.12.2004	18,5	19,1	0,36	
13	30.12.2004	19,4	18,9	0,25	
14	31.12.2004	12,3	12,9	0,36	
15	01.01.2005	30,2	30	0,04	
16	02.01.2005	6,9	10,1	10,24	
17	03.01.2005	18	17,4	0,36	
18	04.01.2005	17	18	1,00	
19	05.01.2005	21,9	22,9	1,00	
20	06.01.2005	21,2	22,4	1,44	
21	07.01.2005	19,1	20,3	1,44	
22	08.01.2005	15,3	16,2	0,81	

Appendix 14

Comparability of the instruments in the field test

Sheet 2 of 13

Manufacturer	OP SIS AB				Measured variable: Suspended particulates PM10, outdoor ambient air Site: Furulund
Measuring range	0 ... 1000 µg/m³				
Instrument type	SM 200				
Serial No.	SN 1110 & SN 1112 (New)				
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 1112)²	Site
		SN 1110	SN 1112	[µg/m³]²	
23	09.01.2005	19,2	19,3	0,01	Furulund
24	10.01.2005	22,7	21,1	2,56	
25	11.01.2005	27,2	27	0,04	
26	12.01.2005	25	24,5	0,25	
27	13.01.2005	18,9	18,6	0,09	
28	14.01.2005	15,4	17,4	4,00	
29	15.01.2005	12,8	16,1	10,89	
30	16.01.2005	10,3	14,2	15,21	
31	20.01.2005	15,6	14,3	1,69	
32	21.01.2005	7,5	7,9	0,16	
33	22.01.2005	4,6	5,6	1,00	
34	23.01.2005	5,9	6,8	0,81	
35	24.01.2005	3,1	4,8	2,89	
36	25.01.2005	4,7	6,4	2,89	
37	26.01.2005	17	17,3	0,09	
38	27.01.2005	23,5	23,1	0,16	
39	28.01.2005	16,8	16,6	0,04	
40	29.01.2005	19,2	20,3	1,21	
41	30.01.2005	31,1	31,7	0,36	
42	31.01.2005	13	13,1	0,01	
43	01.02.2005	10	8,9	1,21	
44	02.02.2005	16,4	16,1	0,09	

Appendix 14

Comparability of the instruments in the field test

Sheet 3 of 13

Manufacturer

OPSIS AB

Measuring range

0 ... 1000 µg/m³

Instrument type

SM 200

Serial No.

SN 1110 & SN 1112 (New)

Average concentration levels ≤ 100 µg/m³

Measured variable: Suspended particulates PM10, outdoor ambient air

Site: Furulund

No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 1112)² [µg/m³]²	Site
		SN 1110	SN 1112		
45	03.02.2005	9,4	10,7	1,69	Furulund
46	04.02.2005	22,7	23	0,09	
47	05.02.2005	33	34,3	1,69	
48	06.02.2005	43,8	45,7	3,61	
49	07.02.2005	50,1	47,3	7,84	
50	08.02.2005	58	57,1	0,81	
51	09.02.2005	61,8	63	1,44	
52	10.02.2005	39,4	38,1	1,69	
53	11.02.2005	13,4	12,7	0,49	
54	12.02.2005	16,7	16,5	0,04	
55	13.02.2005	1,5	2,8	1,69	
56	14.02.2005	1,5	7,9	40,96	
57	19.02.2005	24,8	24	0,64	
58	20.02.2005	24,4	24,6	0,04	
59	21.02.2005	15,9	16,7	0,64	
60	22.02.2005	12,3	12,1	0,04	
61	23.02.2005	17,8	17,7	0,01	
62	24.02.2005	12,4	13,6	1,44	
63	25.02.2005	23,1	21,7	1,96	
64	26.02.2005	11,5	11,7	0,04	
65	27.02.2005	2,8	4	1,44	
66	28.02.2005	17,8	17,5	0,09	

Appendix 14

Comparability of the instruments in the field test

Sheet 4 of 13

Manufacturer	OP SIS AB			Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range	0 ... 1000 µg/m³			Site: Furulund	
Instrument type	SM 200				
Serial No.	SN 1110 & SN 1112 (New)				
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 1112)² [µg/m³]²	Site
		SN 1110	SN 1112		
67	01.03.2005	10,6	13,3	7,29	Furulund
68	02.03.2005	11	12,9	3,61	
69	03.03.2005	12,6	14,6	4,00	
70	04.03.2005	22,3	20,8	2,25	
71	05.03.2005	13,4	13,1	0,09	
72	06.03.2005	22,4	21,2	1,44	
73	07.03.2005	15,2	14,5	0,49	
74	08.03.2005	9,4	9,4	0,00	
75	09.03.2005	5,5	8	6,25	
76	10.03.2005	16,4	17,4	1,00	
77	11.03.2005	13,6	13,4	0,04	
78	12.03.2005	9,2	7,5	2,89	
79	13.03.2005	9,2	9,3	0,01	
80	14.03.2005	13,3	13,1	0,04	
81	15.03.2005	17,1	16,3	0,64	
82	16.03.2005	40,5	42	2,25	
83	17.03.2005	26,2	26,1	0,01	
84	18.03.2005	6,3	7,8	2,25	
85	19.03.2005	0	7,1	50,41	
86	25.03.2005	23,8	21,3	6,25	
87	26.03.2005	7,4	7,7	0,09	
88	27.03.2005	6,9	8,1	1,44	

Appendix 14

Comparability of the instruments in the field test

Sheet 5 of 13

Manufacturer	OP SIS AB			Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range	0 ... 1000 µg/m³			Site: Furulund	
Instrument type	SM 200				
Serial No.	SN 1110 & SN 1112 (New)				
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 1112)² [µg/m³]²	Site
		SN 1110	SN 1112		
89	28.03.2005	9,3	10	0,49	Furulund
90	29.03.2005	9,9	9,8	0,01	
91	30.03.2005	10,4	14,1	13,69	
92	31.03.2005	22,9	22,6	0,09	
93	01.04.2005	37,2	36,9	0,09	
94	02.04.2005	42,8	41,8	1,00	
95	03.04.2005	55,1	56,1	1,00	
96	04.04.2005	49,6	49	0,36	
97	05.04.2005	17,9	16,7	1,44	
98	06.04.2005	33,6	36	5,76	
99	07.04.2005	25,6	26,2	0,36	
100	08.04.2005	17,5	18,3	0,64	
101	09.04.2005	12,2	10,6	2,56	
102	10.04.2005	14,3	14,3	0,00	
103	11.04.2005	19,4	20,7	1,69	
104	12.04.2005	22,6	23,9	1,69	

Appendix 14

Comparability of the instruments in the field test

Sheet 6 of 13

Manufacturer

OP SIS AB

Measuring range

0 ... 1000 µg/m³

Instrument type

SM 200

Serial No.

SN 1110 (New) & SN 276 (Old)

Average concentration levels ≤ 100 µg/m³

Measured variable: Suspended particulates PM10, outdoor ambient air

Site: Furulund

No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 276)² [µg/m³]²	Site
		SN 1110	SN 276		
1	18.12.2004	6,5	8,6	4,36	Furulund
2	19.12.2004	4,6	7,8	10,16	
3	20.12.2004	7,1	10,1	9,25	
4	21.12.2004	18,1	18,7	0,41	
5	22.12.2004	19,6	16,8	7,88	
6	23.12.2004	15,8	11,3	20,46	
7	24.12.2004	7,9	5,9	4,20	
8	25.12.2004	8,5	10,3	3,13	
9	26.12.2004	10,4	15,7	28,00	
10	27.12.2004	4,5	7,4	8,51	
11	28.12.2004	14,7	13,6	1,19	
12	29.12.2004	18,5	22,2	13,82	
13	30.12.2004	19,4	18,0	2,10	
14	31.12.2004	12,3	14,3	3,98	
15	01.01.2005	30,2	27,4	7,79	
16	02.01.2005	6,9	7,2	0,09	
17	03.01.2005	18	17,1	0,84	
18	04.01.2005	17	16,1	0,83	
19	05.01.2005	21,9	22,0	0,02	
20	06.01.2005	21,2	22,1	0,78	
21	07.01.2005	19,1	19,9	0,67	
22	08.01.2005	15,3	16,0	0,50	

Appendix 14

Comparability of the instruments in the field test

Sheet 7 of 13

Manufacturer	OPSIS AB				Measured variable: Suspended particulates PM10, outdoor ambient air Site: Furulund
Measuring range	0 ... 1000 µg/m³				
Instrument type	SM 200				
Serial No.	SN 1110 (New) & SN 276 (Old)				
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 276)² [µg/m³]²	Site
		SN 1110	SN 276		
23	09.01.2005	19,2	20,3	1,28	Furulund
24	10.01.2005	22,7	21,1	2,52	
25	11.01.2005	27,2	24,8	5,57	
26	12.01.2005	25	21,2	14,66	
27	13.01.2005	18,9	17,0	3,46	
28	14.01.2005	15,4	17,5	4,27	
29	15.01.2005	12,8	17,0	17,26	
30	16.01.2005	10,3	16,6	40,03	
31	01.02.2005	10	12,5	6,39	
32	02.02.2005	16,4	18,4	4,03	
33	03.02.2005	9,4	11,9	6,23	
34	04.02.2005	22,7	23,5	0,57	
35	05.02.2005	33	34,1	1,12	
36	06.02.2005	43,8	46,1	5,20	
37	19.02.2005	24,8	23,6	1,49	
38	20.02.2005	24,4	26,4	3,87	
39	21.02.2005	15,9	18,4	6,25	
40	22.02.2005	12,3	13,1	0,67	
41	23.02.2005	17,8	17,7	0,02	
42	24.02.2005	12,4	14,4	4,01	
43	25.02.2005	23,1	22,4	0,54	
44	26.02.2005	11,5	11,3	0,03	

Appendix 14

Comparability of the instruments in the field test

Sheet 8 of 13

Manufacturer	OP SIS AB			Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range	0 ... 1000 µg/m³			Site: Furulund	
Instrument type	SM 200				
Serial No.	SN 1110 (New) & SN 276 (Old)				
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 276)² [µg/m³]²	Site
		SN 1110	SN 276		
45	27.02.2005	2,8	4,6	3,31	Furulund
46	28.02.2005	17,8	13,6	17,46	
47	01.03.2005	10,6	8,1	6,34	
48	02.03.2005	11	15,1	16,93	
49	03.03.2005	12,6	15,9	11,17	
50	04.03.2005	22,3	22,6	0,09	
51	05.03.2005	13,4	14,4	1,01	
52	06.03.2005	22,4	20,2	4,76	
53	07.03.2005	15,2	13,8	1,84	
54	08.03.2005	9,4	9,1	0,07	
55	11.03.2005	13,6	9,6	16,27	
56	12.03.2005	9,2	9,3	0,02	
57	13.03.2005	9,2	11,1	3,58	
58	14.03.2005	13,3	15,8	6,18	
59	15.03.2005	17,1	20,2	9,61	
60	16.03.2005	40,5	42,5	4,16	
61	17.03.2005	26,2	29,4	10,48	
62	25.03.2005	23,8	22,0	3,24	
63	26.03.2005	7,4	6,6	0,61	
64	27.03.2005	6,9	8,0	1,29	
65	28.03.2005	9,3	11,0	3,06	
66	29.03.2005	9,9	11,7	3,11	

Appendix 14

Comparability of the instruments in the field test

Sheet 9 of 13

Manufacturer	OP SIS AB			Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range	0 ... 1000 µg/m³			Site: Furulund	
Instrument type	SM 200				
Serial No.	SN 1110 & SN 276 (Old)				
Average concentration levels ≤ 100 µg/m³					

No.	Date	24-hr. readings [µg/m³]		(SN 1110 - SN 276)² [µg/m³]²	Site
		SN 1110	SN 276		
67	30.03.2005	10,4	13,1	7,27	Furulund
68	31.03.2005	22,9	24,7	3,27	
69	01.04.2005	37,2	36,0	1,33	
70	02.04.2005	42,8	40,8	4,18	
71	03.04.2005	55,1	53,2	3,72	
72	04.04.2005	49,6	46,5	9,86	
73	05.04.2005	17,9	20,3	5,76	
74	06.04.2005	33,6	34,6	0,90	
75	07.04.2005	25,6	23,6	3,84	
76	08.04.2005	17,5	20,7	10,25	
77	09.04.2005	12,2	12,2	0,00	
78	10.04.2005	14,3	15,7	2,00	
79	11.04.2005	19,4	20,9	2,18	
80	12.04.2005	22,6	22,1	0,24	

Appendix 14

Comparability of the instruments in the field test

Sheet 10 of 13

Manufacturer OPSIS AB		Measured variable: Suspended particulates PM10, outdoor ambient air			
Measuring range 0 ... 1000 µg/m³		Site: Furulund			
Instrument type SM 200					
Serial No. SN 1112 (New) & SN 276 (Old)					
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1112 - SN 276)² [µg/m³]²	Site
		SN 1112	SN 276		
1	18.12.2004	7,1	8,6	2,22	Furulund
2	19.12.2004	5,6	7,8	4,78	
3	20.12.2004	6,7	10,1	11,84	
4	21.12.2004	18	18,7	0,54	
5	22.12.2004	19,3	16,8	6,29	
6	23.12.2004	13,8	11,3	6,37	
7	24.12.2004	8,1	5,9	5,06	
8	25.12.2004	8,8	10,3	2,16	
9	26.12.2004	11,9	15,7	14,37	
10	27.12.2004	7,3	7,4	0,01	
11	28.12.2004	15,5	13,6	3,58	
12	29.12.2004	19,1	22,2	9,72	
13	30.12.2004	18,9	18,0	0,90	
14	31.12.2004	12,9	14,3	1,95	
15	01.01.2005	30	27,4	6,72	
16	02.01.2005	10,1	7,2	8,44	
17	03.01.2005	17,4	17,1	0,10	
18	04.01.2005	18	16,1	3,64	
19	05.01.2005	22,9	22,0	0,76	
20	06.01.2005	22,4	22,1	0,10	
21	07.01.2005	20,3	19,9	0,15	
22	08.01.2005	16,2	16,0	0,04	

Appendix 14

Comparability of the instruments in the field test

Sheet 11 of 13

Manufacturer	OP SIS AB			Measured variable: Suspended particulates PM10, outdoor ambient air	
Measuring range	0 ... 1000 µg/m³			Site: Furulund	
Instrument type	SM 200				
Serial No.	SN 1112 (New) & SN 276 (Old)				
Average concentration levels ≤ 100 µg/m³					
No.	Date	24-hr. readings [µg/m³]		(SN 1112 - SN 276)² [µg/m³]²	Site
		SN 1112	SN 276		
23	09.01.2005	19,3	20,3	1,06	Furulund
24	10.01.2005	21,1	21,1	0,00	
25	11.01.2005	27	24,8	4,66	
26	12.01.2005	24,5	21,2	11,08	
27	13.01.2005	18,6	17,0	2,44	
28	14.01.2005	17,4	17,5	0,00	
29	15.01.2005	16,1	17,0	0,73	
30	16.01.2005	14,2	16,6	5,89	
31	01.02.2005	8,9	12,5	13,16	
32	02.02.2005	16,1	18,4	5,32	
33	03.02.2005	10,7	11,9	1,43	
34	04.02.2005	23	23,5	0,21	
35	05.02.2005	34,3	34,1	0,06	
36	06.02.2005	45,7	46,1	0,14	
37	19.02.2005	24	23,6	0,18	
38	20.02.2005	24,6	26,4	3,12	
39	21.02.2005	16,7	18,4	2,89	
40	22.02.2005	12,1	13,1	1,04	
41	23.02.2005	17,7	17,7	0,00	
42	24.02.2005	13,6	14,4	0,64	
43	25.02.2005	21,7	22,4	0,44	
44	26.02.2005	11,7	11,3	0,14	

Appendix 14

Comparability of the instruments in the field test

Sheet 12 of 13

Manufacturer OPSIS AB Measuring range 0 ... 1000 µg/m³ Instrument type SM 200 Serial No. SN 1112 (New) & SN 276 (Old) Average concentration levels ≤ 100 µg/m³ Measured variable: Suspended particulates PM10, outdoor ambient air Site: Furulund					
No.	Date	24-hr. readings [µg/m³]		(SN 1112 - SN 276)²	Site
		SN 1112	SN 276	[µg/m³]²	
45	27.02.2005	4	4,6	0,38	Furulund
46	28.02.2005	17,5	13,6	15,05	
47	01.03.2005	13,3	8,1	27,24	
48	02.03.2005	12,9	15,1	4,90	
49	03.03.2005	14,6	15,9	1,80	
50	04.03.2005	20,8	22,6	3,24	
51	05.03.2005	13,1	14,4	1,70	
52	06.03.2005	21,2	20,2	0,96	
53	07.03.2005	14,5	13,8	0,43	
54	08.03.2005	9,4	9,1	0,07	
55	11.03.2005	13,4	9,6	14,69	
56	12.03.2005	7,5	9,3	3,39	
57	13.03.2005	9,3	11,1	3,21	
58	14.03.2005	13,1	15,8	7,22	
59	15.03.2005	16,3	20,2	15,21	
60	16.03.2005	42	42,5	0,29	
61	17.03.2005	26,1	29,4	11,14	
62	25.03.2005	21,3	22,0	0,49	
63	26.03.2005	7,7	6,6	1,17	
64	27.03.2005	8,1	8,0	0,00	
65	28.03.2005	10	11,0	1,10	
66	29.03.2005	9,8	11,7	3,47	

Appendix 14

Comparability of the instruments in the field test

Sheet 13 of 13

Manufacturer

OPSIS AB

Measuring range

0 ... 1000 µg/m³

Instrument type

SM 200

Serial No.

SN 1112 (New) & SN 276 (Old)

Average concentration levels ≤ 100 µg/m³

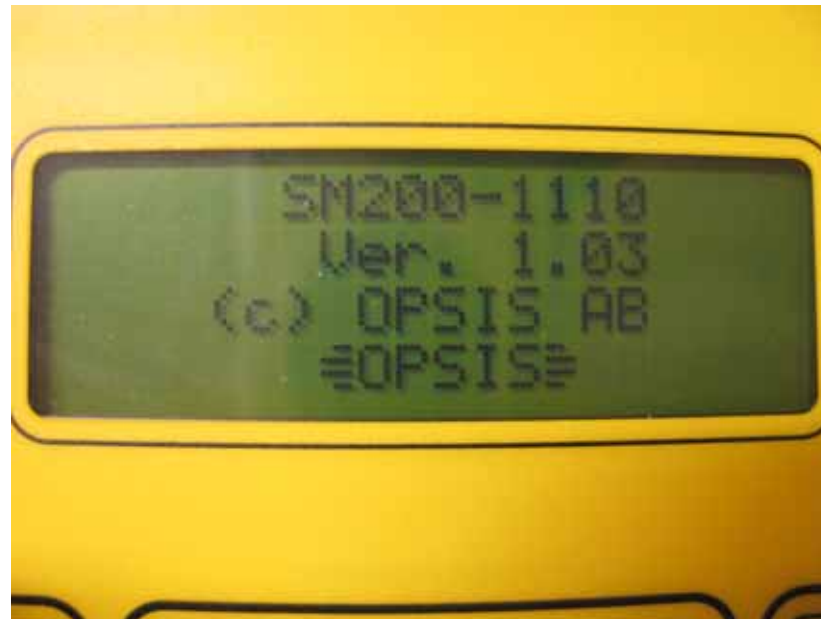
Measured variable: Suspended particulates PM10, outdoor ambient air

Site: Furulund

No.	Date	24-hr. readings [µg/m³]		(SN 1112 - SN 276)² [µg/m³]²	Site
		SN 1112	SN 276		
67	30.03.2005	14,1	13,1	1,01	Furulund
68	31.03.2005	22,6	24,7	4,44	
69	01.04.2005	36,9	36,0	0,73	
70	02.04.2005	41,8	40,8	1,09	
71	03.04.2005	56,1	53,2	8,57	
72	04.04.2005	49	46,5	6,45	
73	05.04.2005	16,7	20,3	12,97	
74	06.04.2005	36	34,6	2,10	
75	07.04.2005	26,2	23,6	6,55	
76	08.04.2005	18,3	20,7	5,77	
77	09.04.2005	10,6	12,2	2,41	
78	10.04.2005	14,3	15,7	2,00	
79	11.04.2005	20,7	20,9	0,03	
80	12.04.2005	23,9	22,1	3,20	

Appendix 15: Start-up screen OPSIS SM 200 (New)

Sheet 1 of 1





TÜV Rheinland Group

Seite 208 von 306

TÜV Immissionsschutz und Energiesysteme GmbH
Luftreinhaltung

Report on the Supplementary Test of the OPSIS SM 200 Emission Measuring
Instrument with PM10 Pre-separator marketed by OPSIS AB
for the PM10, Berichts-Nr.: 936/21201592/A

Annex 2

Manual